NON-WSQ COURSE

AI for Product Development

Discover → design → deploy → learn

C840 | v1.0 | 2 days / 15 hours

Tertiary Infotech Academy Pte. Ltd.
UEN 201200696W

Mechanism-led courseware • Twelve self-contained activities



Trainer profile · General

?

Complete this profile before facilitating the agentic AI product-development course.

1. **Name:** _____
2. Title / designation: _____
3. Qualifications: _____
4. AI product and facilitation experience: _____

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Dr Alfred Ang

AA

Principal Trainer, Tertiary Infotech Academy

1. **Doctorate-level applied technology educator**
2. AI, data, cloud and automation curriculum leadership
3. Industry-oriented implementation practice
4. Focus: explainable, governed technical outcomes

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Learner introduction

YOU

Connect your existing role to one software or knowledge-work automation problem.

1. **Role and organisation context**
2. Current development or operations workflow
3. One repeated process bottleneck
4. One risk you want the agent to control

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Working agreements

RULE

Use governed experimentation throughout the course.

1. **Use synthetic or approved data only**
2. Never paste live secrets into prompts or files
3. Keep consequential actions behind approval
4. Challenge completion claims with observable evidence

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Course facts

C840

Non-WSQ course by Tertiary Infotech Academy

1. **Course code: C840**
2. Duration: 2 days / 15 instructional hours
3. Daily schedule: 9:30am – 5:30pm
4. Version: v1.0 / 1 October 2026

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Day 1 lesson plan

D1

Evaluate opportunities and design evidence-grounded agentic product workflows.

1. **Product opportunity and JTBD framing**
2. Builder comparison and readiness
3. Research evidence and persona assumptions
4. Activities 01–06: discovery through AI-ready PRD

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Day 2 lesson plan

D2

Optimise, deploy and govern an agentic product solution.

1. **Prompt contracts, evaluation sets and iteration**
2. Deployment architecture, security and human gates
3. KPI feedback loops and risk-adjusted decisions
4. Activities 07–12: evaluation set through executive recommendation

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Download course material

Use the LMS course record as the single verified entry point for learner materials.

1. **Sign in at lms-tms.tertiaryinfotech.com**
2. Open AI for Product Development
3. Select Course Material and confirm C840
4. Download slides, Learner Guide and activity files

https://lms-tms.tertiaryinfotech.com/

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Access the activities

ACT

Twelve self-contained activity folders carry the complete end-to-end workflow.

1. **Open the individual activity README/PDF**
2. Use only the supplied mock data and YAML brief
3. Follow detailed steps in the Learner Guide
4. Record evidence in the folder checklist

https://github.com/tertiarycourses/C840-AI-for-Product-Development

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1

TOPIC 1

Evaluating Agentic AI Tools for Product Research and Development

Identify a product opportunity, map it to a suitable agentic pattern, and compare builders using evidence, readiness and total-cost criteria.

1 CONTROL

2 EVIDENCE

3 AUTOMATION

4 DECISION

Every product decision ends in observable evidence—not an agent's assertion.

01 · Agentic AI versus assistants: execution path

AGENTIC PRODUCT LOOP

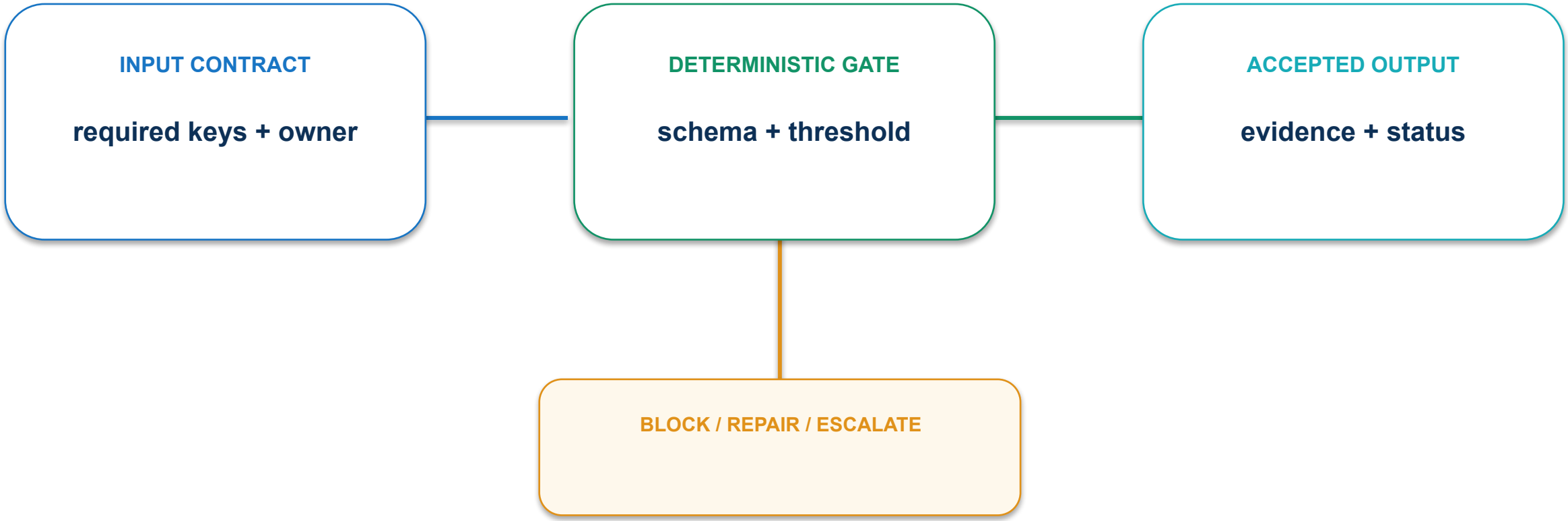
1	DISCOVER	Evidence and opportunity scope
2	DEFINE	Outcome, users and constraints
3	BUILD	Thin prototype and workflow
4	LEARN	Evaluation and next decision



CONCEPT VISUAL • LABELLED

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01 · Agentic AI versus assistants: contract and control



Require a multi-step loop, tool boundary and stop condition before calling a system agentic.

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01 · Agentic AI versus assistants: evidence and decision

MEASURE

Goal completion rate with valid evidence

ACCEPTANCE EVIDENCE

Trace showing plan, tool result and final decision

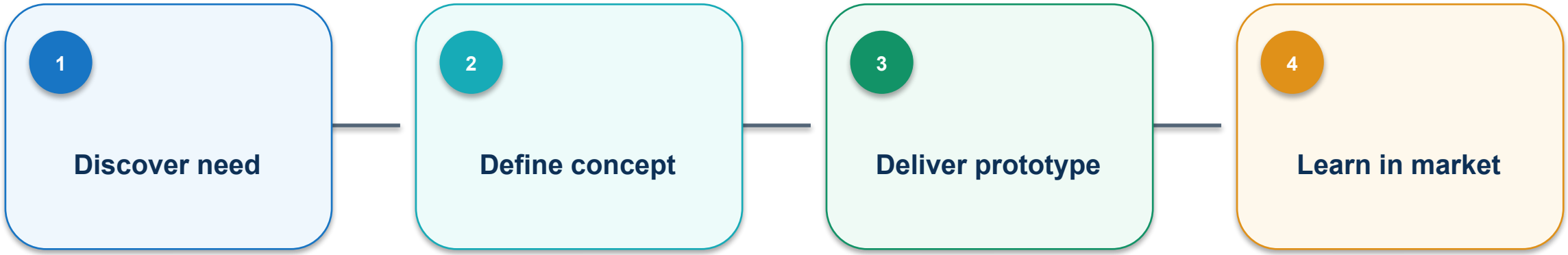
OBSERVED	THRESHOLD	DECISION	OWNER
machine value	approved rule	PASS / REPAIR	named human

Completion principle: evidence travels with the product hypothesis, changed state and decision—not in an unsupported claim.

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MECHANISM

02 · Product lifecycle application map: execution path



STATE PASSED BETWEEN STAGES

insight

hypothesis

artifact

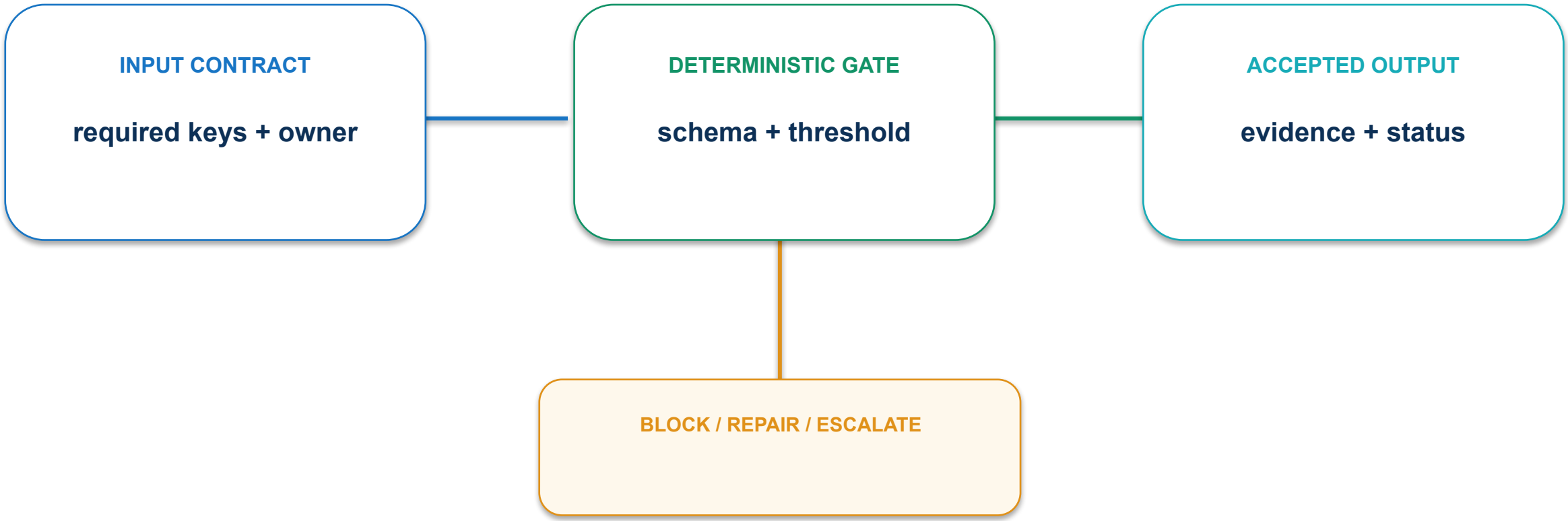
feedback

FAILURE MODE

Automation is applied everywhere without a clear decision or owner.

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02 · Product lifecycle application map: contract and control



Map each agent task to one lifecycle decision and accountable role.

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02 · Product lifecycle application map: evidence and decision

MEASURE

% lifecycle stages with a named human owner

ACCEPTANCE EVIDENCE

Lifecycle map with task-owner pairs

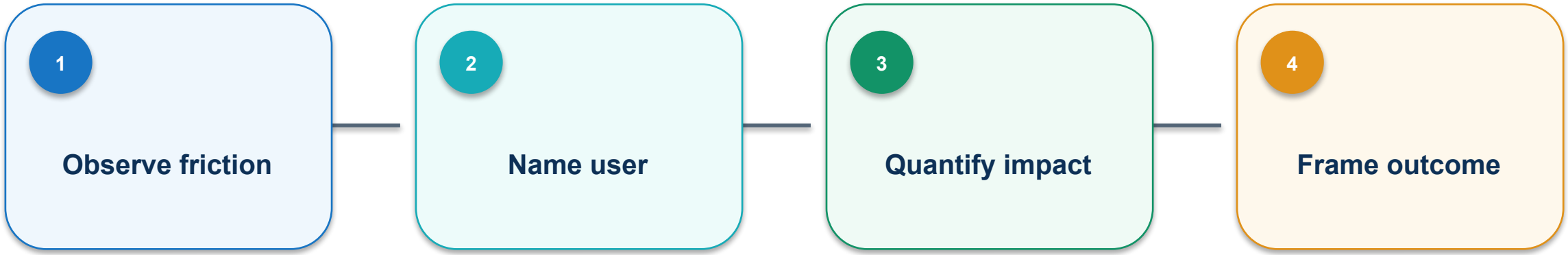
OBSERVED	THRESHOLD	DECISION	OWNER
machine value	approved rule	PASS / REPAIR	named human

Completion principle: evidence travels with the product hypothesis, changed state and decision—not in an unsupported claim.

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MECHANISM

03 · Opportunity framing: execution path



STATE PASSED BETWEEN STAGES

problem

user

baseline

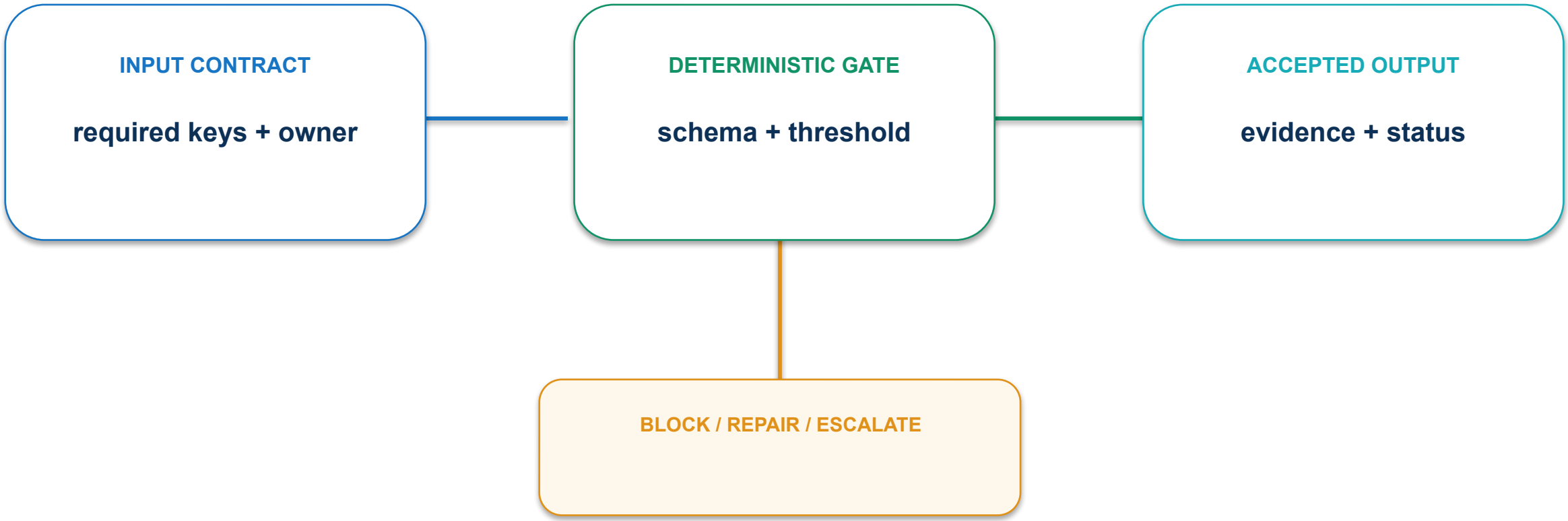
target

FAILURE MODE

The team automates an interesting task that is not a material user problem.

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03 · Opportunity framing: contract and control



Start with observable friction and a measurable outcome, not a tool demonstration.

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03 · Opportunity framing: evidence and decision

MEASURE

Validated problem statements / total ideas

ACCEPTANCE EVIDENCE

Opportunity brief with baseline and target

OBSERVED

machine value

THRESHOLD

approved rule

DECISION

PASS / REPAIR

OWNER

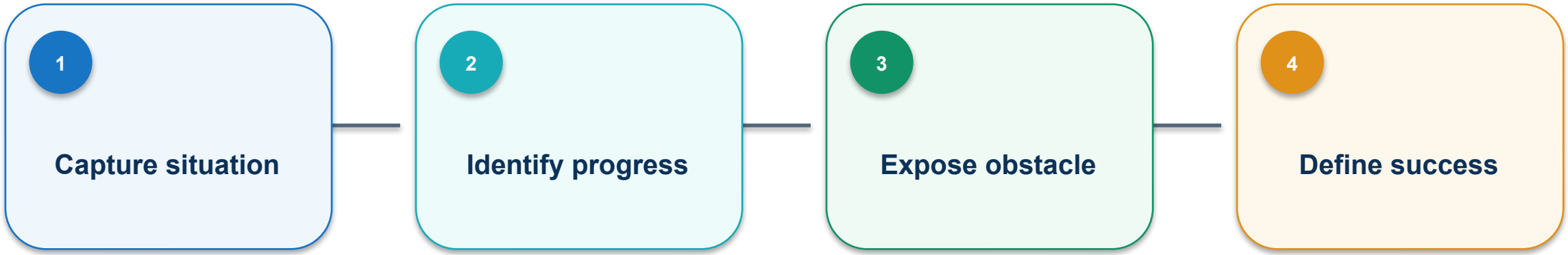
named human

Completion principle: evidence travels with the product hypothesis, changed state and decision—not in an unsupported claim.

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MECHANISM

04 · Jobs-to-be-done lens: execution path



STATE PASSED BETWEEN STAGES

situation

job

barrier

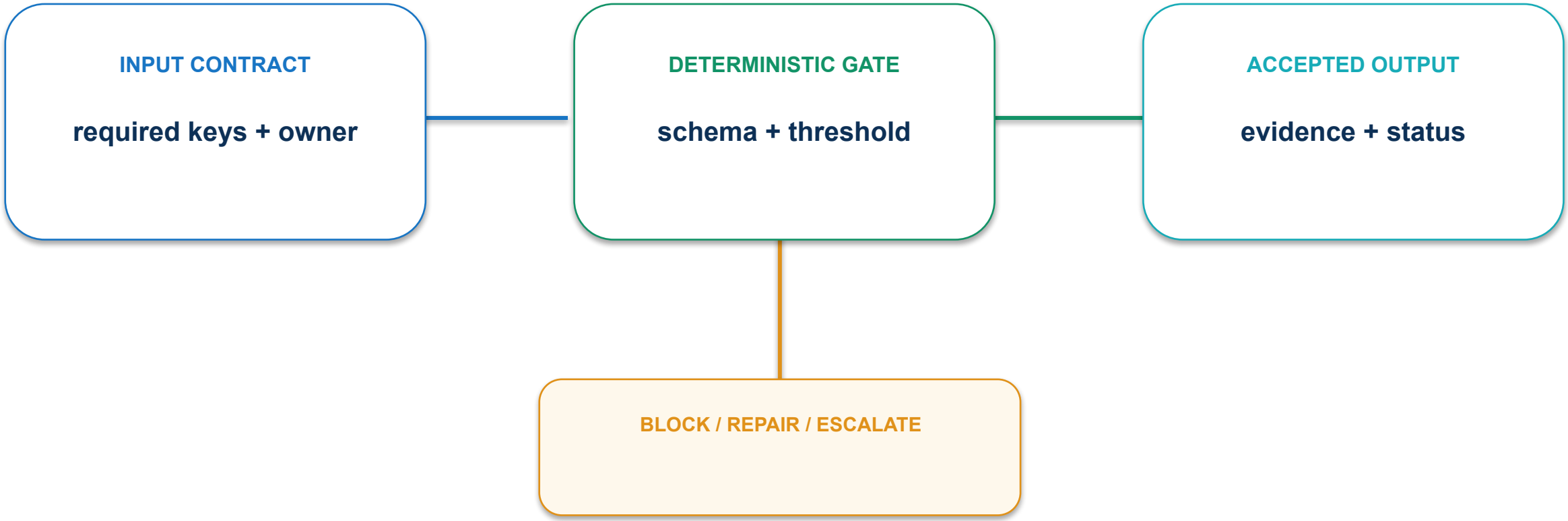
outcome

FAILURE MODE

Personas become demographics without decision-relevant needs.

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04 · Jobs-to-be-done lens: contract and control



Write needs as progress users seek in a real context.

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04 · Jobs-to-be-done lens: evidence and decision

MEASURE

% research observations linked to a user job

ACCEPTANCE EVIDENCE

JTBD statement linked to verbatim research notes

OBSERVED

machine value

THRESHOLD

approved rule

DECISION

PASS / REPAIR

OWNER

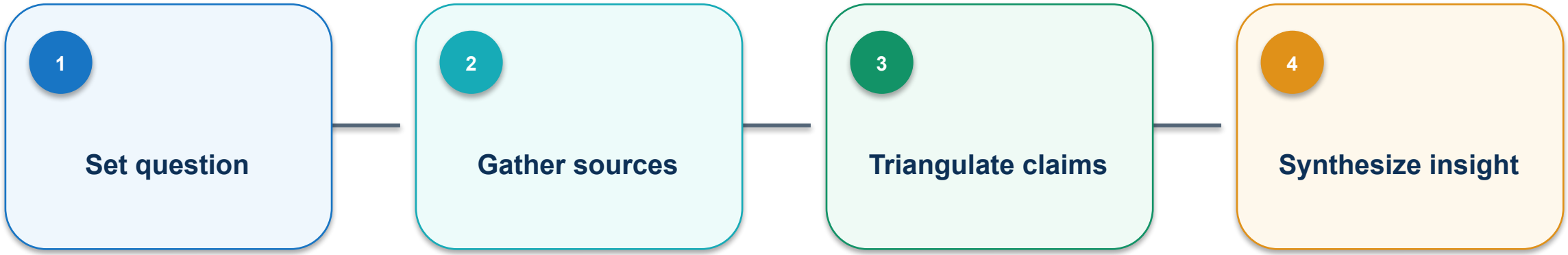
named human

Completion principle: evidence travels with the product hypothesis, changed state and decision—not in an unsupported claim.

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MECHANISM

05 · Research-agent pattern: execution path



STATE PASSED BETWEEN STAGES

question

source

claim

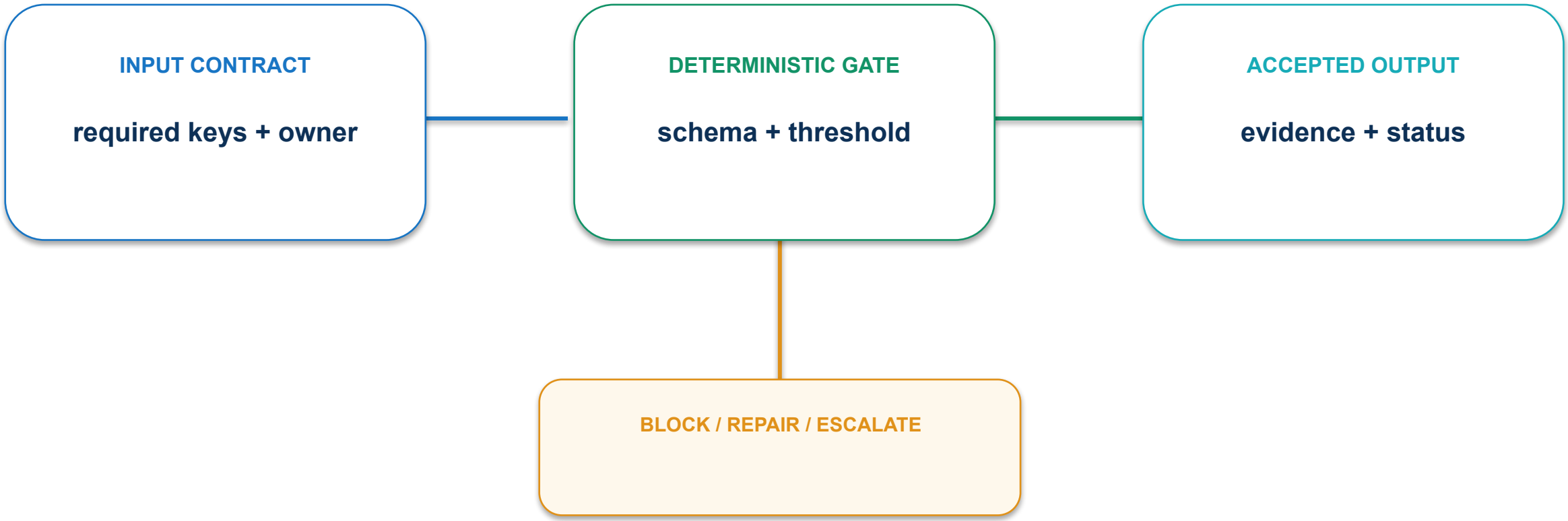
confidence

FAILURE MODE

The agent produces polished synthesis from weak or circular evidence.

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05 · Research-agent pattern: contract and control



Require source provenance, claim-level support and uncertainty labels.

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05 · Research-agent pattern: evidence and decision

MEASURE

% material claims supported by two independent sources

ACCEPTANCE EVIDENCE

Research ledger and cited insight brief

OBSERVED

machine value

THRESHOLD

approved rule

DECISION

PASS / REPAIR

OWNER

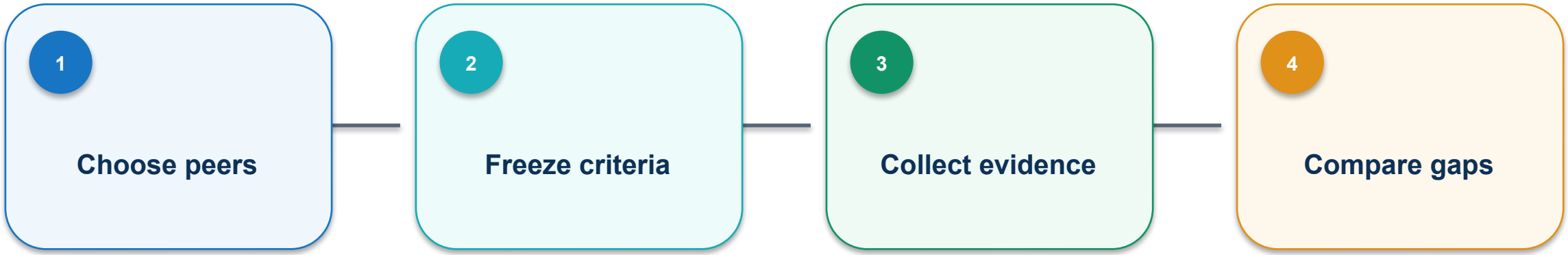
named human

Completion principle: evidence travels with the product hypothesis, changed state and decision—not in an unsupported claim.

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MECHANISM

06 · Competitor intelligence workflow: execution path



STATE PASSED BETWEEN STAGES

peer

criterion

evidence

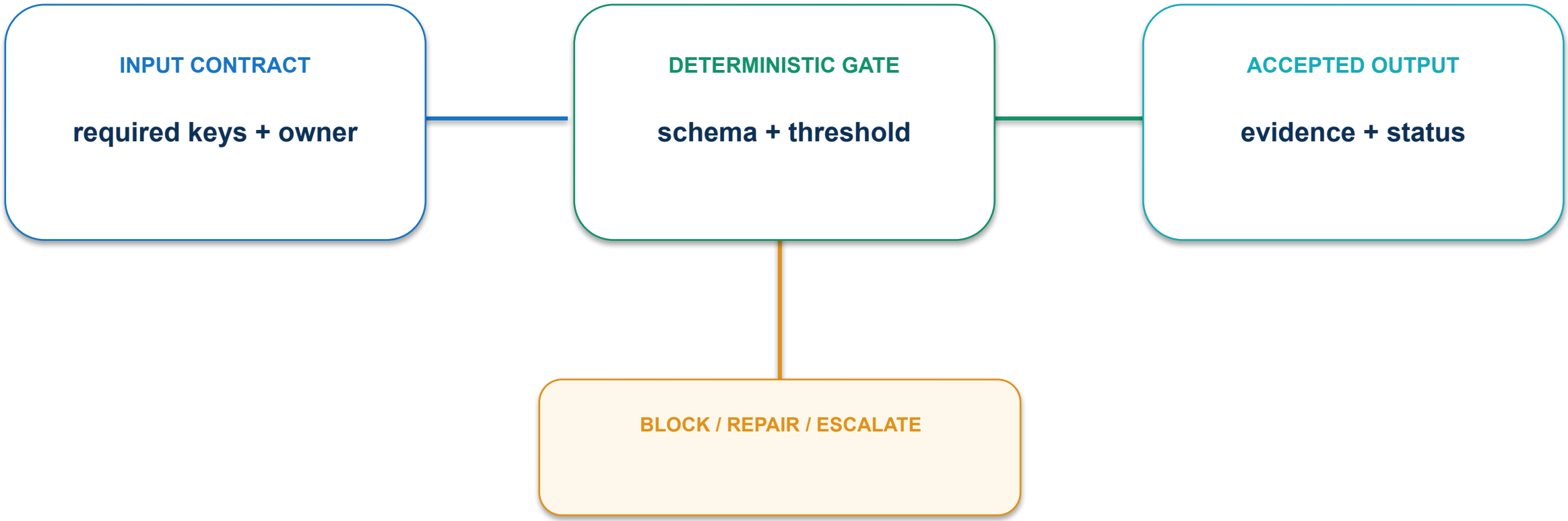
gap

FAILURE MODE

Different evidence standards make the comparison misleading.

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06 · Competitor intelligence workflow: contract and control



Freeze the same criteria and observation date before collection.

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06 · Competitor intelligence workflow: evidence and decision

MEASURE

Comparable criteria populated per competitor

ACCEPTANCE EVIDENCE

Timestamped comparison matrix

OBSERVED

machine value

THRESHOLD

approved rule

DECISION

PASS / REPAIR

OWNER

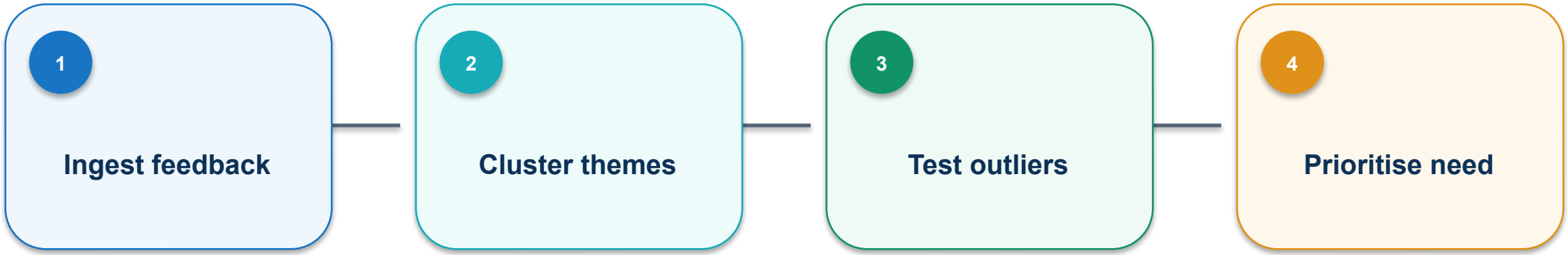
named human

Completion principle: evidence travels with the product hypothesis, changed state and decision—not in an unsupported claim.

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MECHANISM

07 · Voice-of-customer synthesis: execution path



STATE PASSED BETWEEN STAGES

feedback_id

theme

outlier

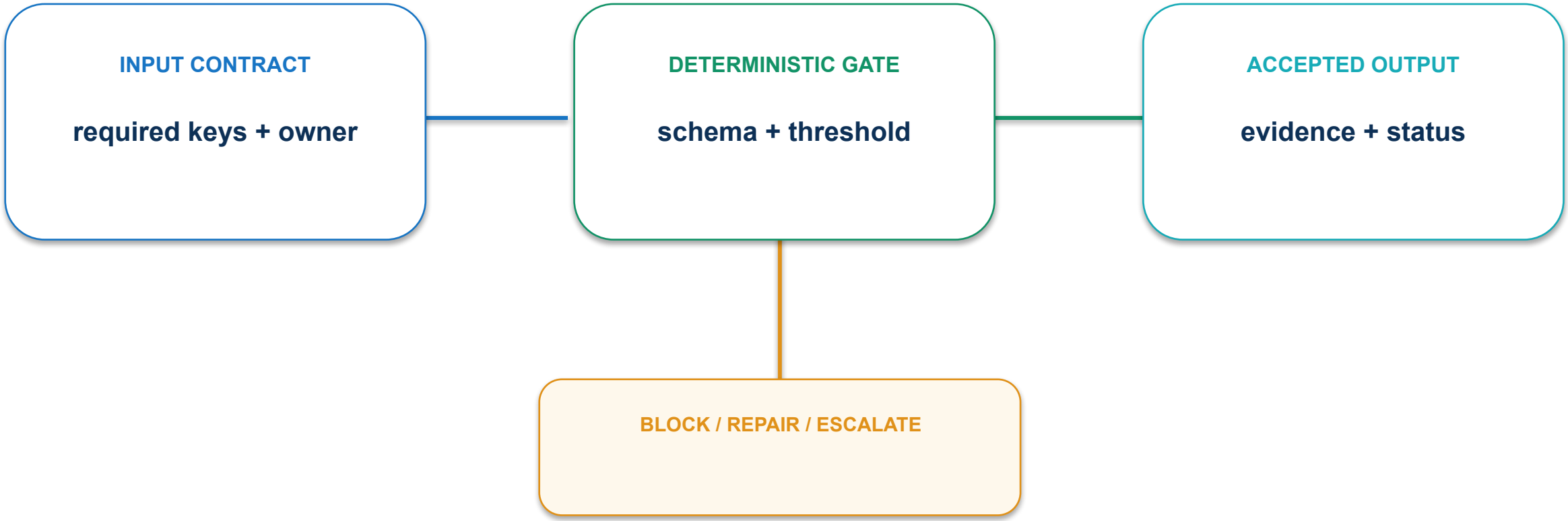
priority

FAILURE MODE

A dominant-sounding summary hides minority but critical needs.

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07 · Voice-of-customer synthesis: contract and control



Retain row-level traceability and inspect negative cases separately.

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07 · Voice-of-customer synthesis: evidence and decision

MEASURE

% themes traceable to source rows

ACCEPTANCE EVIDENCE

Theme table with supporting and contradicting excerpts

OBSERVED

machine value

THRESHOLD

approved rule

DECISION

PASS / REPAIR

OWNER

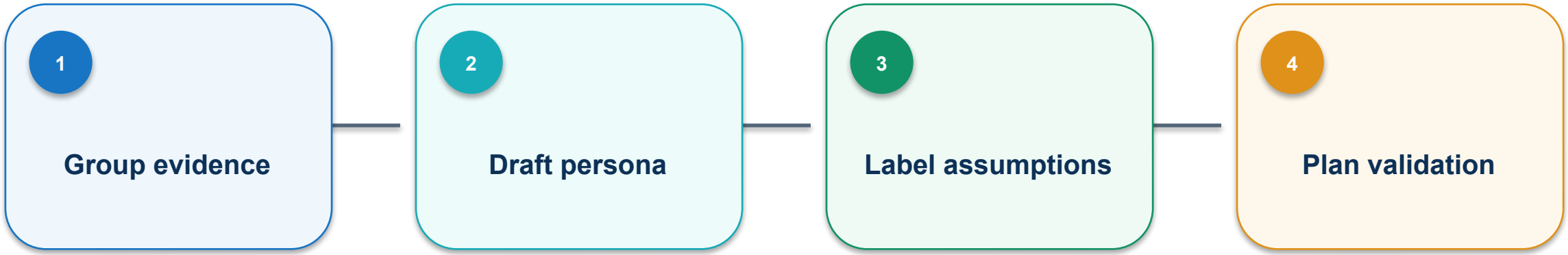
named human

Completion principle: evidence travels with the product hypothesis, changed state and decision—not in an unsupported claim.

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MECHANISM

08 · Persona and assumption map: execution path



STATE PASSED BETWEEN STAGES

segment

need

assumption

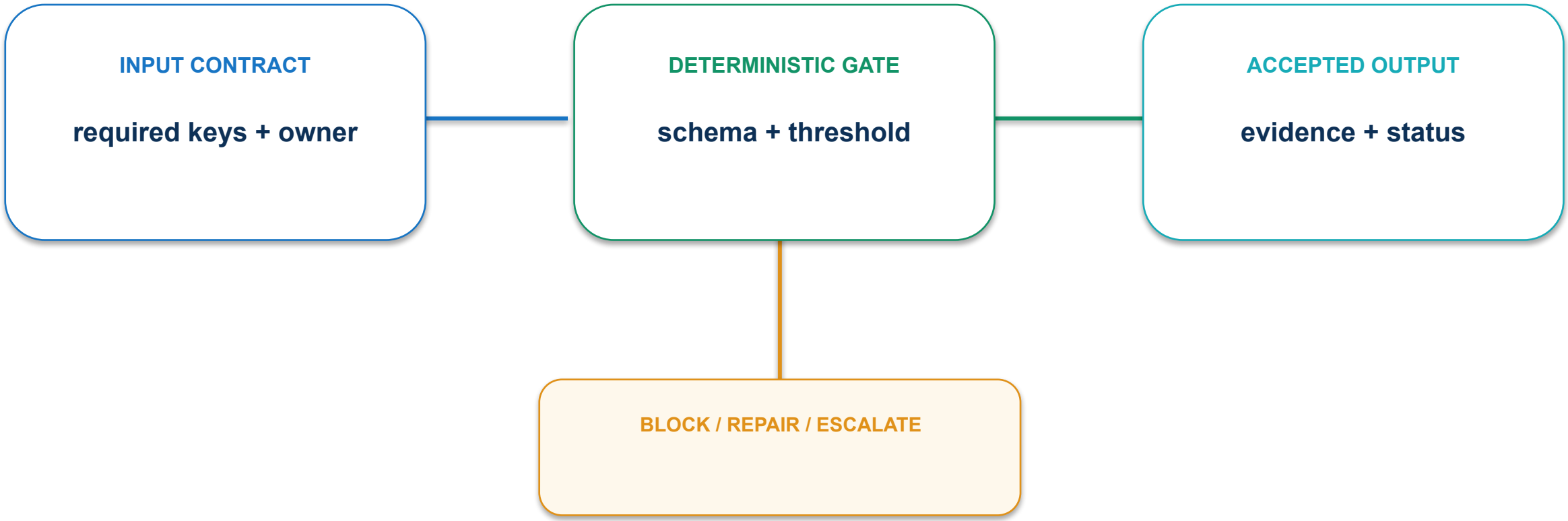
test

FAILURE MODE

Generated personas are treated as facts about real users.

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08 · Persona and assumption map: contract and control



Mark inferred attributes and separate evidence from hypothesis.

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08 · Persona and assumption map: evidence and decision

MEASURE

High-risk assumptions with a validation test

ACCEPTANCE EVIDENCE

Persona card plus assumption register

OBSERVED

machine value

THRESHOLD

approved rule

DECISION

PASS / REPAIR

OWNER

named human

Completion principle: evidence travels with the product hypothesis, changed state and decision—not in an unsupported claim.

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MECHANISM

09 · Agentic builder landscape: execution path

1. Define use case
2. Shortlist builders
3. Run common task
4. Score fit

STATE PASSED BETWEEN STAGES

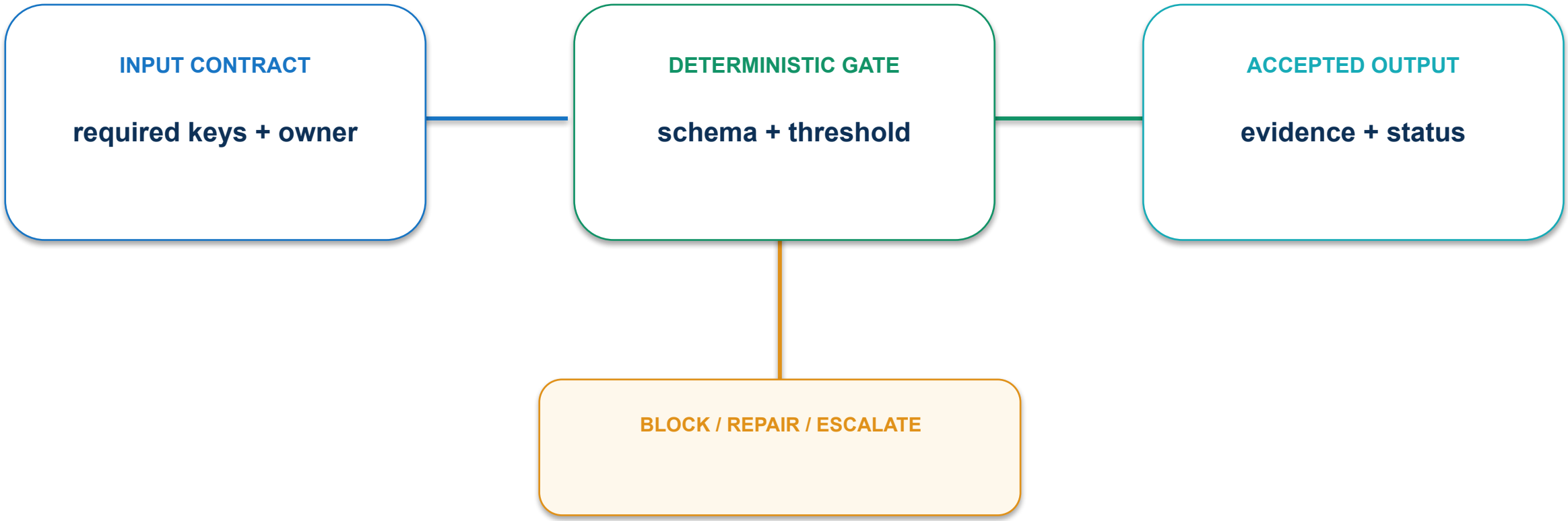
- use_case
- builder
- test_case
- score

FAILURE MODE

Feature lists replace direct comparison on the intended task.

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09 · Agentic builder landscape: contract and control



Run the same input, output and constraints across every candidate.

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09 · Agentic builder landscape: evidence and decision

MEASURE

Weighted score on a common golden task

ACCEPTANCE EVIDENCE

Paired trial results and recommendation

OBSERVED

machine value

THRESHOLD

approved rule

DECISION

PASS / REPAIR

OWNER

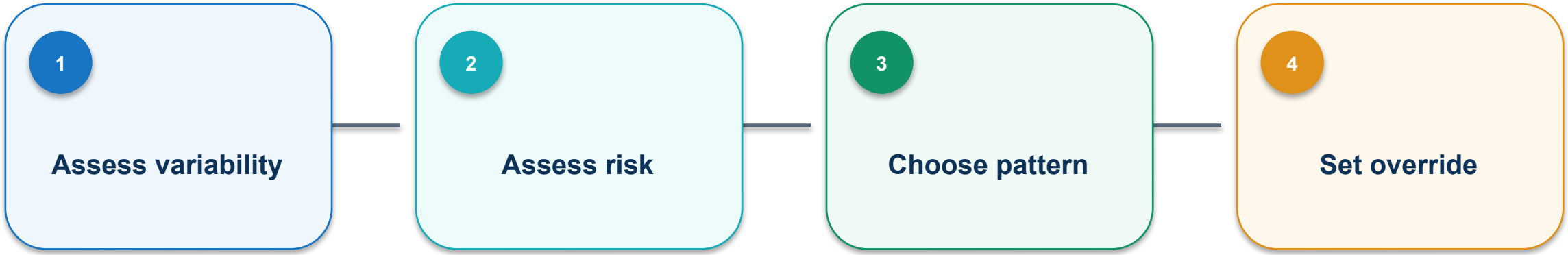
named human

Completion principle: evidence travels with the product hypothesis, changed state and decision—not in an unsupported claim.

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MECHANISM

10 · Workflow versus autonomous agent: execution path



STATE PASSED BETWEEN STAGES

variability

risk

pattern

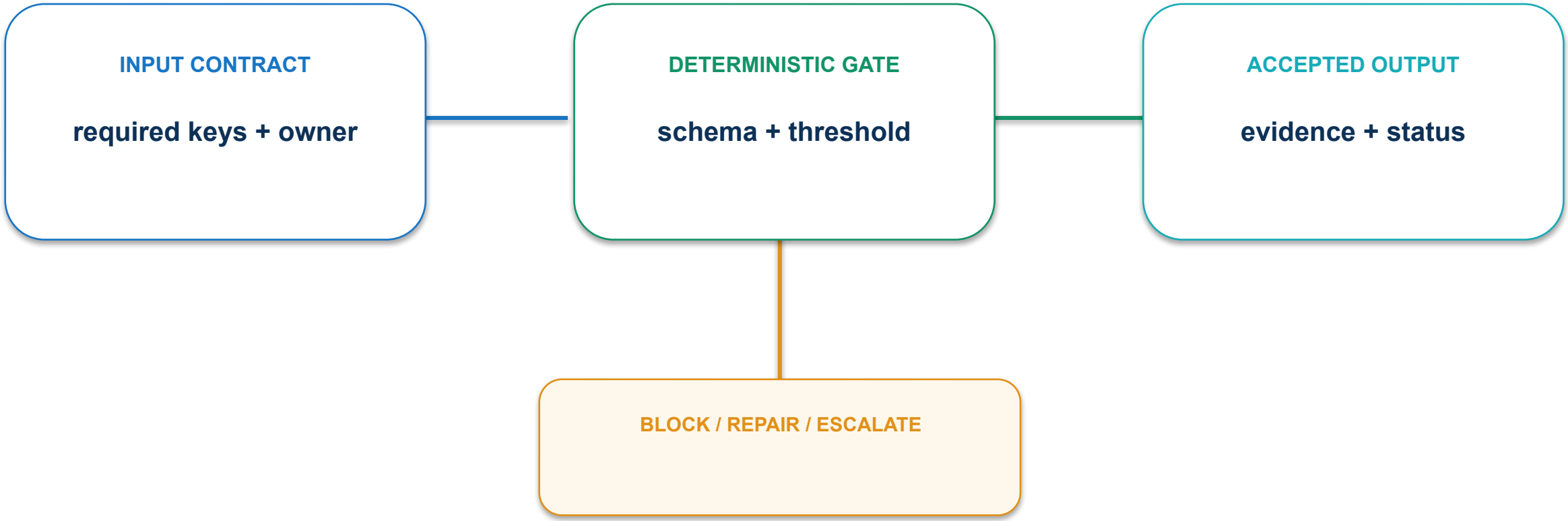
override

FAILURE MODE

Autonomy is added where a predictable workflow would be safer and cheaper.

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10 · Workflow versus autonomous agent: contract and control



Use the simplest architecture that satisfies variability and judgment needs.

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10 · Workflow versus autonomous agent: evidence and decision

MEASURE

% high-risk steps on deterministic paths

ACCEPTANCE EVIDENCE

Pattern decision with rejected alternatives

OBSERVED

machine value

THRESHOLD

approved rule

DECISION

PASS / REPAIR

OWNER

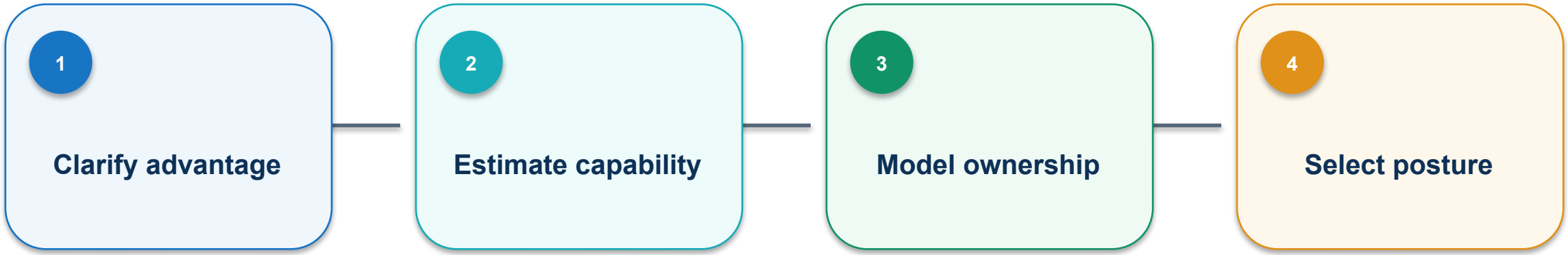
named human

Completion principle: evidence travels with the product hypothesis, changed state and decision—not in an unsupported claim.

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MECHANISM

11 · Build-buy-partner decision: execution path



STATE PASSED BETWEEN STAGES

advantage

capability

ownership_cost

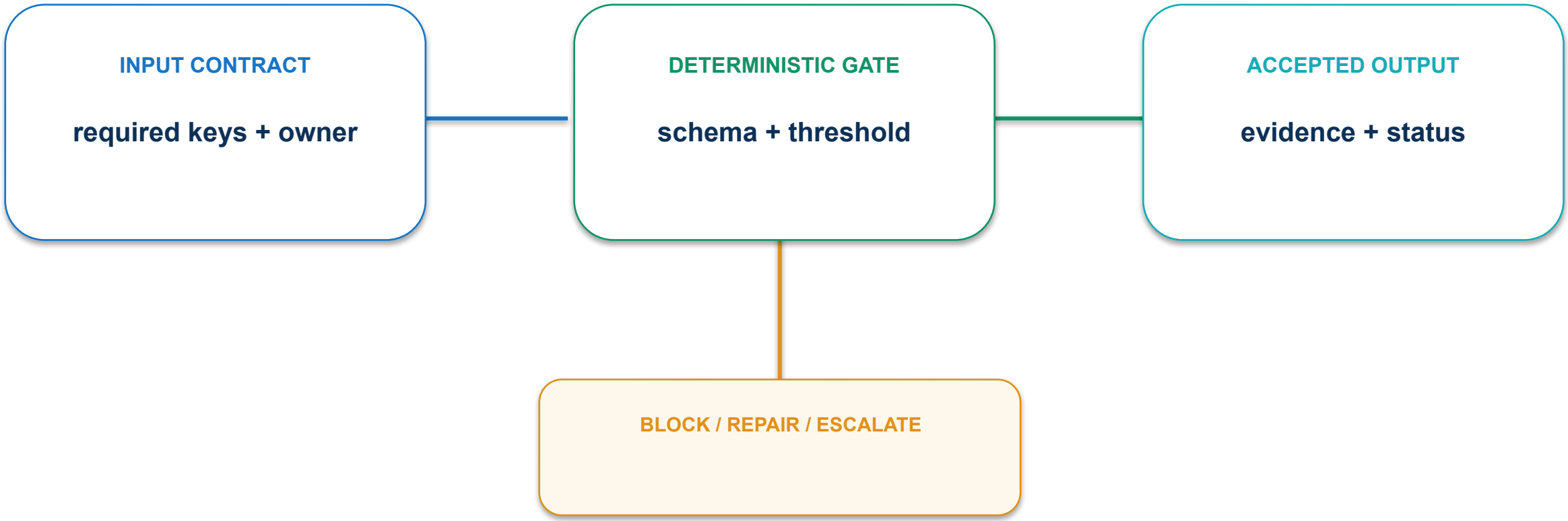
posture

FAILURE MODE

Subscription price is compared with in-house build cost while integration and governance are ignored.

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11 · Build-buy-partner decision: contract and control



Include change, integration, assurance and exit costs.

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11 · Build-buy-partner decision: evidence and decision

MEASURE

Risk-adjusted value over total cost of ownership

ACCEPTANCE EVIDENCE

Three-option decision table with sensitivity range

OBSERVED

machine value

THRESHOLD

approved rule

DECISION

PASS / REPAIR

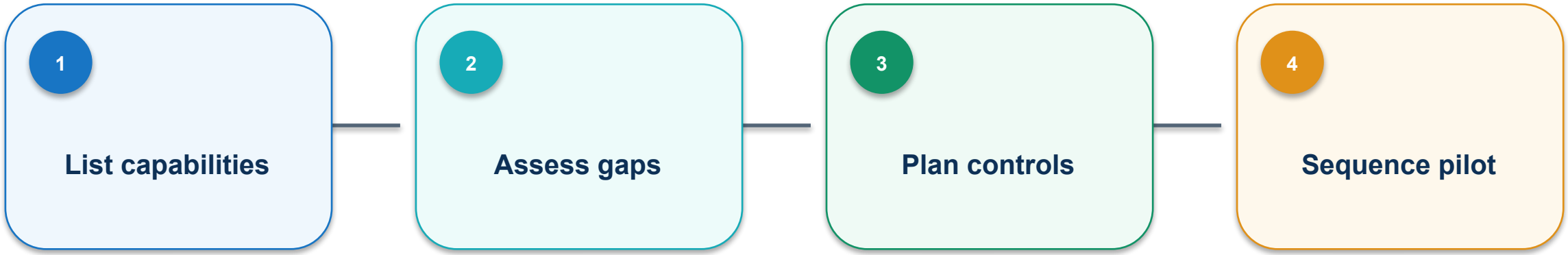
OWNER

named human

Completion principle: evidence travels with the product hypothesis, changed state and decision—not in an unsupported claim.

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12 · Resource and skills readiness: execution path



STATE PASSED BETWEEN STAGES

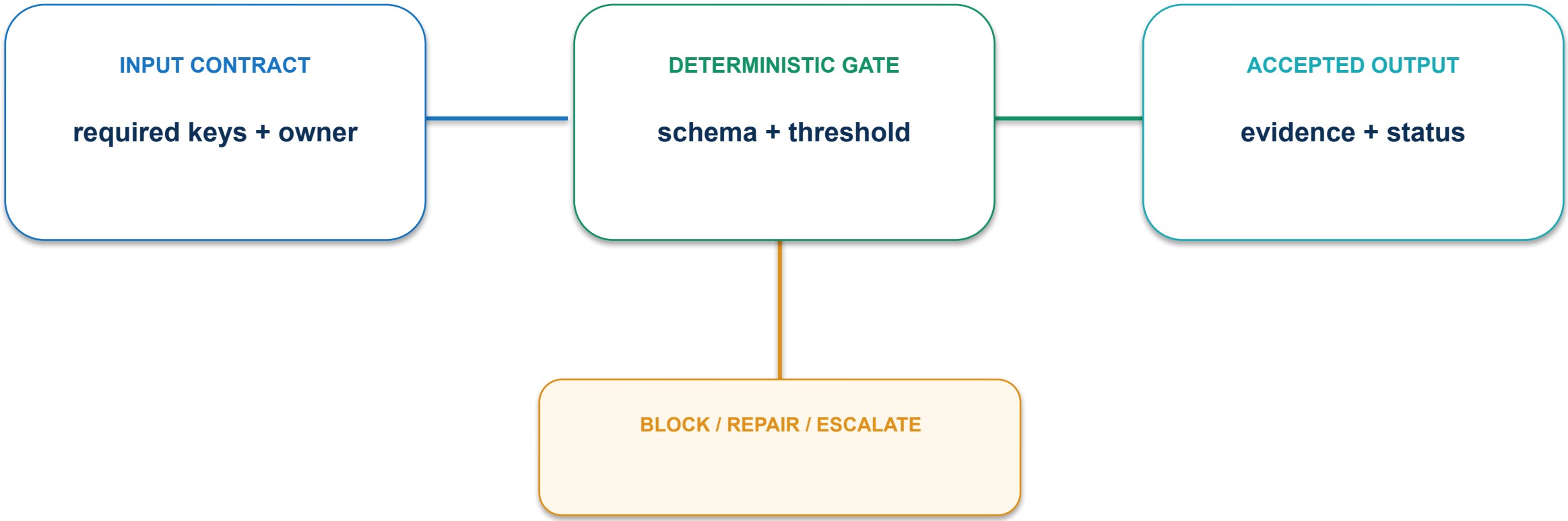
- **data**
- **people**
- **platform**
- **governance**

FAILURE MODE

A tool is selected before data access, ownership and evaluation skills exist.

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12 · Resource and skills readiness: contract and control



Gate pilot entry on minimum data, product, technical and risk capabilities.

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12 · Resource and skills readiness: evidence and decision

MEASURE

Critical readiness gaps closed before pilot

ACCEPTANCE EVIDENCE

Readiness heatmap and action owners

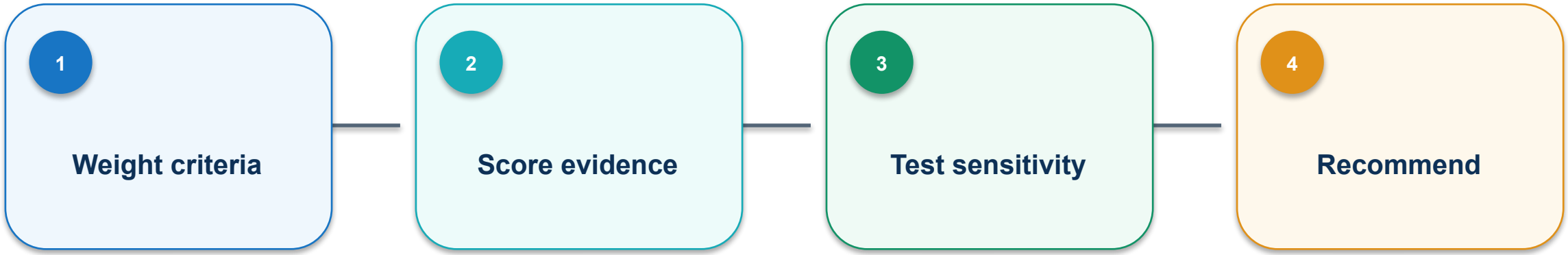
OBSERVED	THRESHOLD	DECISION	OWNER
machine value	approved rule	PASS / REPAIR	named human

Completion principle: evidence travels with the product hypothesis, changed state and decision—not in an unsupported claim.

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MECHANISM

13 · Evaluation decision matrix: execution path



STATE PASSED BETWEEN STAGES

weight

score

sensitivity

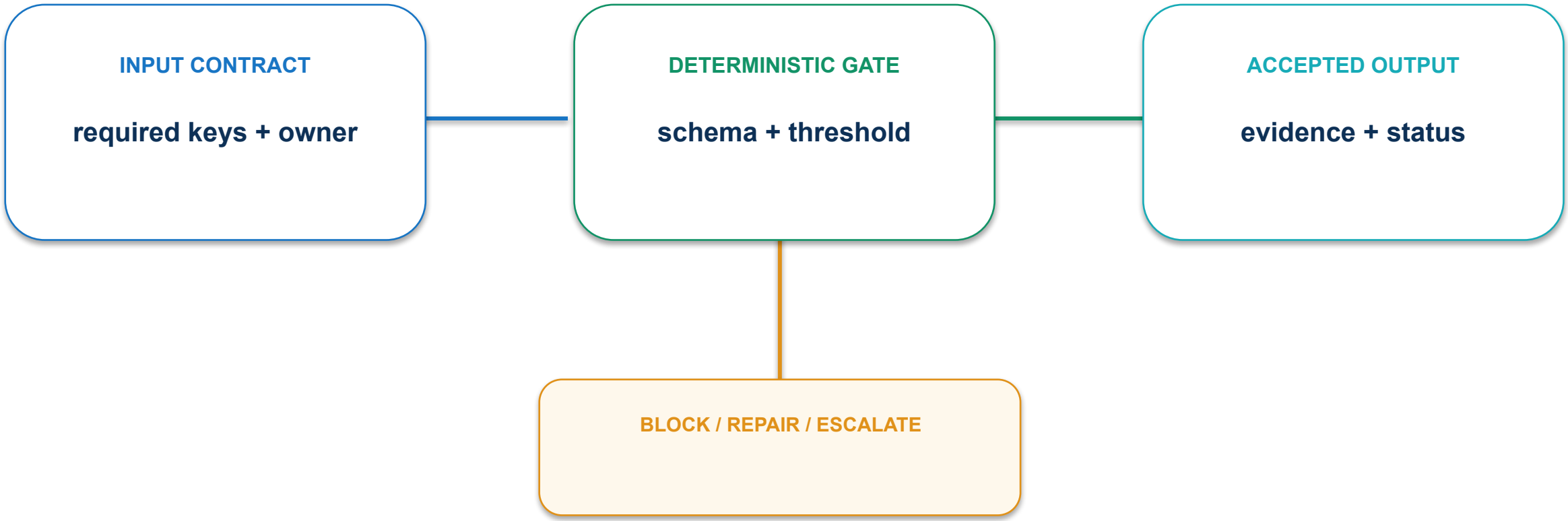
decision

FAILURE MODE

A predetermined winner is justified with arbitrary weights.

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13 · Evaluation decision matrix: contract and control



Agree criteria before testing and show how weight changes affect the result.

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13 · Evaluation decision matrix: evidence and decision

MEASURE

Recommendation stability across plausible weights

ACCEPTANCE EVIDENCE

Weighted matrix, caveats and decision owner

OBSERVED	THRESHOLD	DECISION	OWNER
machine value	approved rule	PASS / REPAIR	named human

Completion principle: evidence travels with the product hypothesis, changed state and decision—not in an unsupported claim.

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Discovery boundary: evidence, agent synthesis and accountable human

1

RESEARCH

Collect and triangulate evidence

2

AGENT

Synthesize and expose uncertainty

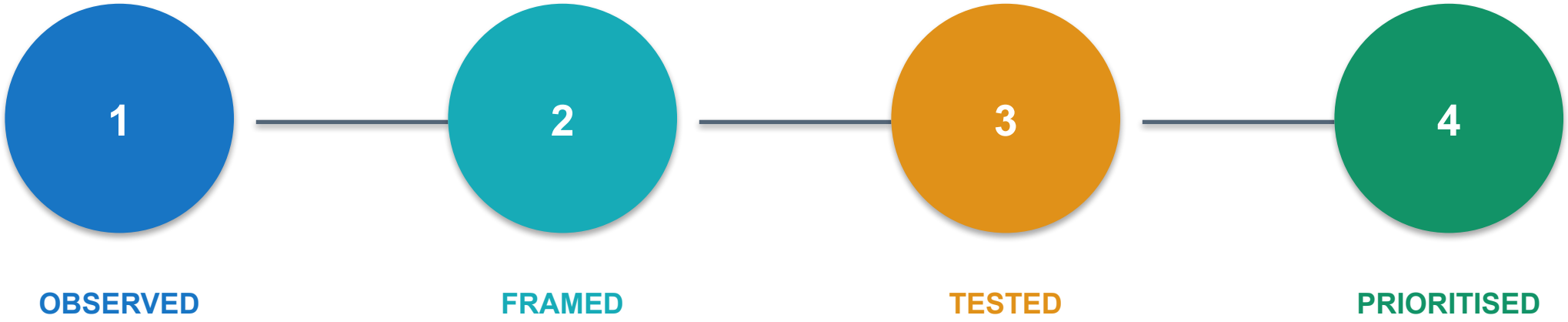
3

PRODUCT

Prioritise need and own decision

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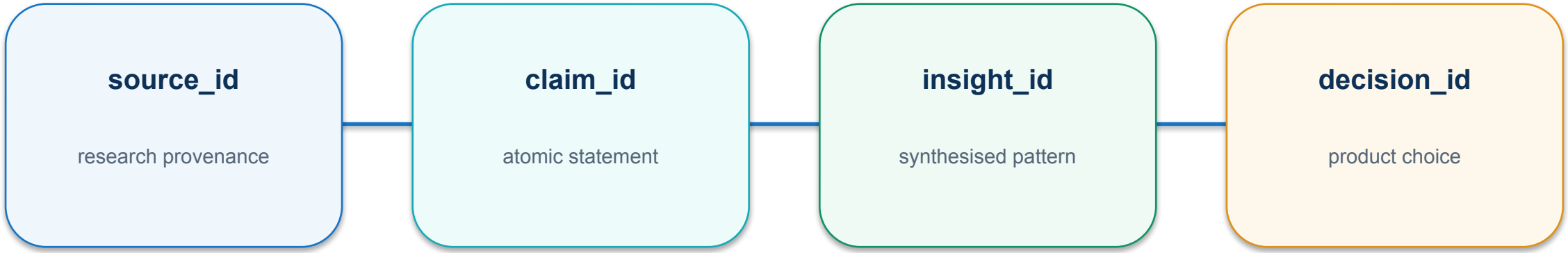
Opportunity state: OBSERVED, FRAMED, TESTED and PRIORITISED



PRIORITISED means the opportunity is supported by evidence, not presentation polish.

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Research evidence chain: source → claim → insight → decision



A discovery decision is explainable when each insight resolves to source evidence.

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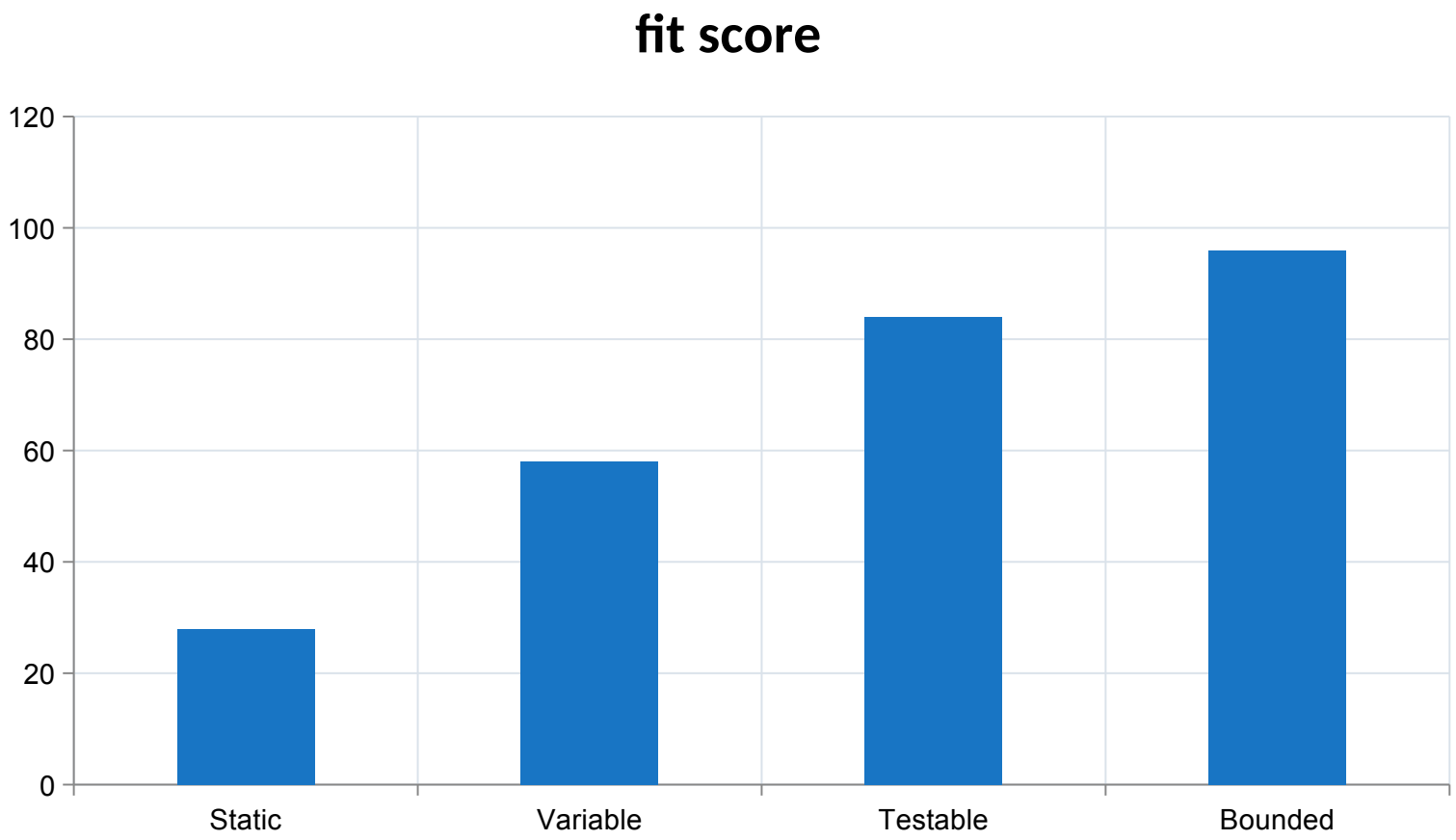
Discovery failure taxonomy: bias, gaps, conflict and overclaim

BIAS	broaden sampling and segment results
GAP	collect missing evidence
CONFLICT	retain competing explanations
OVERCLAIM	lower confidence and validate

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EDITABLE NATIVE CHART

Agent fit rises with variability and verifiability



INSIGHT

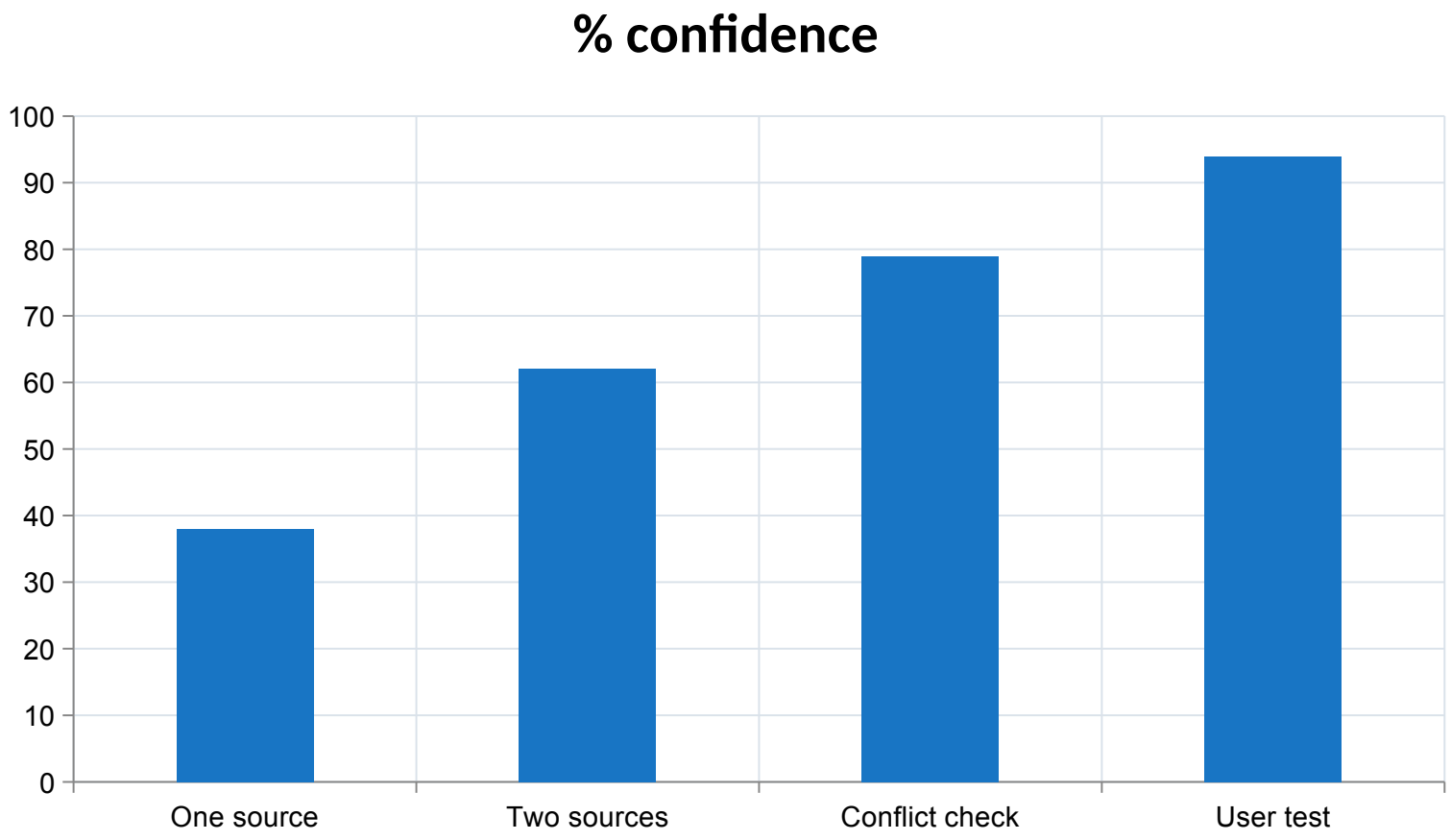
Agentic AI fits work that is variable enough to need judgment but bounded enough to verify.

Illustrative synthetic values for mechanism explanation.

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EDITABLE NATIVE CHART

Evidence coverage raises discovery confidence



INSIGHT

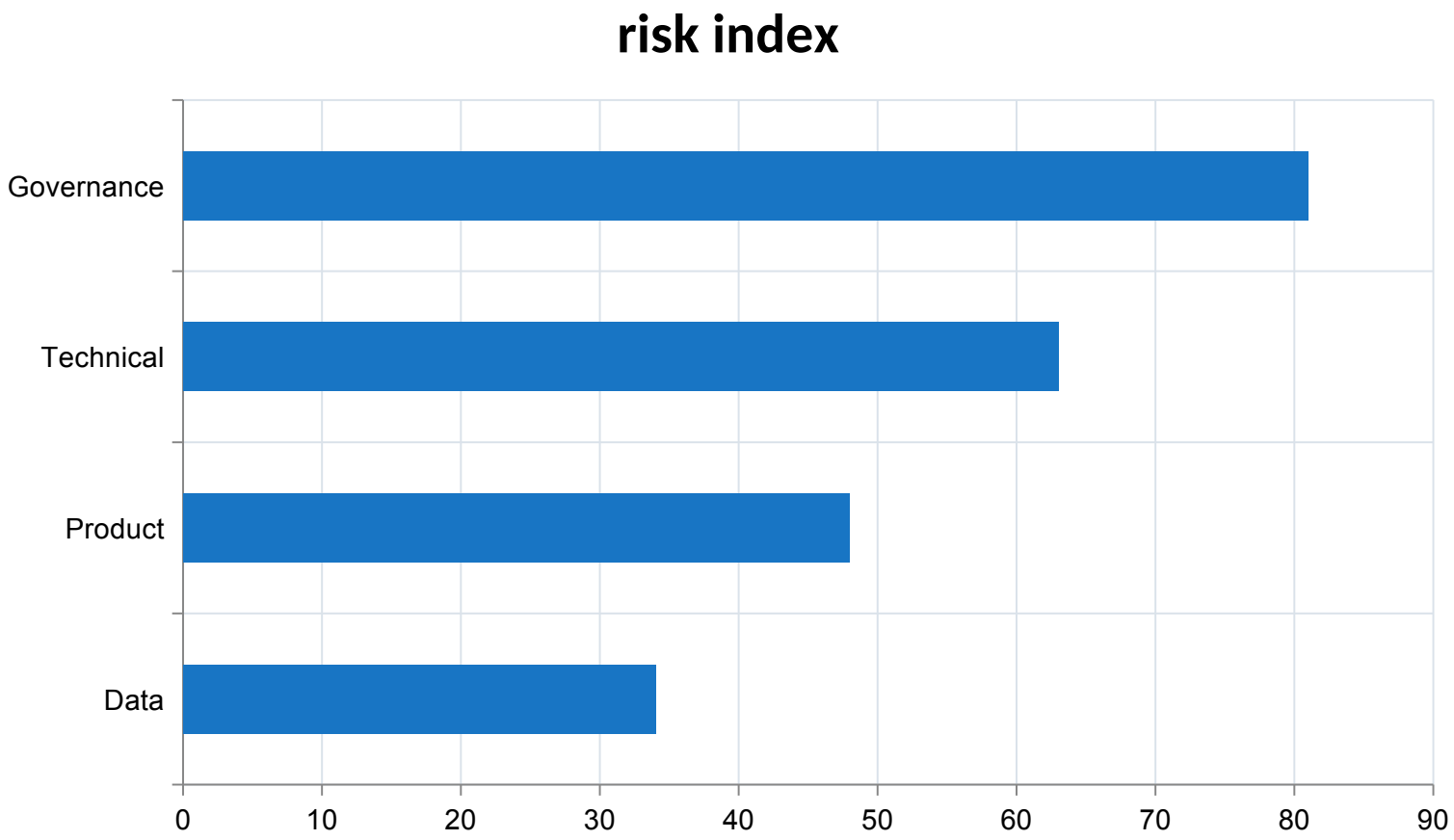
Discovery confidence rises when claims survive triangulation, conflict review and user validation.

Illustrative synthetic values for mechanism explanation.

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EDITABLE NATIVE CHART

Readiness gaps increase pilot exposure



INSIGHT

Builder selection depends on organisational readiness as much as product features.

Illustrative synthetic values for mechanism explanation.

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ACTIVITY BRIEF

Activity 01 · Frame a Product Opportunity

SCENARIO

Turn synthetic stakeholder notes into a measurable agentic-AI opportunity brief.

EXPECTED OUTPUT

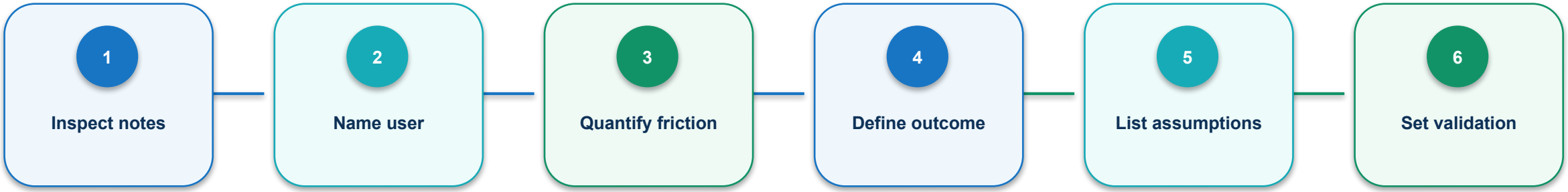
Completed opportunity brief and assumption register

Detailed procedures, prompts, troubleshooting and acceptance checks are in the Learner Guide and activity PDF.

Self-contained folder: activities/activity-01-opportunity-brief/

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Activity 01 · agent workflow contract



Consequential actions require a current diff, passing checks and a named decision.

HUMAN GATE

Approval resumes only when the reviewed diff and current state still match.

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Activity 01 · evidence and failure gates

PASS EVIDENCE

Every claim traces to a source row; the outcome is measurable; high-risk assumptions have named tests.

FAIL CLOSED WHEN

a protected boundary is crossed; evidence is missing; approval is absent; or the current diff no longer matches review.

README.pdf

detailed procedure

mock-data/

synthetic source state

templates/

learner working files

evidence/

acceptance record

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ACTIVITY BRIEF

Activity 02 · Compare Agentic AI Builders

SCENARIO

Run a common product-research task across three synthetic builder profiles and select a fit-for-purpose option.

EXPECTED OUTPUT

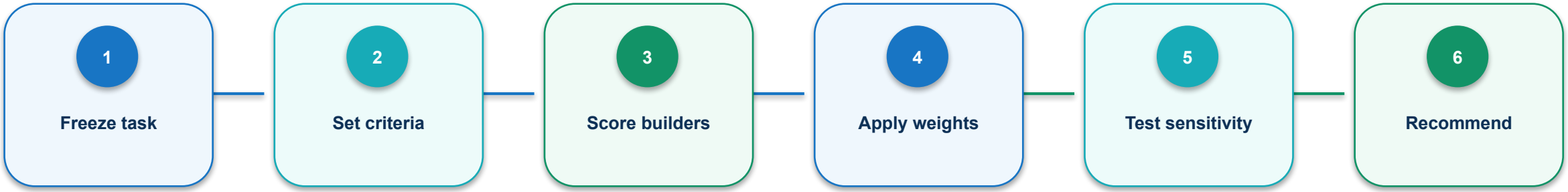
Weighted builder comparison with sensitivity check

Detailed procedures, prompts, troubleshooting and acceptance checks are in the Learner Guide and activity PDF.

Self-contained folder: activities/activity-02-builder-comparison/

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Activity 02 · agent workflow contract



Consequential actions require a current diff, passing checks and a named decision.

HUMAN GATE

Approval resumes only when the reviewed diff and current state still match.

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Activity 02 · evidence and failure gates

PASS EVIDENCE

Scores use one common task, weights total 100%, and the recommendation survives or explains sensitivity.

FAIL CLOSED WHEN

a protected boundary is crossed; evidence is missing; approval is absent; or the current diff no longer matches review.

README.pdf

detailed procedure

mock-data/

synthetic source state

templates/

learner working files

evidence/

acceptance record

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ACTIVITY BRIEF

Activity 03 · Build a Research Evidence Pack

SCENARIO

Transform mock market and competitor records into a traceable, uncertainty-aware insight brief.

EXPECTED OUTPUT

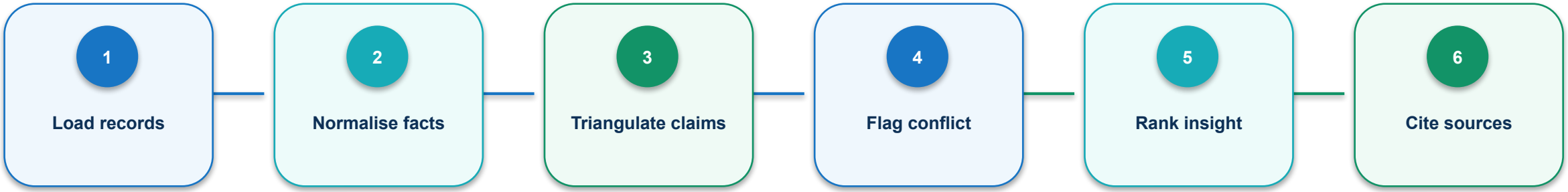
Claim ledger, competitor matrix and three prioritised insights

Detailed procedures, prompts, troubleshooting and acceptance checks are in the Learner Guide and activity PDF.

Self-contained folder: activities/activity-03-research-evidence-pack/

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Activity 03 · agent workflow contract



Each node consumes named keys, emits a status, and leaves retrievable execution evidence.

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Activity 03 · evidence and failure gates

PASS EVIDENCE

Material insights cite at least two records or explicitly state single-source uncertainty.

FAIL CLOSED WHEN

a protected boundary is crossed; evidence is missing; approval is absent; or the current diff no longer matches review.

README.pdf

detailed procedure

mock-data/

synthetic source state

templates/

learner working files

evidence/

acceptance record

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ACTIVITY BRIEF

Activity 04 · Create Persona and JTBD Map

SCENARIO

Derive evidence-based personas and jobs-to-be-done without presenting generated assumptions as facts.

EXPECTED OUTPUT

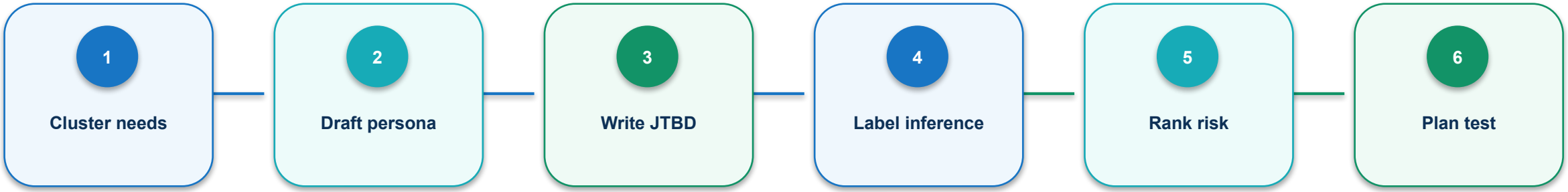
Two persona cards, JTBD statements and validation plan

Detailed procedures, prompts, troubleshooting and acceptance checks are in the Learner Guide and activity PDF.

Self-contained folder: activities/activity-04-persona-jtbd-map/

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Activity 04 · agent workflow contract



Consequential actions require a current diff, passing checks and a named decision.

HUMAN GATE

Approval resumes only when the reviewed diff and current state still match.

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Activity 04 · evidence and failure gates

PASS EVIDENCE

Persona needs trace to source IDs; assumptions are labelled; each risky assumption has a feasible validation test.

FAIL CLOSED WHEN

a protected boundary is crossed; evidence is missing; approval is absent; or the current diff no longer matches review.

README.pdf

detailed procedure

mock-data/

synthetic source state

templates/

learner working files

evidence/

acceptance record

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CHECKPOINT

Topic 1 technical checkpoint

1. Which state field prevents a silent downstream assumption?

2. What evidence proves the mechanism completed successfully?

3. Which failure must stop rather than retry?

4. Who owns the consequential decision?

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CHECKPOINT

Topic 1 transfer challenge

1. Change one input constraint and predict the affected nodes.
2. Name the identifier that preserves provenance across the change.
3. Define one machine threshold and one human decision.
4. Describe the minimum evidence required to resume the run.

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2

TOPIC 2

Building and Optimising Agentic Product Development Workflows

Translate product intent into bounded agent roles, specifications, hand-offs, evaluation sets and an iterative prototype workflow that improves with evidence.

1 CONTROL

2 EVIDENCE

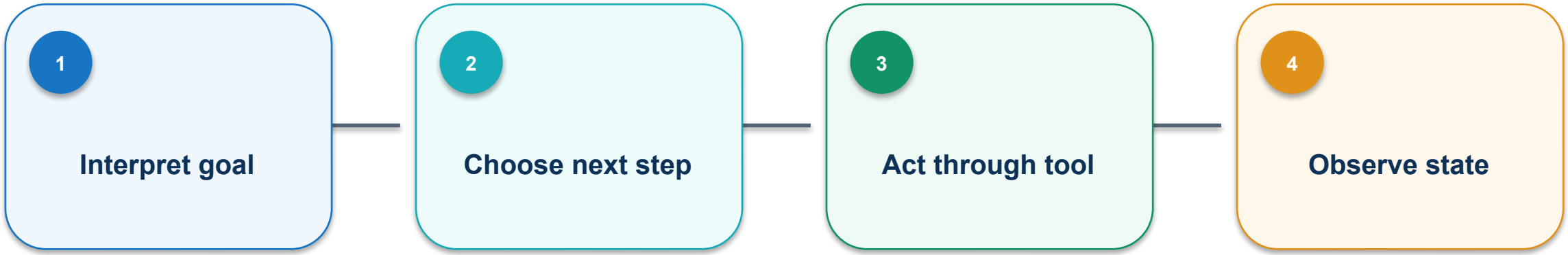
3 AUTOMATION

4 DECISION

Every product decision ends in observable evidence—not an agent's assertion.

MECHANISM

01 · Goal-plan-act-observe loop: execution path



STATE PASSED BETWEEN STAGES

goal

next_action

tool_result

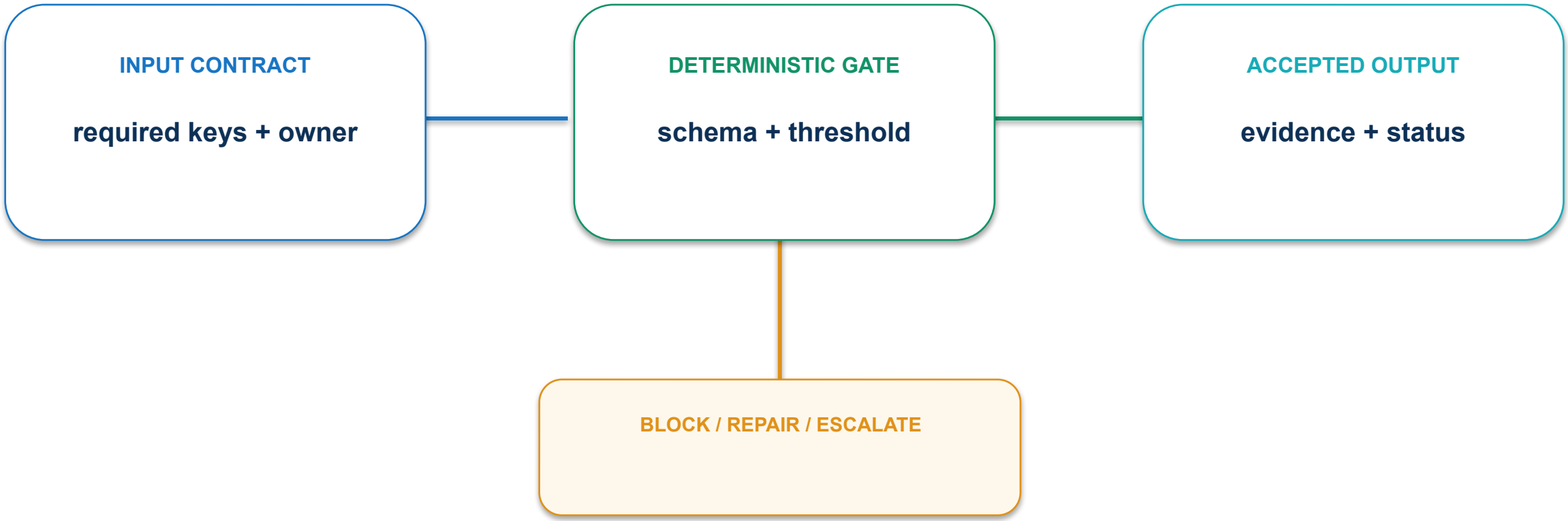
status

FAILURE MODE

The agent loops without progress or stops on a plausible narrative.

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01 · Goal-plan-act-observe loop: contract and control



Use explicit budgets, stop conditions and observable state transitions.

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01 · Goal-plan-act-observe loop: evidence and decision

MEASURE

% loops ending in a verified stop state

ACCEPTANCE EVIDENCE

Loop trace with terminal status

OBSERVED

machine value

THRESHOLD

approved rule

DECISION

PASS / REPAIR

OWNER

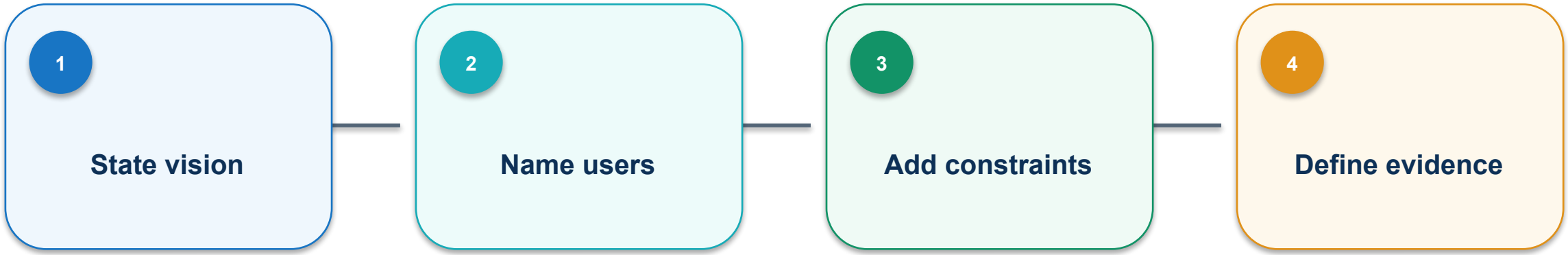
named human

Completion principle: evidence travels with the product hypothesis, changed state and decision—not in an unsupported claim.

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MECHANISM

02 · Product context packet: execution path



STATE PASSED BETWEEN STAGES

vision

users

constraints

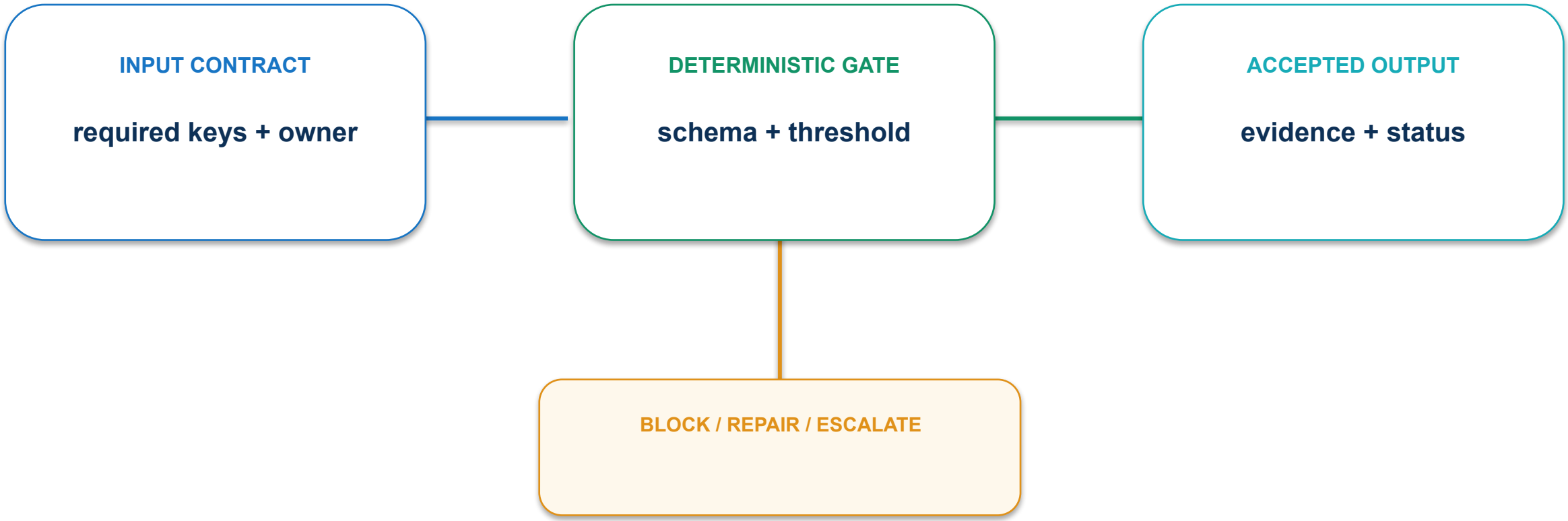
acceptance

FAILURE MODE

The agent fills missing business context with generic assumptions.

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02 · Product context packet: contract and control



Version a compact context packet and expose unknowns explicitly.

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02 · Product context packet: evidence and decision

MEASURE

Required context fields present before execution

ACCEPTANCE EVIDENCE

Context file and assumption log

OBSERVED

machine value

THRESHOLD

approved rule

DECISION

PASS / REPAIR

OWNER

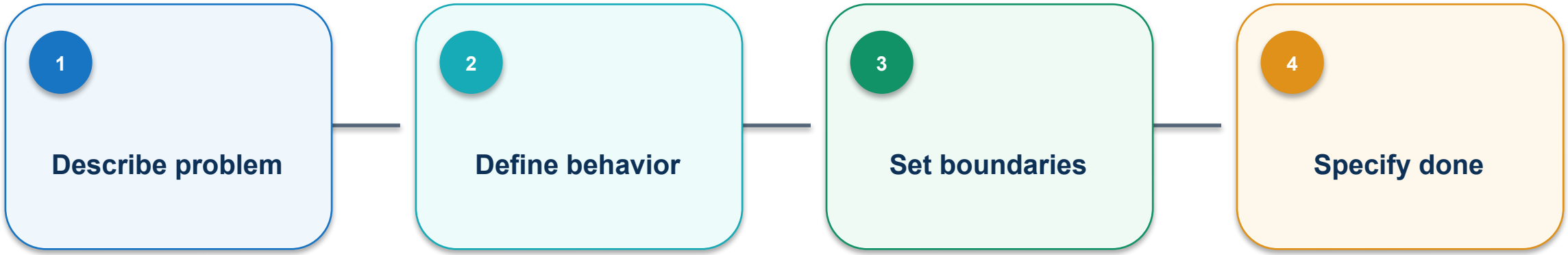
named human

Completion principle: evidence travels with the product hypothesis, changed state and decision—not in an unsupported claim.

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MECHANISM

03 · Outcome-focused specification: execution path



STATE PASSED BETWEEN STAGES

problem

behavior

non_goals

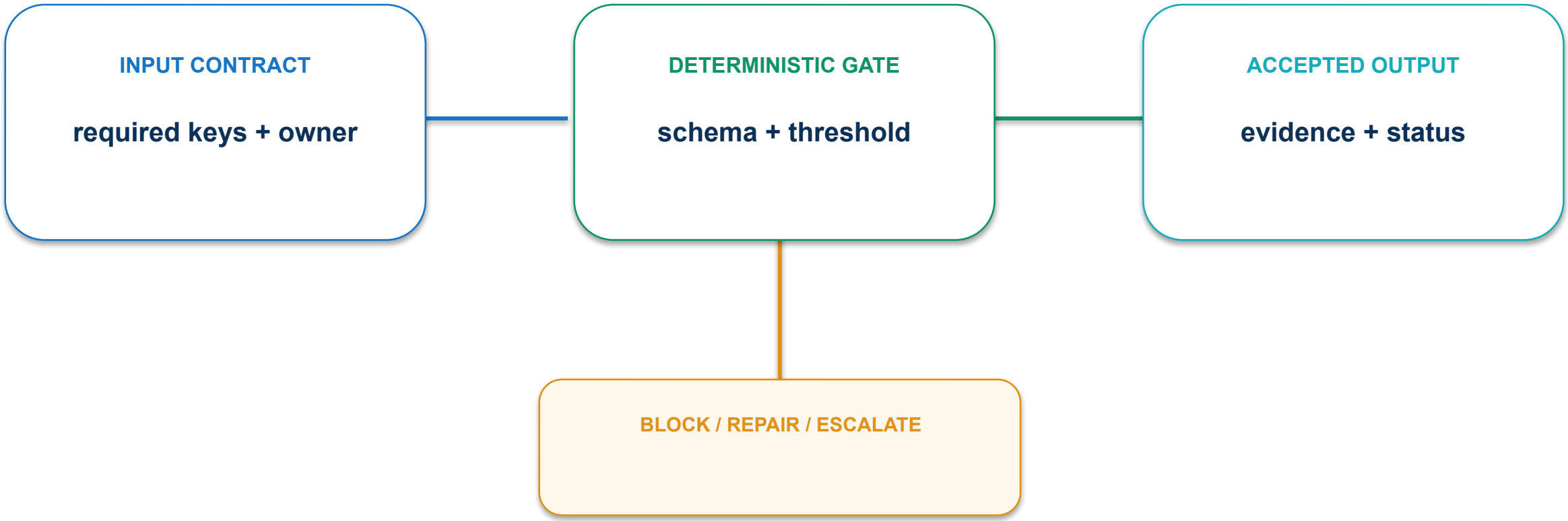
acceptance

FAILURE MODE

The specification dictates implementation while leaving success ambiguous.

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03 · Outcome-focused specification: contract and control



Describe what must be true and what must not change.

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03 · Outcome-focused specification: evidence and decision

MEASURE

% acceptance criteria observable by reviewer

ACCEPTANCE EVIDENCE

PRD section with testable criteria

OBSERVED

machine value

THRESHOLD

approved rule

DECISION

PASS / REPAIR

OWNER

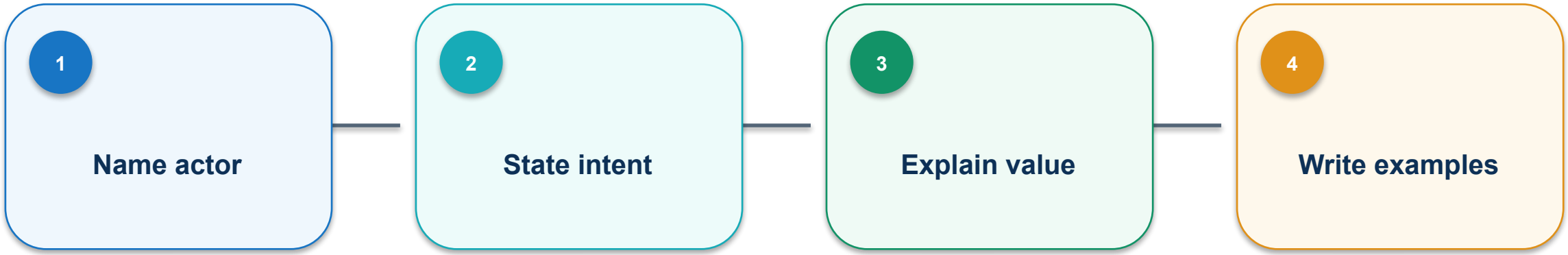
named human

Completion principle: evidence travels with the product hypothesis, changed state and decision—not in an unsupported claim.

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MECHANISM

04 · User stories and acceptance criteria: execution path



STATE PASSED BETWEEN STAGES

actor **intent**

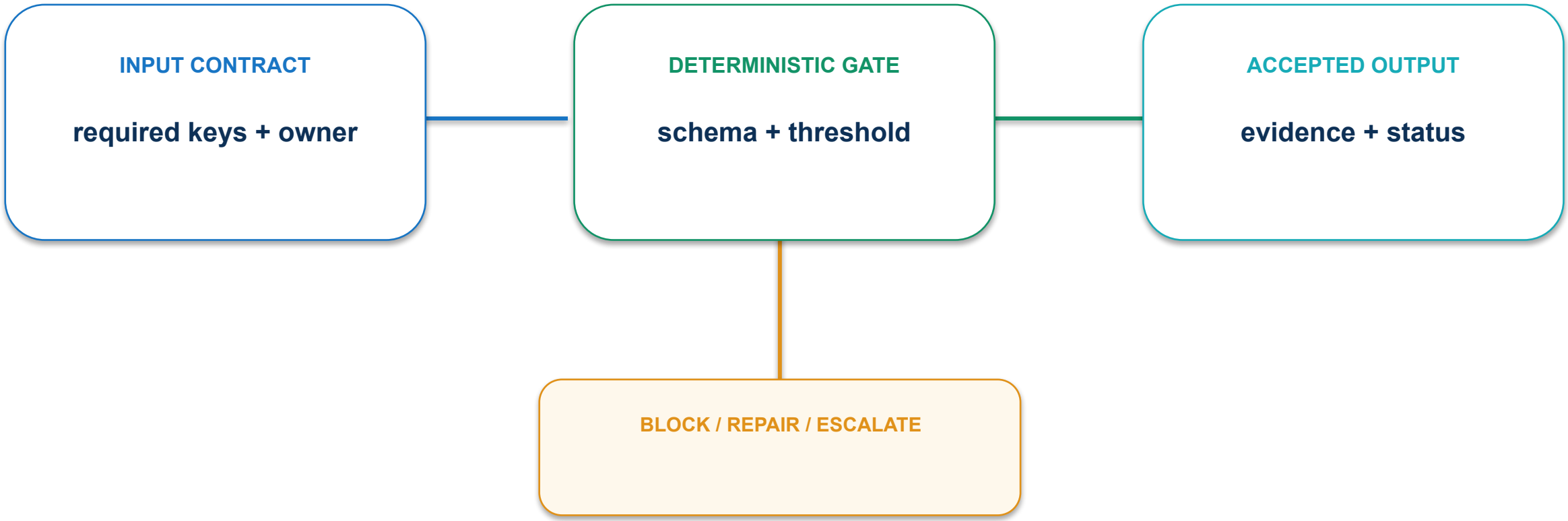
value **examples**

FAILURE MODE

A story is marked complete when only the happy path works.

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04 · User stories and acceptance criteria: contract and control



Attach concrete examples and negative conditions to each story.

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04 · User stories and acceptance criteria: evidence and decision

MEASURE

% stories with positive, edge and failure examples

ACCEPTANCE EVIDENCE

Story map with example-based acceptance

OBSERVED

machine value

THRESHOLD

approved rule

DECISION

PASS / REPAIR

OWNER

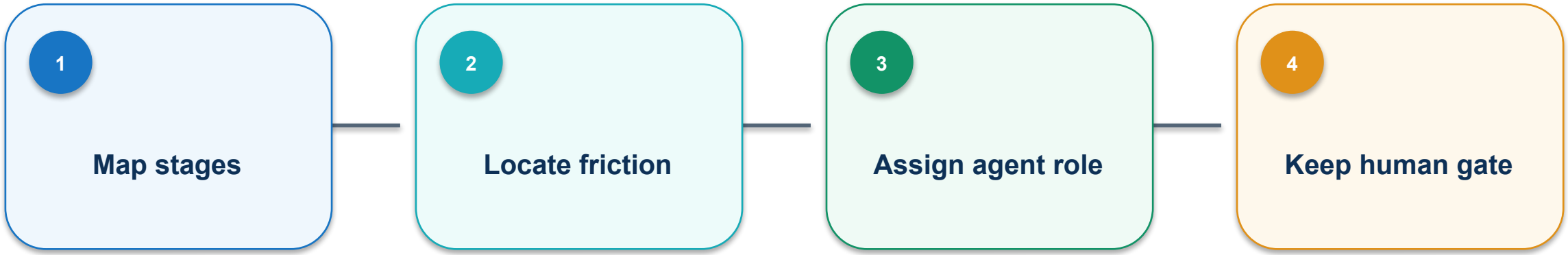
named human

Completion principle: evidence travels with the product hypothesis, changed state and decision—not in an unsupported claim.

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MECHANISM

05 · Journey-map orchestration: execution path



STATE PASSED BETWEEN STAGES

stage

friction

agent_role

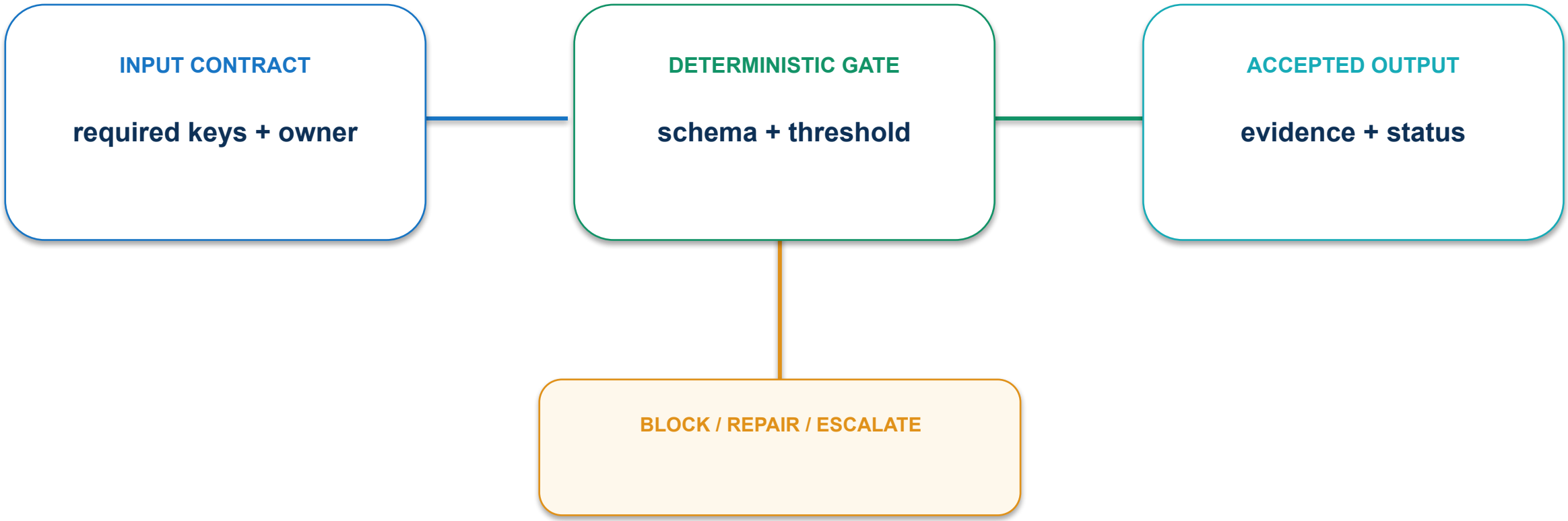
gate

FAILURE MODE

The workflow optimises speed while weakening trust at critical moments.

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05 · Journey-map orchestration: contract and control



Preserve recovery, explanation and control at consequential journey points.

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05 · Journey-map orchestration: evidence and decision

MEASURE

% high-consequence moments with human control

ACCEPTANCE EVIDENCE

Journey map with agent and human swimlanes

OBSERVED

machine value

THRESHOLD

approved rule

DECISION

PASS / REPAIR

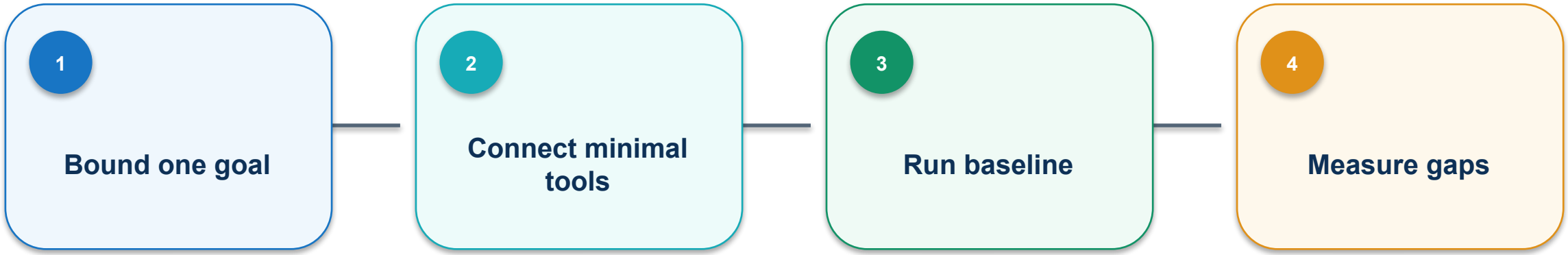
OWNER

named human

Completion principle: evidence travels with the product hypothesis, changed state and decision—not in an unsupported claim.

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06 · Single-agent baseline: execution path



STATE PASSED BETWEEN STAGES

goal

tools

baseline

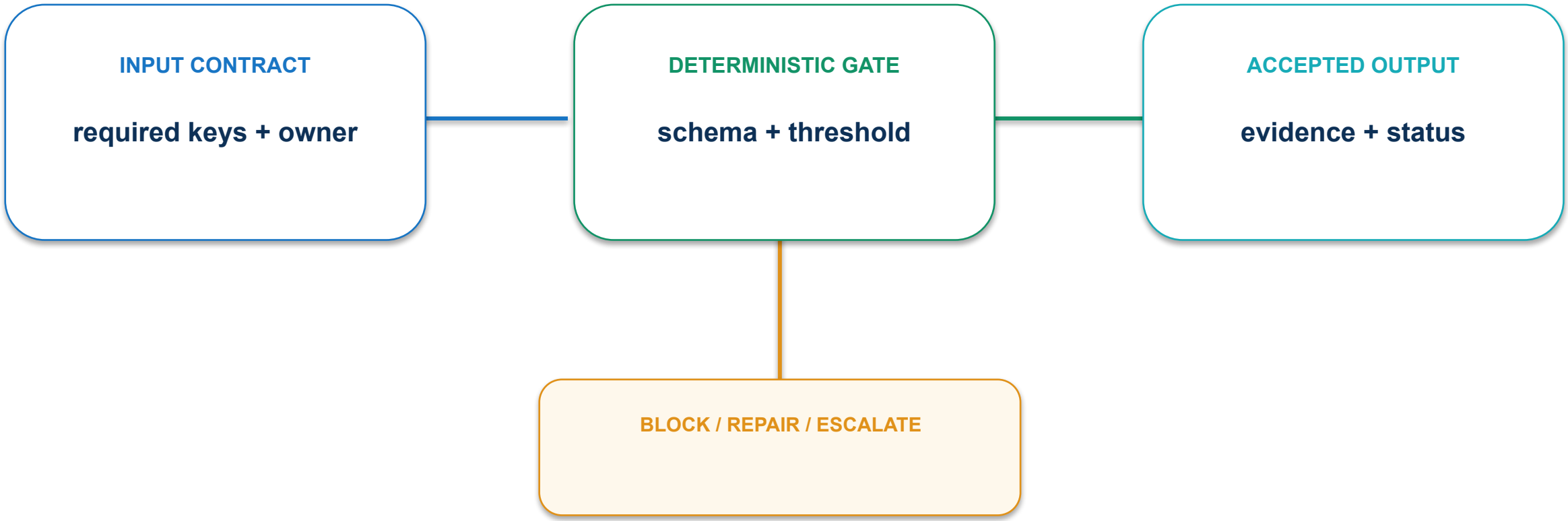
gaps

FAILURE MODE

A multi-agent design is introduced before a simpler baseline is measured.

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06 · Single-agent baseline: contract and control



Prove the minimum viable agent before adding roles.

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06 · Single-agent baseline: evidence and decision

MEASURE

Accepted tasks per 20-case evaluation set

ACCEPTANCE EVIDENCE

Baseline run log and failure summary

OBSERVED

machine value

THRESHOLD

approved rule

DECISION

PASS / REPAIR

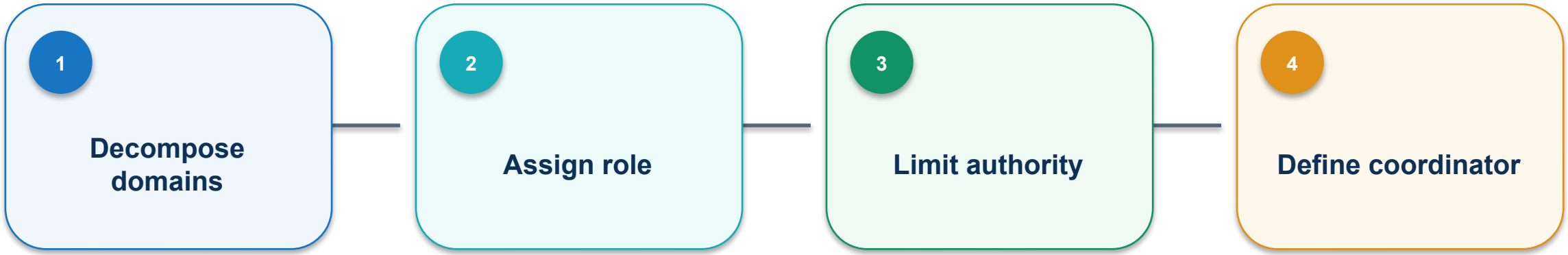
OWNER

named human

Completion principle: evidence travels with the product hypothesis, changed state and decision—not in an unsupported claim.

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07 · Multi-agent role design: execution path



STATE PASSED BETWEEN STAGES

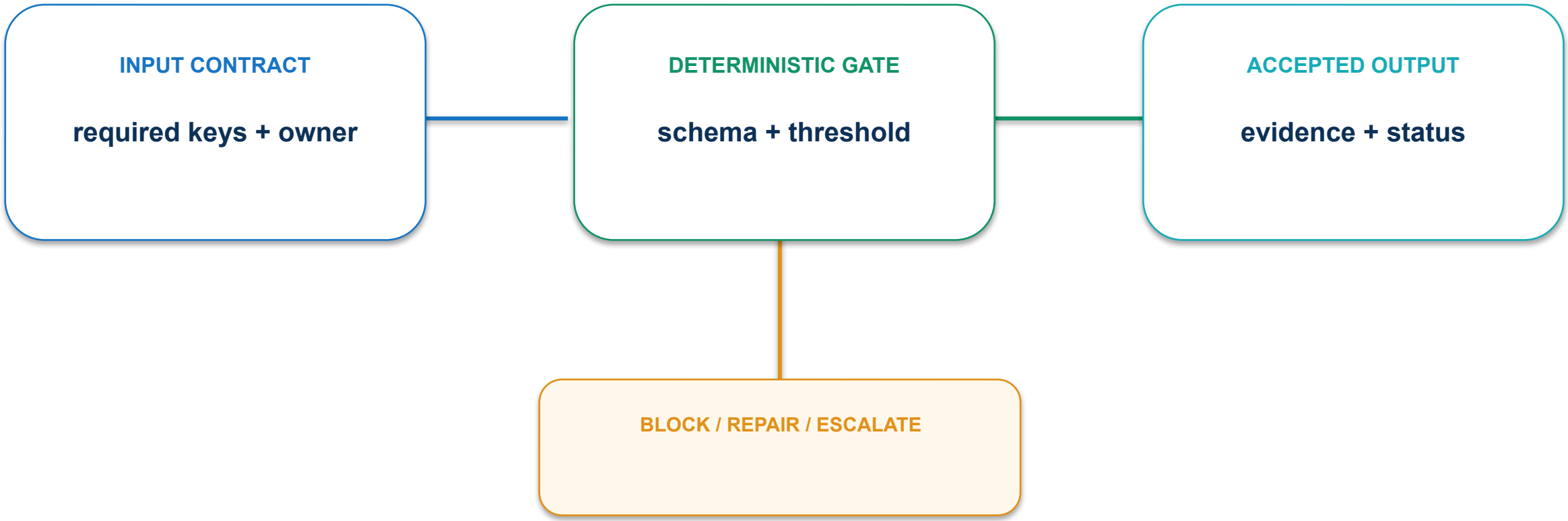
- **domain**
- **role**
- **authority**
- **coordinator**

FAILURE MODE

Multiple agents duplicate work and amplify inconsistent assumptions.

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07 · Multi-agent role design: contract and control



Give each role distinct inputs, output schema and decision rights.

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07 · Multi-agent role design: evidence and decision

MEASURE

Handoffs that add unique value / total handoffs

ACCEPTANCE EVIDENCE

Role charter and responsibility matrix

OBSERVED

machine value

THRESHOLD

approved rule

DECISION

PASS / REPAIR

OWNER

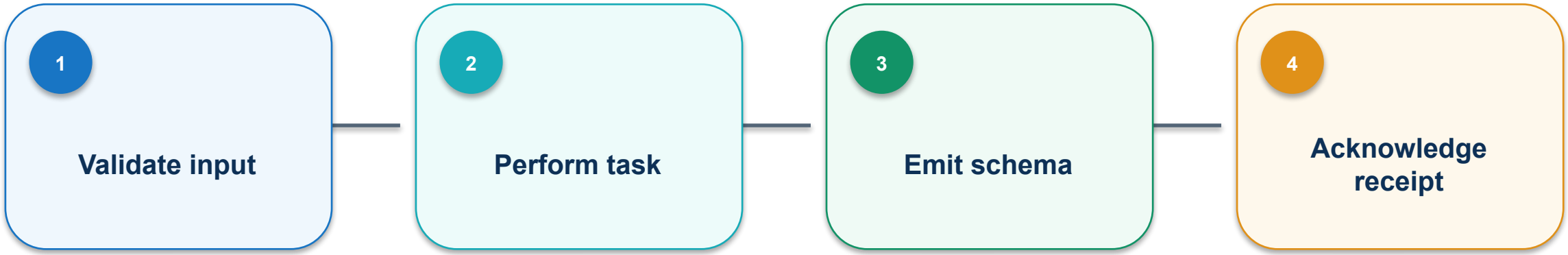
named human

Completion principle: evidence travels with the product hypothesis, changed state and decision—not in an unsupported claim.

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MECHANISM

08 · Handoff contract: execution path



STATE PASSED BETWEEN STAGES

input_schema

task

output_schema

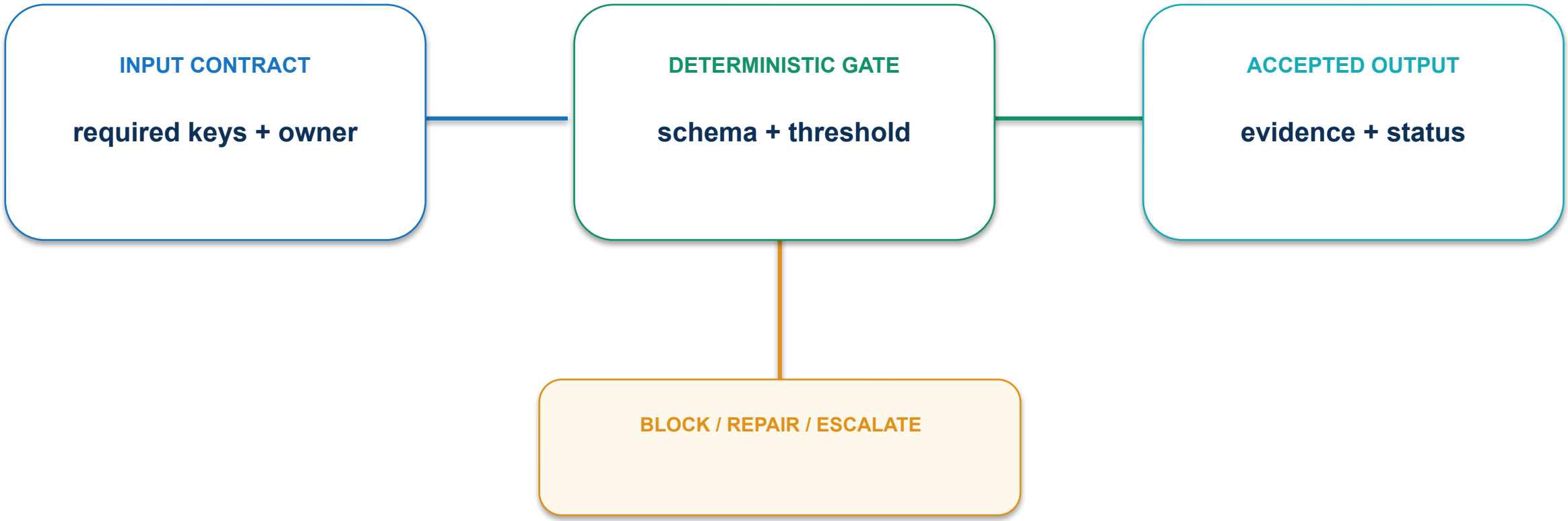
ack

FAILURE MODE

Free-form outputs break downstream steps or lose provenance.

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08 · Handoff contract: contract and control



Use stable identifiers, structured outputs and fail-closed validation.

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08 · Handoff contract: evidence and decision

MEASURE

% handoffs passing schema validation

ACCEPTANCE EVIDENCE

Handoff record with validation status

OBSERVED

machine value

THRESHOLD

approved rule

DECISION

PASS / REPAIR

OWNER

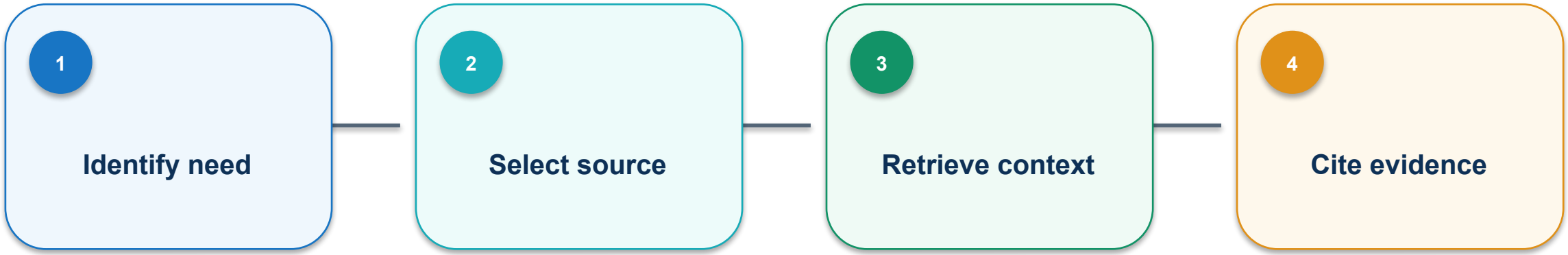
named human

Completion principle: evidence travels with the product hypothesis, changed state and decision—not in an unsupported claim.

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MECHANISM

09 · Tool and retrieval grounding: execution path



STATE PASSED BETWEEN STAGES

need

source

context

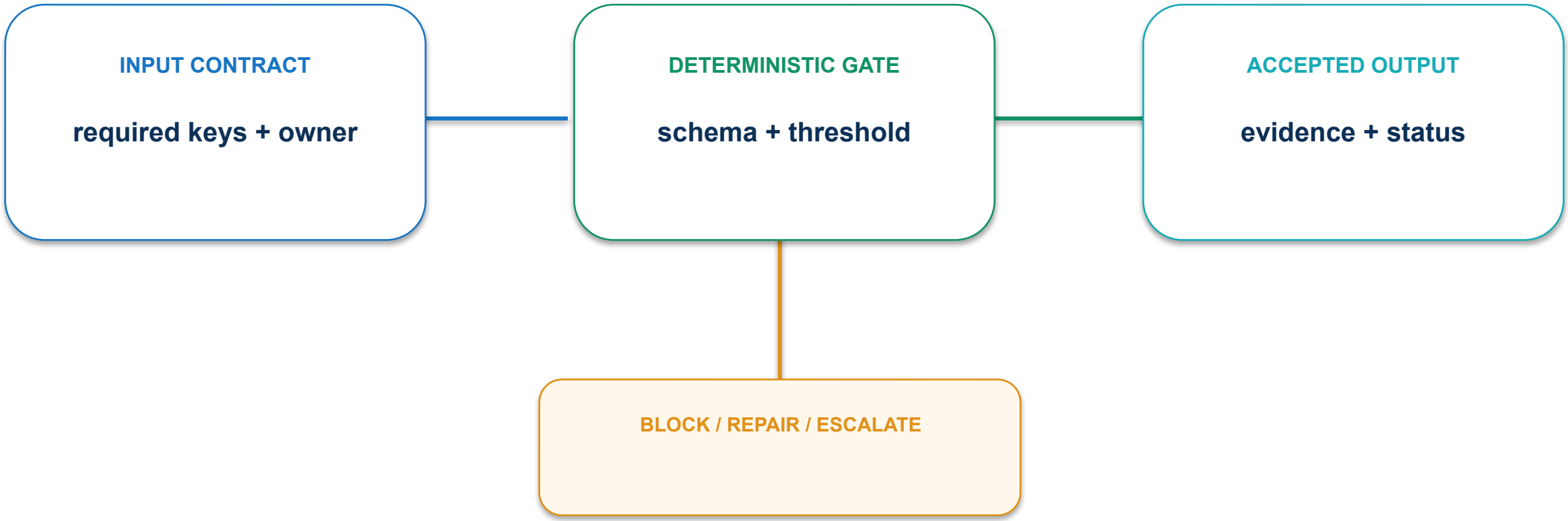
citation

FAILURE MODE

The agent uses stale, irrelevant or untrusted material as product truth.

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09 · Tool and retrieval grounding: contract and control



Constrain sources, retain citations and test retrieval misses.

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09 · Tool and retrieval grounding: evidence and decision

MEASURE

% answer claims grounded in approved sources

ACCEPTANCE EVIDENCE

Retrieved evidence linked to output claims

OBSERVED

machine value

THRESHOLD

approved rule

DECISION

PASS / REPAIR

OWNER

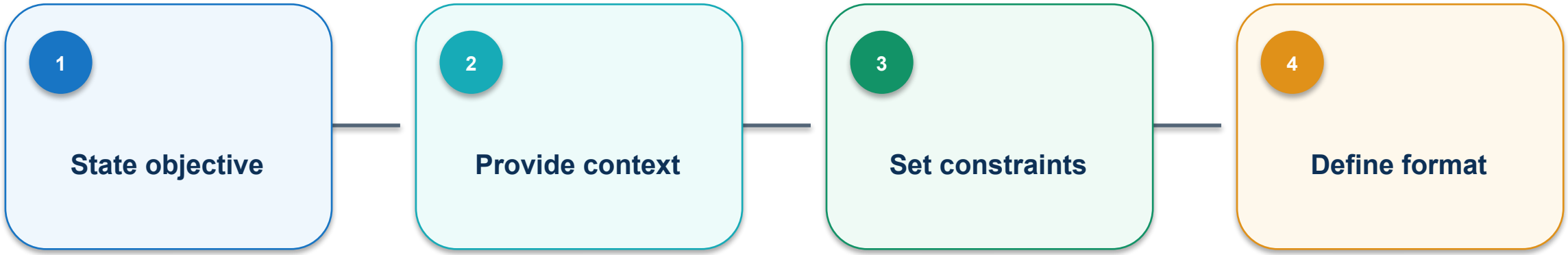
named human

Completion principle: evidence travels with the product hypothesis, changed state and decision—not in an unsupported claim.

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MECHANISM

10 · Prompt contract design: execution path



STATE PASSED BETWEEN STAGES

objective

context

constraints

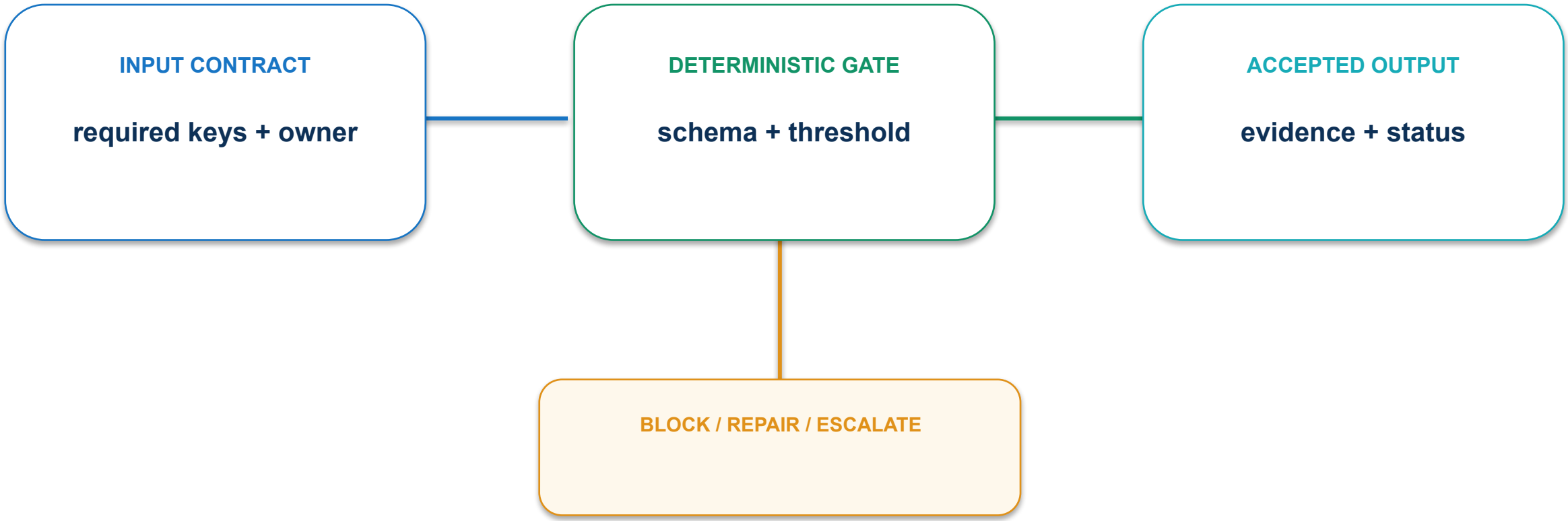
output_schema

FAILURE MODE

A long prompt hides conflicting instructions and undefined priorities.

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10 · Prompt contract design: contract and control



Separate goal, context, rules, examples and output contract.

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10 · Prompt contract design: evidence and decision

MEASURE

First-pass outputs meeting all contract fields

ACCEPTANCE EVIDENCE

Prompt version and conformance checklist

OBSERVED

machine value

THRESHOLD

approved rule

DECISION

PASS / REPAIR

OWNER

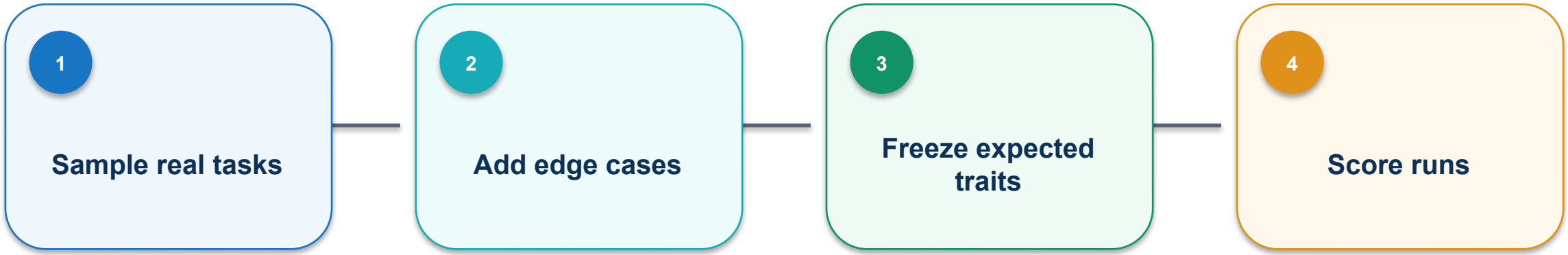
named human

Completion principle: evidence travels with the product hypothesis, changed state and decision—not in an unsupported claim.

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MECHANISM

11 · Evaluation-set engineering: execution path



STATE PASSED BETWEEN STAGES

task_id

input

expected

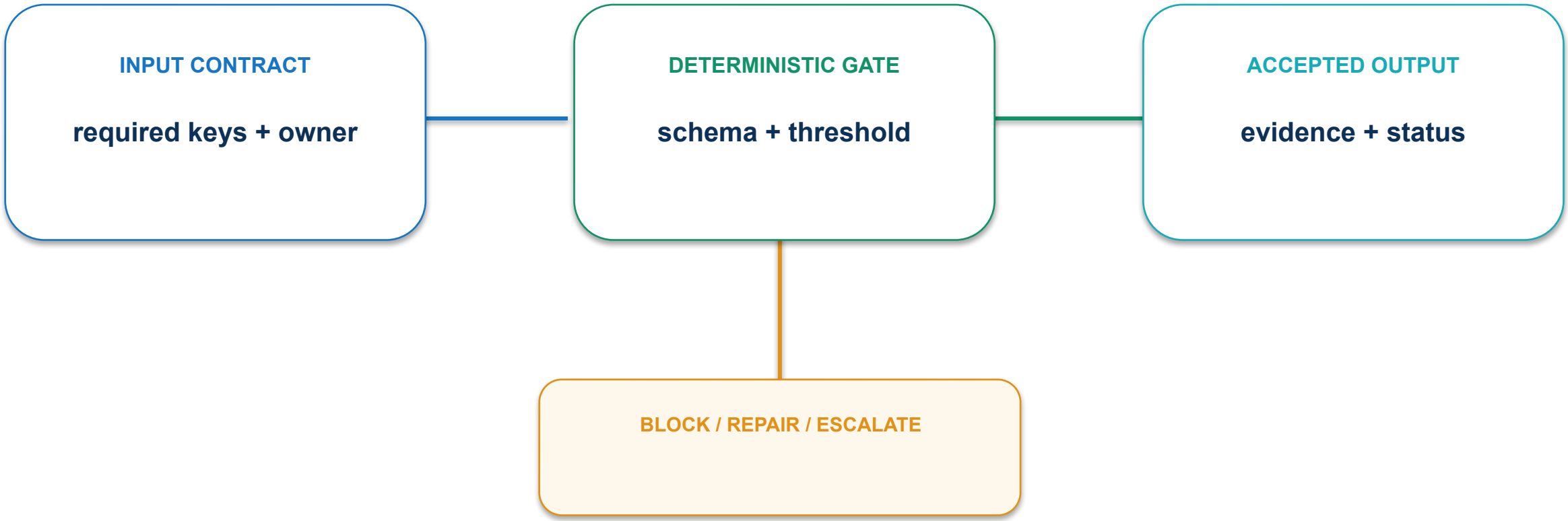
score

FAILURE MODE

Teams judge prototypes from one impressive demonstration.

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11 · Evaluation-set engineering: contract and control



Build a representative frozen set before optimisation.

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11 · Evaluation-set engineering: evidence and decision

MEASURE

Coverage across core, edge and adversarial cases

ACCEPTANCE EVIDENCE

Versioned evaluation CSV and score report

OBSERVED

machine value

THRESHOLD

approved rule

DECISION

PASS / REPAIR

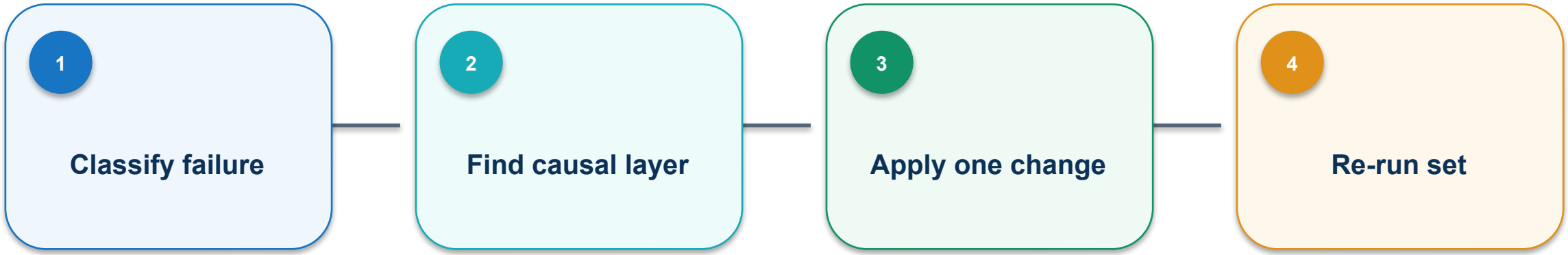
OWNER

named human

Completion principle: evidence travels with the product hypothesis, changed state and decision—not in an unsupported claim.

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12 · Iteration and error taxonomy: execution path



STATE PASSED BETWEEN STAGES

failure_type

layer

change

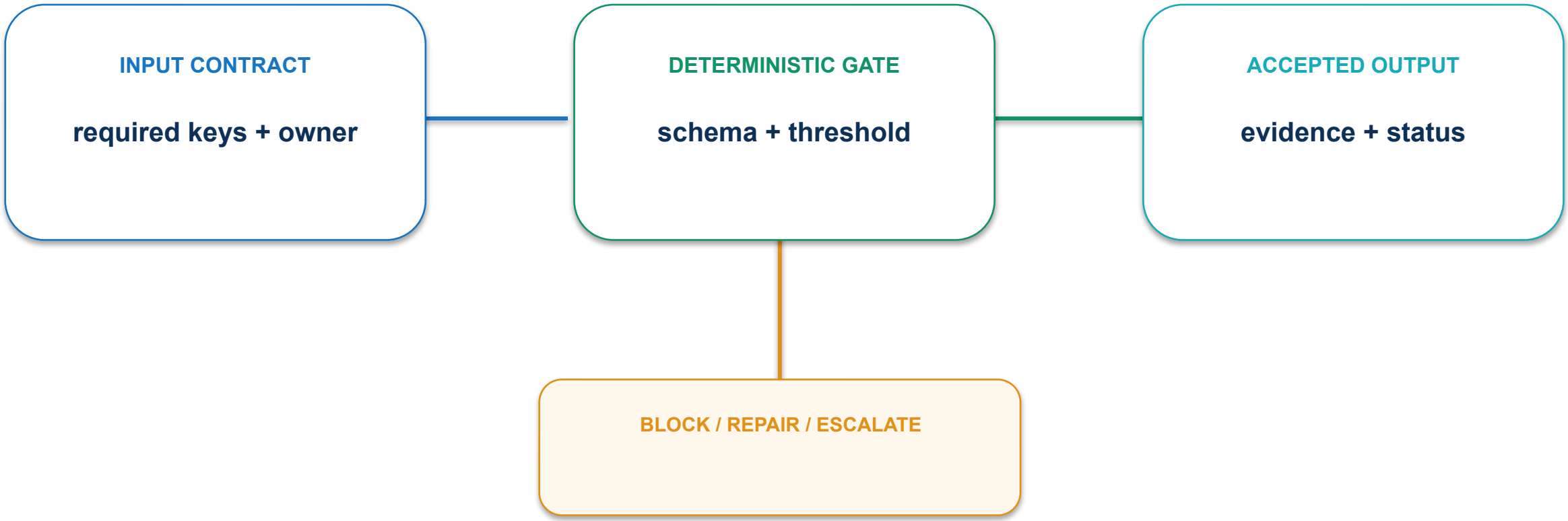
delta

FAILURE MODE

Several prompt, model and data changes are bundled, obscuring causality.

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12 · Iteration and error taxonomy: contract and control



Change one causal layer per experiment and re-run the full frozen set.

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12 · Iteration and error taxonomy: evidence and decision

MEASURE

Improvement on failed cases without regression

ACCEPTANCE EVIDENCE

Experiment ledger with before-after deltas

OBSERVED

machine value

THRESHOLD

approved rule

DECISION

PASS / REPAIR

OWNER

named human

Completion principle: evidence travels with the product hypothesis, changed state and decision—not in an unsupported claim.

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13 · Prototype-to-learning loop: execution path

PROTOTYPE LEARNING SURFACE

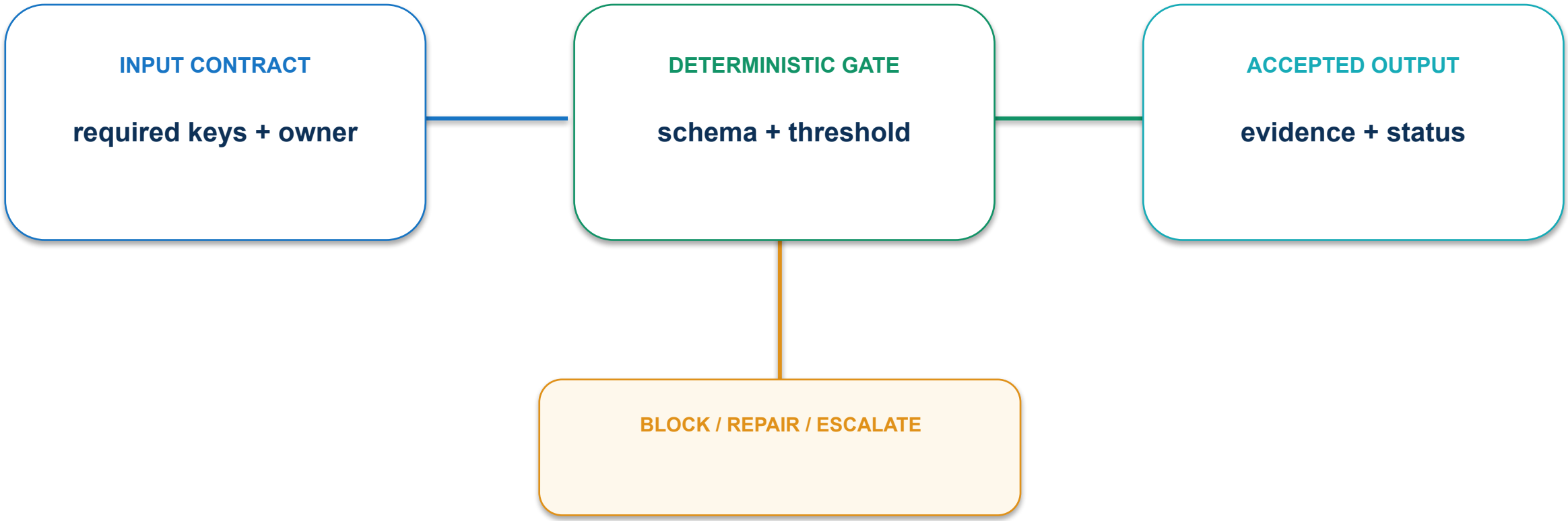
1	BUILD	Create the thinnest useful slice
2	TEST	Observe representative users
3	LEARN	Separate signal from opinion
4	DECIDE	Continue, pivot or stop



CONCEPT VISUAL • LABELLED

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13 · Prototype-to-learning loop: contract and control



Time-box builds and define the decision each prototype must unlock.

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13 · Prototype-to-learning loop: evidence and decision

MEASURE

Validated assumptions per prototype cycle

ACCEPTANCE EVIDENCE

Experiment brief and go-pivot-stop decision

OBSERVED

machine value

THRESHOLD

approved rule

DECISION

PASS / REPAIR

OWNER

named human

Completion principle: evidence travels with the product hypothesis, changed state and decision—not in an unsupported claim.

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Workflow boundary: context, agent roles and human gates

1

CONTEXT

Define outcome and constraints

2

WORKFLOW

Coordinate roles and tools

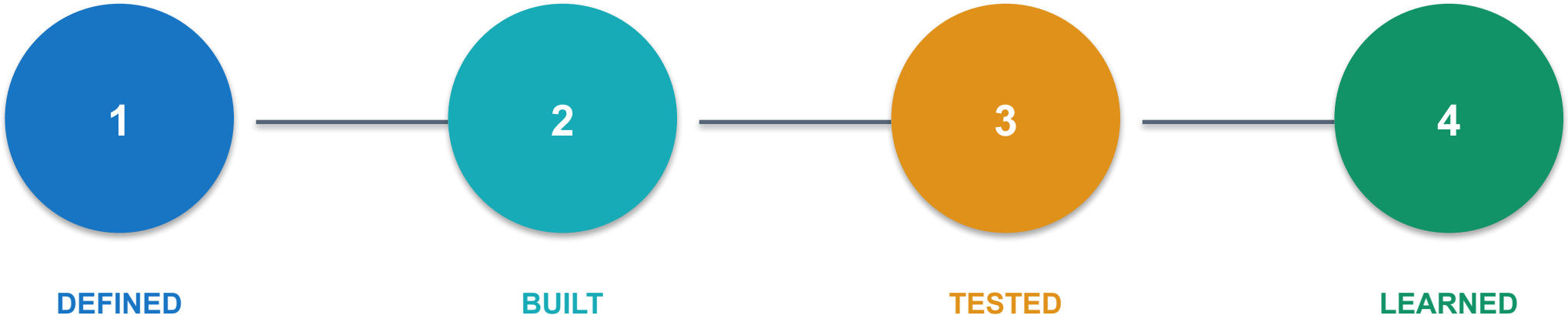
3

HUMAN GATE

Review evidence and consequence

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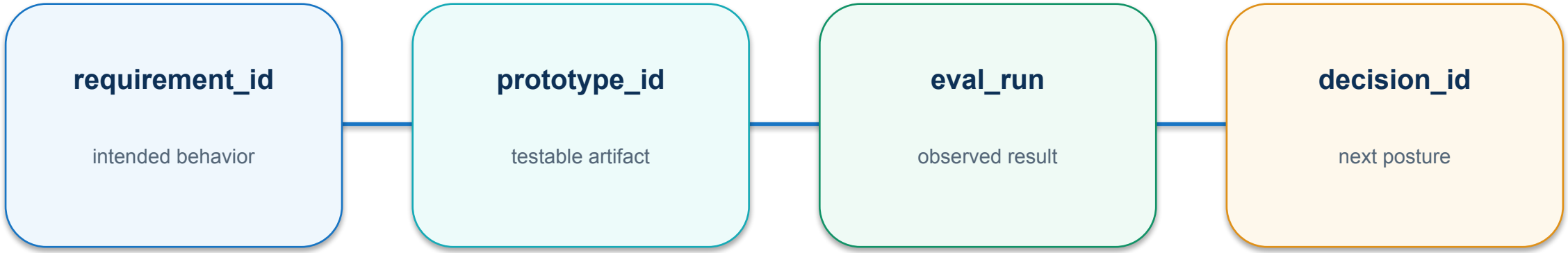
Prototype state: DEFINED, BUILT, TESTED and LEARNED



LEARNED means the prototype changed or confirmed a product hypothesis.

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Product evidence chain: requirement → prototype → evaluation → decision



A prototype decision is credible when requirements, tests and learning remain connected.

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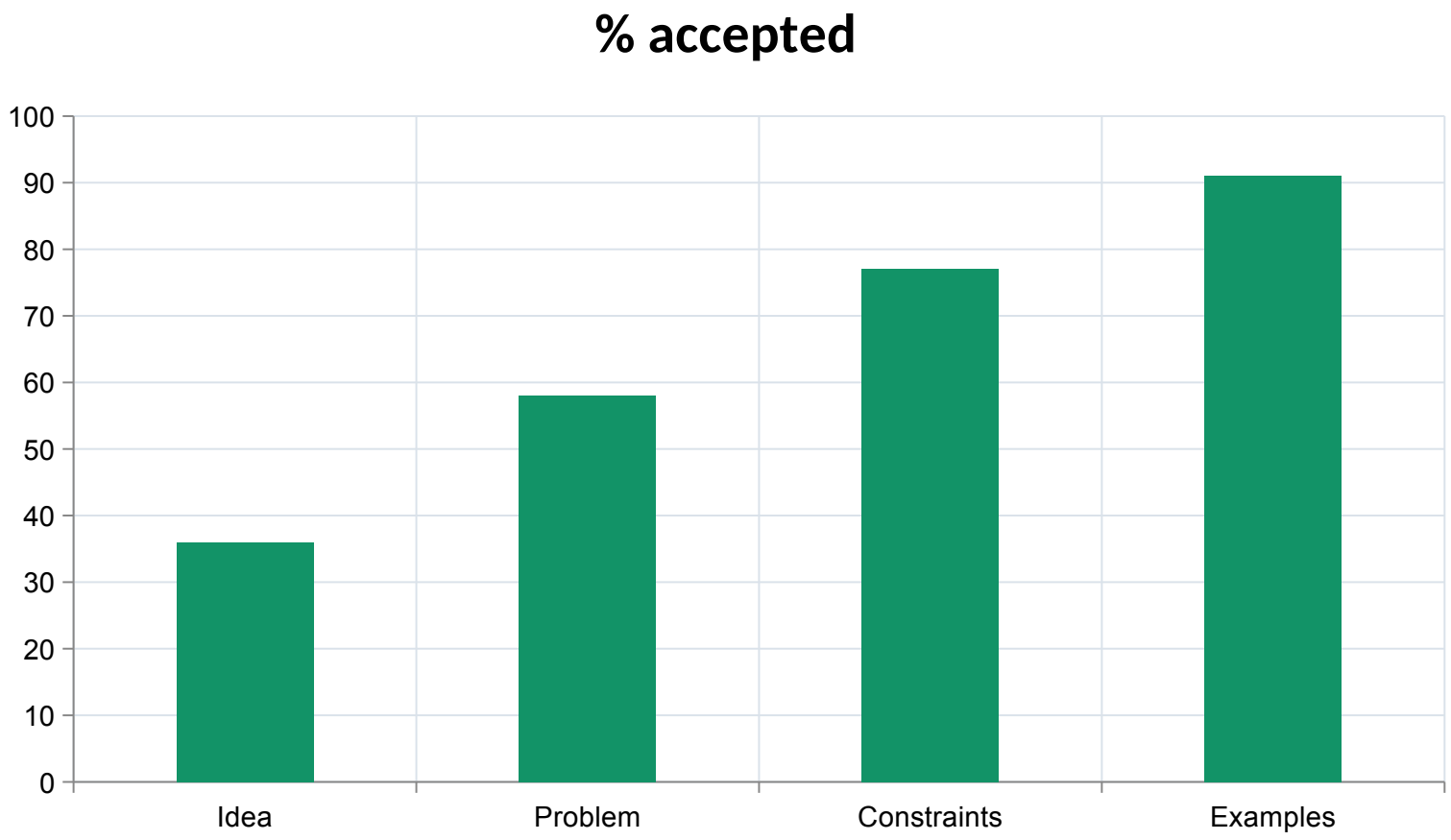
Optimisation taxonomy: context, handoff, quality and regression

CONTEXT	repair missing product constraints
HANDOFF	restore schema and provenance
QUALITY	target the causal prompt or data layer
REGRESSION	re-run the frozen evaluation set

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EDITABLE NATIVE CHART

Structured context improves first-pass quality



INSIGHT

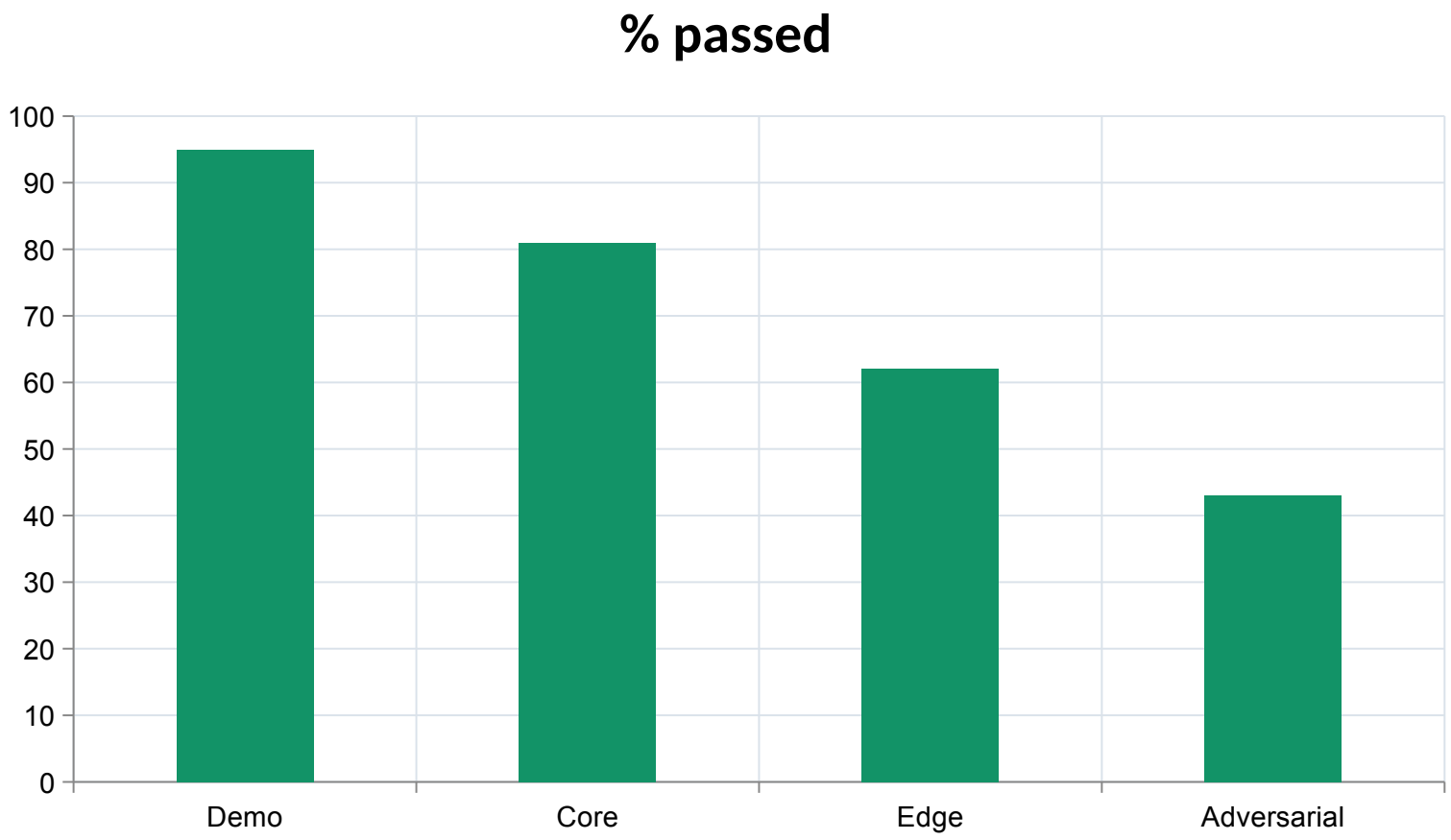
Product context, constraints and examples improve first-pass usefulness.

Illustrative synthetic values for mechanism explanation.

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EDITABLE NATIVE CHART

Frozen evaluations expose real improvement



INSIGHT

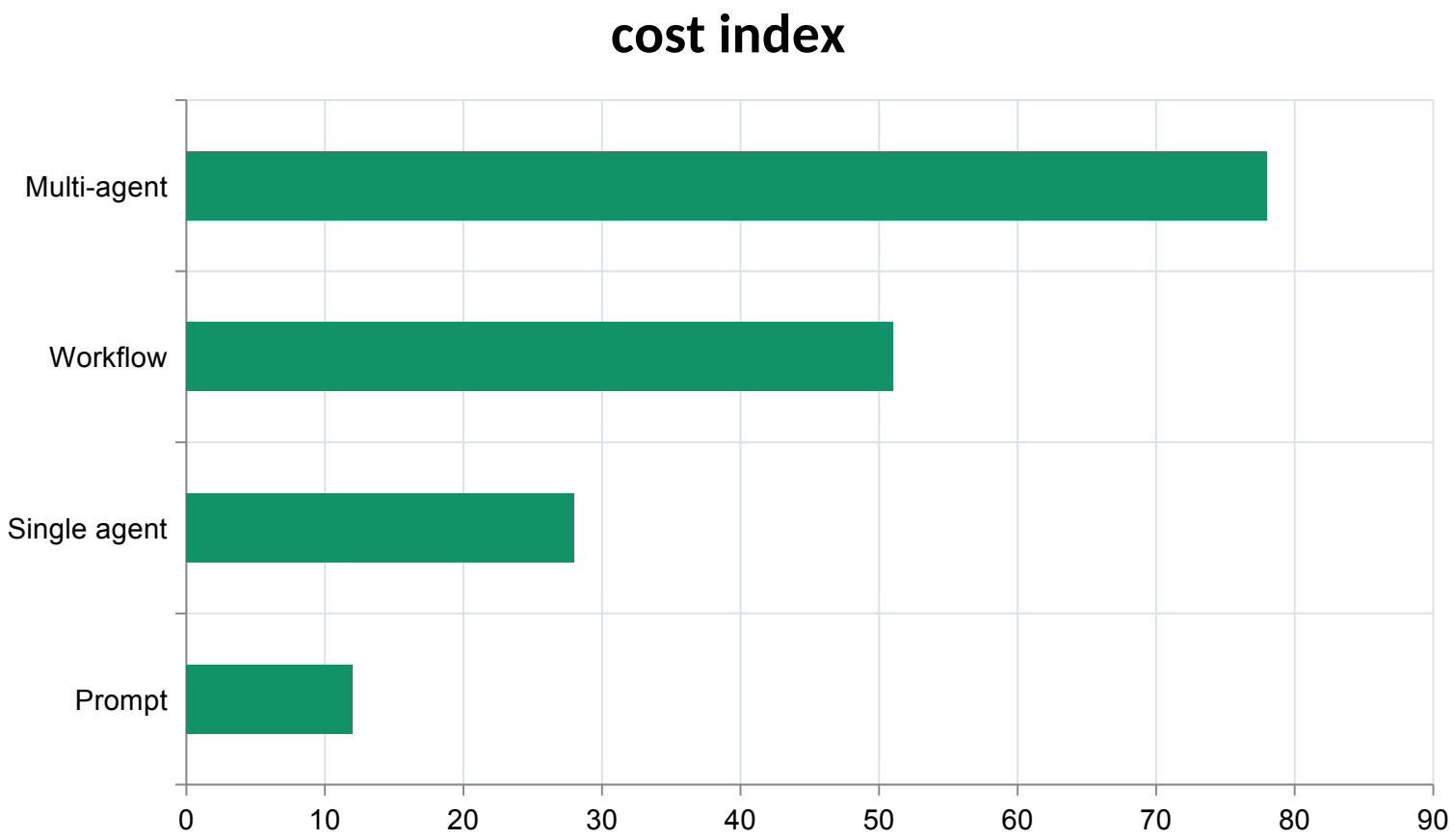
A frozen evaluation set prevents teams from moving the goalposts during optimisation.

Illustrative synthetic values for mechanism explanation.

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EDITABLE NATIVE CHART

Workflow complexity expands coordination cost



INSIGHT

More agent roles add coordination cost and must earn their place through measured value.

Illustrative synthetic values for mechanism explanation.

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ACTIVITY BRIEF

Activity 05 · Design an Agent Workflow Canvas

SCENARIO

Map a product-discovery task into an agentic loop with tools, state, stop conditions and human gates.

EXPECTED OUTPUT

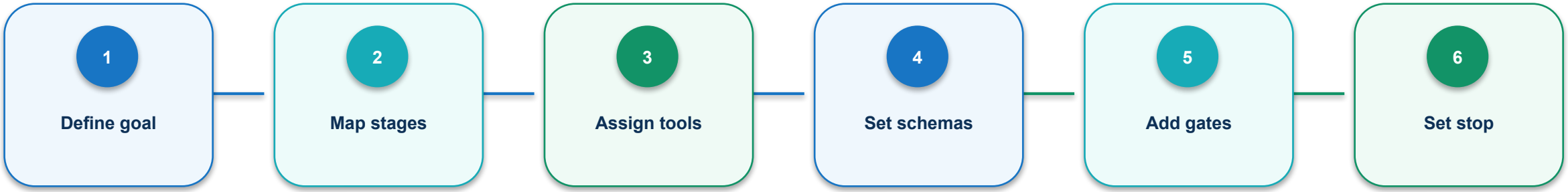
Workflow canvas and responsibility matrix

Detailed procedures, prompts, troubleshooting and acceptance checks are in the Learner Guide and activity PDF.

Self-contained folder: activities/activity-05-agent-workflow-canvas/

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Activity 05 · agent workflow contract



Consequential actions require a current diff, passing checks and a named decision.

HUMAN GATE

Approval resumes only when the reviewed diff and current state still match.

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Activity 05 · evidence and failure gates

PASS EVIDENCE

Every stage has an owner and observable output; consequential actions stop for human review.

FAIL CLOSED WHEN

a protected boundary is crossed; evidence is missing; approval is absent; or the current diff no longer matches review.

README.pdf

detailed procedure

mock-data/

synthetic source state

templates/

learner working files

evidence/

acceptance record

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ACTIVITY BRIEF

Activity 06 · Create an AI-Ready PRD

SCENARIO

Convert an opportunity brief into outcome-focused requirements, user stories and acceptance examples.

EXPECTED OUTPUT

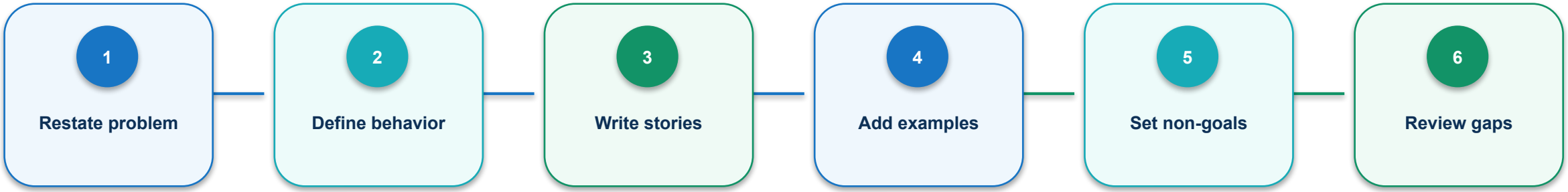
Lean PRD, story map and acceptance set

Detailed procedures, prompts, troubleshooting and acceptance checks are in the Learner Guide and activity PDF.

Self-contained folder: activities/activity-06-prd-and-user-stories/

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Activity 06 · agent workflow contract



Consequential actions require a current diff, passing checks and a named decision.

HUMAN GATE

Approval resumes only when the reviewed diff and current state still match.

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Activity 06 · evidence and failure gates

PASS EVIDENCE

Requirements are testable, non-goals are explicit, and every priority story has three acceptance examples.

FAIL CLOSED WHEN

a protected boundary is crossed; evidence is missing; approval is absent; or the current diff no longer matches review.

README.pdf

detailed procedure

mock-data/

synthetic source state

templates/

learner working files

evidence/

acceptance record

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ACTIVITY BRIEF

Activity 07 · Engineer a Prompt and Evaluation Set

SCENARIO

Create a structured prompt contract and a frozen evaluation set for a product-feedback agent.

EXPECTED OUTPUT

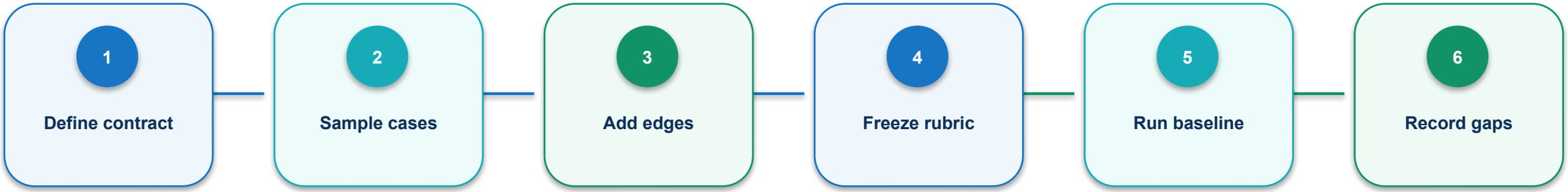
Versioned prompt, 20-case evaluation CSV and scoring rubric

Detailed procedures, prompts, troubleshooting and acceptance checks are in the Learner Guide and activity PDF.

Self-contained folder: activities/activity-07-prompt-evaluation-set/

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Activity 07 · agent workflow contract



Each node consumes named keys, emits a status, and leaves retrievable execution evidence.

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Activity 07 · evidence and failure gates

PASS EVIDENCE

The evaluation set covers all segments; scoring rules are reproducible; baseline failures are retained.

FAIL CLOSED WHEN

a protected boundary is crossed; evidence is missing; approval is absent; or the current diff no longer matches review.

README.pdf

detailed procedure

mock-data/

synthetic source state

templates/

learner working files

evidence/

acceptance record

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ACTIVITY BRIEF

Activity 08 · Run a Prototype Experiment

SCENARIO

Evaluate a synthetic agent prototype, classify failures and recommend one causal improvement.

EXPECTED OUTPUT

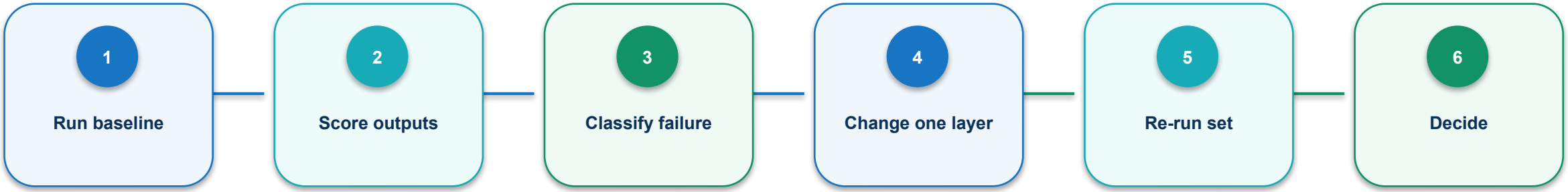
Experiment ledger with before-after scores and decision

Detailed procedures, prompts, troubleshooting and acceptance checks are in the Learner Guide and activity PDF.

Self-contained folder: activities/activity-08-prototype-experiment/

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Activity 08 · agent workflow contract



Consequential actions require a current diff, passing checks and a named decision.

HUMAN GATE

Approval resumes only when the reviewed diff and current state still match.

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Activity 08 · evidence and failure gates

PASS EVIDENCE

The change targets one causal layer, the frozen set is preserved, and regression risk is documented.

FAIL CLOSED WHEN

a protected boundary is crossed; evidence is missing; approval is absent; or the current diff no longer matches review.

README.pdf

detailed procedure

mock-data/

synthetic source state

templates/

learner working files

evidence/

acceptance record

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CHECKPOINT

Topic 2 technical checkpoint

1. Which state field prevents a silent downstream assumption?

2. What evidence proves the mechanism completed successfully?

3. Which failure must stop rather than retry?

4. Who owns the consequential decision?

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CHECKPOINT

Topic 2 transfer challenge

1. Change one input constraint and predict the affected nodes.
2. Name the identifier that preserves provenance across the change.
3. Define one machine threshold and one human decision.
4. Describe the minimum evidence required to resume the run.

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3

TOPIC 3

Deploying and Evaluating Agentic AI-Powered Product Solutions

Design a deployment posture that preserves human control, security, observability, cost discipline and a production feedback loop.

1 CONTROL

2 EVIDENCE

3 AUTOMATION

4 DECISION

Every product decision ends in observable evidence—not an agent's assertion.

01 · Prototype-to-production gate: execution path

PRODUCTION READINESS REVIEW

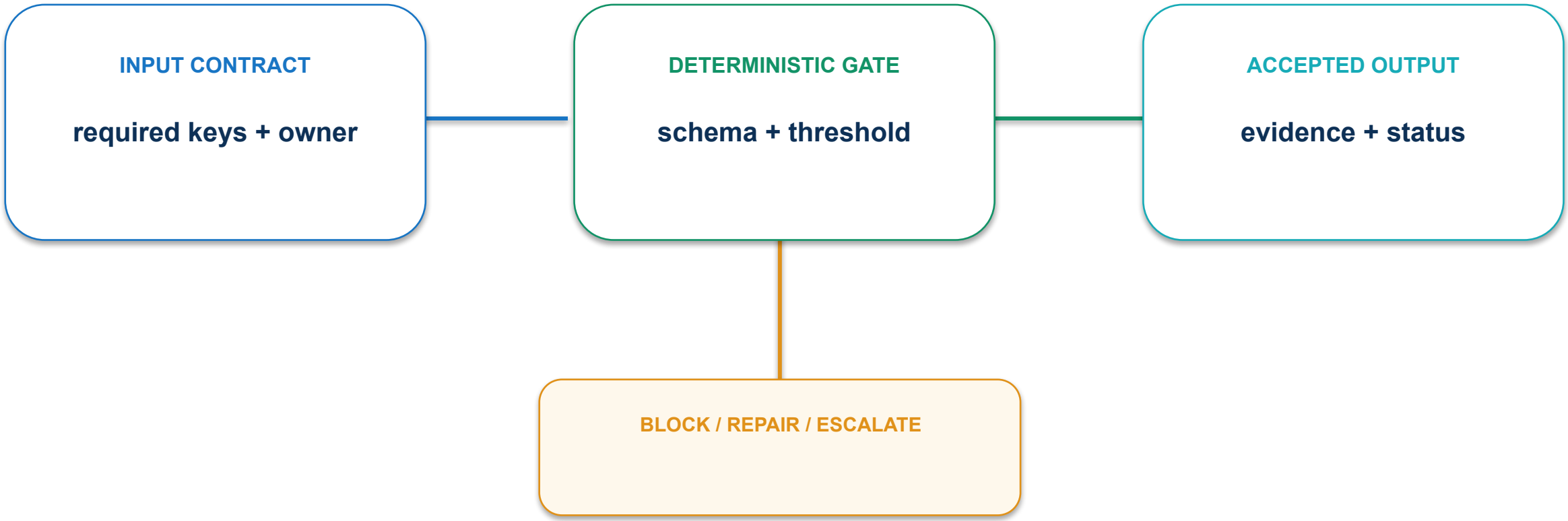
1. **VALUE** — Validate user and business outcome
2. **QUALITY** — Pass frozen and live evaluations
3. **OPERATE** — Instrument cost, latency and risk
4. **GOVERN** — Bind rollout to human ownership



CONCEPT VISUAL • LABELLED

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01 · Prototype-to-production gate: contract and control



Separate learning code from supported production architecture and evidence.

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01 · Prototype-to-production gate: evidence and decision

MEASURE

Required gates passed before production investment

ACCEPTANCE EVIDENCE

Handoff pack with eval set, edge cases and prompt strategy

OBSERVED

machine value

THRESHOLD

approved rule

DECISION

PASS / REPAIR

OWNER

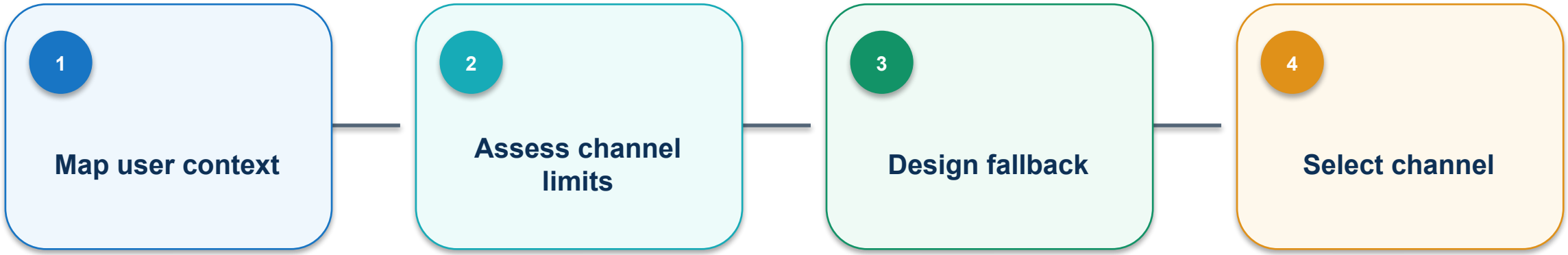
named human

Completion principle: evidence travels with the product hypothesis, changed state and decision—not in an unsupported claim.

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MECHANISM

02 · Deployment channel fit: execution path



STATE PASSED BETWEEN STAGES

context

channel

fallback

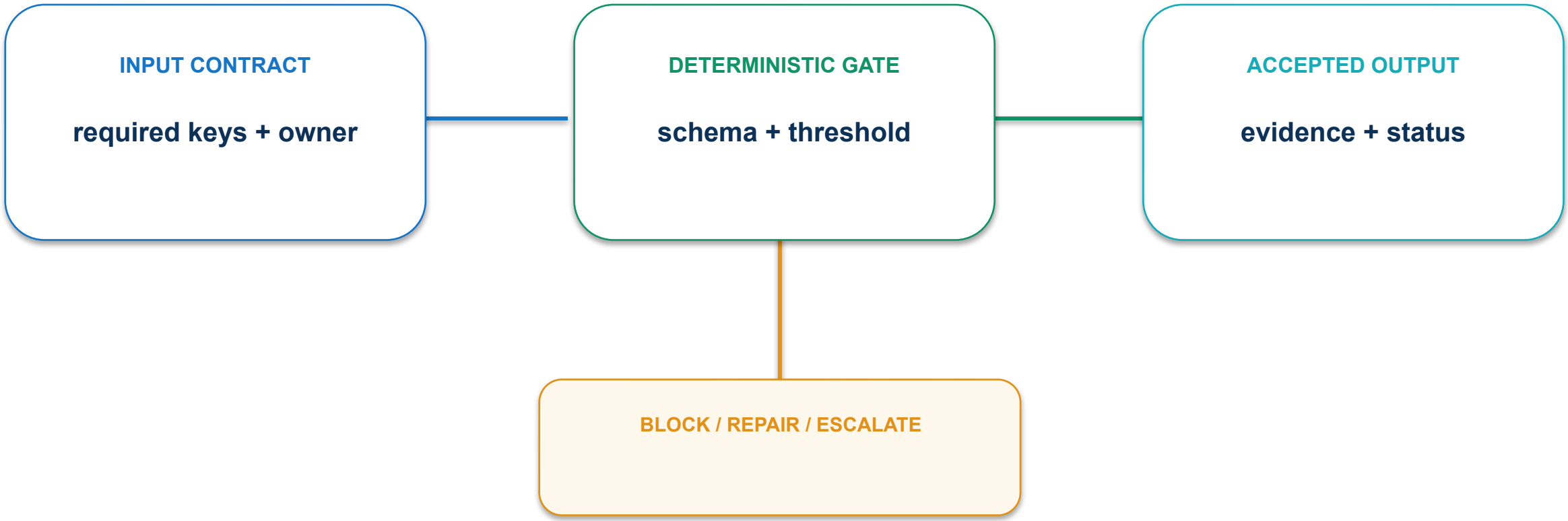
choice

FAILURE MODE

The easiest integration is chosen despite poor user context or control.

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02 · Deployment channel fit: contract and control



Compare web, workspace, messaging and embedded channels on task and risk.

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02 · Deployment channel fit: evidence and decision

MEASURE

Task completion and recovery rate by channel

ACCEPTANCE EVIDENCE

Channel matrix and fallback path

OBSERVED

machine value

THRESHOLD

approved rule

DECISION

PASS / REPAIR

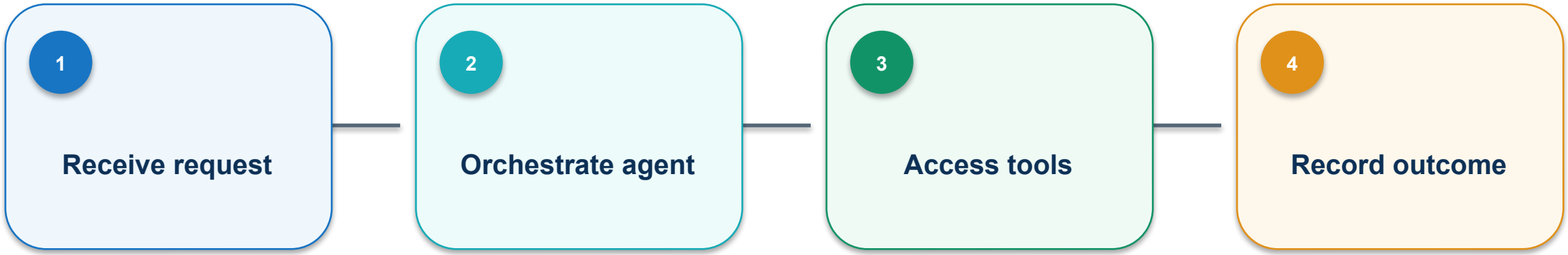
OWNER

named human

Completion principle: evidence travels with the product hypothesis, changed state and decision—not in an unsupported claim.

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03 · Reference deployment architecture: execution path



STATE PASSED BETWEEN STAGES

request_id

orchestrator

tool_policy

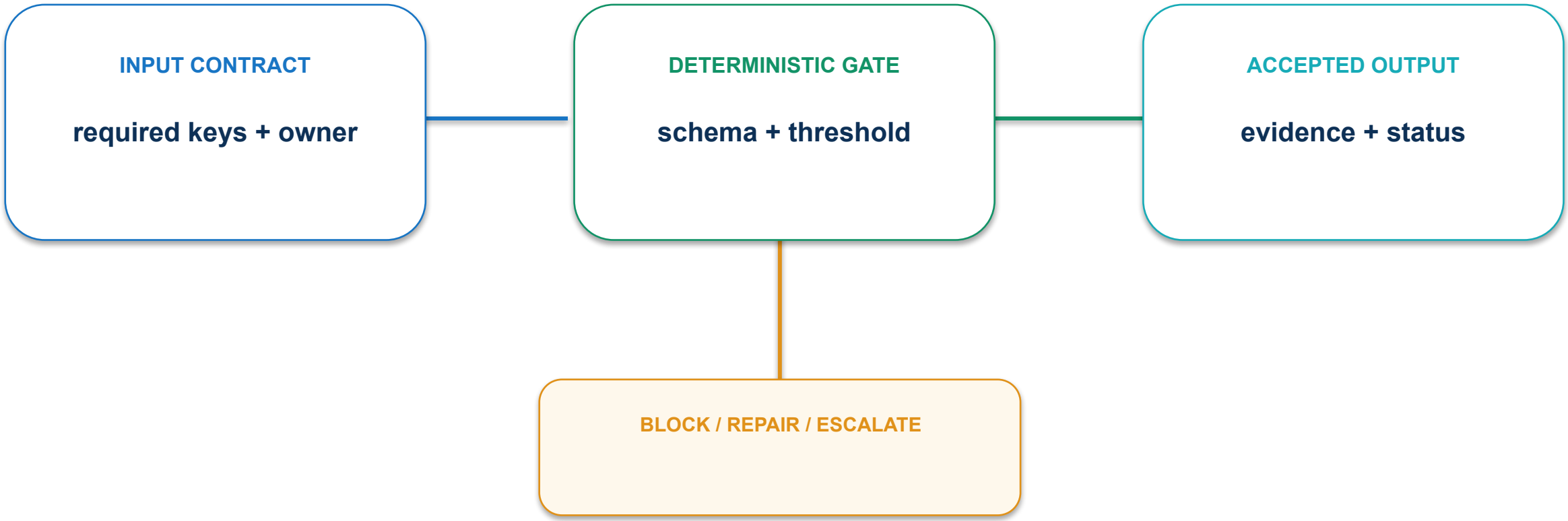
audit_event

FAILURE MODE

Model calls, tool actions and user decisions cannot be correlated.

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03 · Reference deployment architecture: contract and control



Carry a stable request identifier through every component.

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03 · Reference deployment architecture: evidence and decision

MEASURE

% transactions traceable end to end

ACCEPTANCE EVIDENCE

Architecture diagram and correlated trace

OBSERVED

machine value

THRESHOLD

approved rule

DECISION

PASS / REPAIR

OWNER

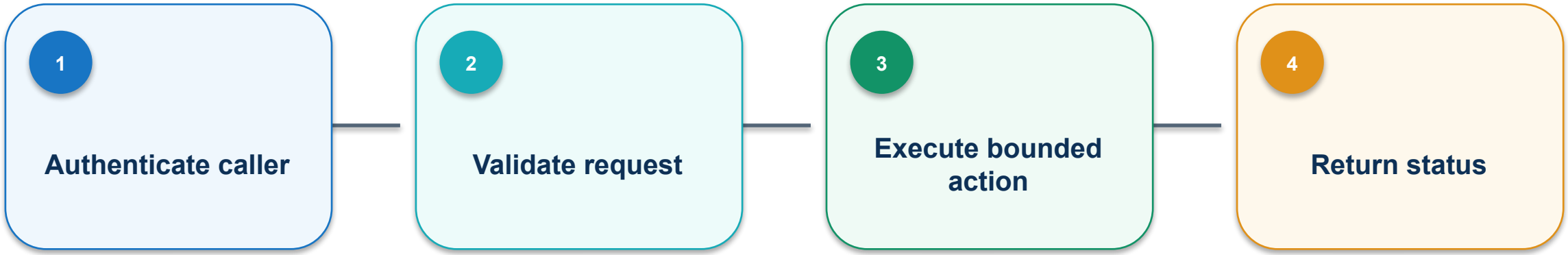
named human

Completion principle: evidence travels with the product hypothesis, changed state and decision—not in an unsupported claim.

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MECHANISM

04 · API and integration contract: execution path



STATE PASSED BETWEEN STAGES

identity

schema

action

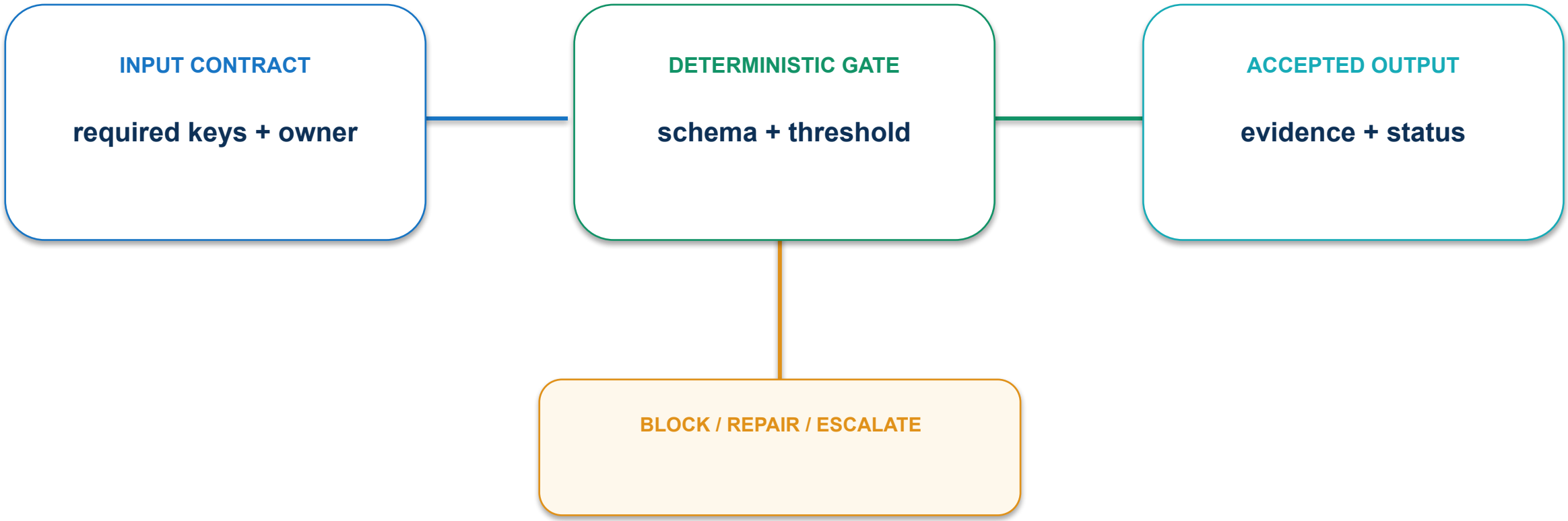
status

FAILURE MODE

Loose integrations permit malformed or over-broad actions.

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04 · API and integration contract: contract and control



Use typed schemas, idempotency and explicit error states.

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04 · API and integration contract: evidence and decision

MEASURE

% requests with schema-valid input and explicit outcome

ACCEPTANCE EVIDENCE

API contract and negative-test results

OBSERVED

machine value

THRESHOLD

approved rule

DECISION

PASS / REPAIR

OWNER

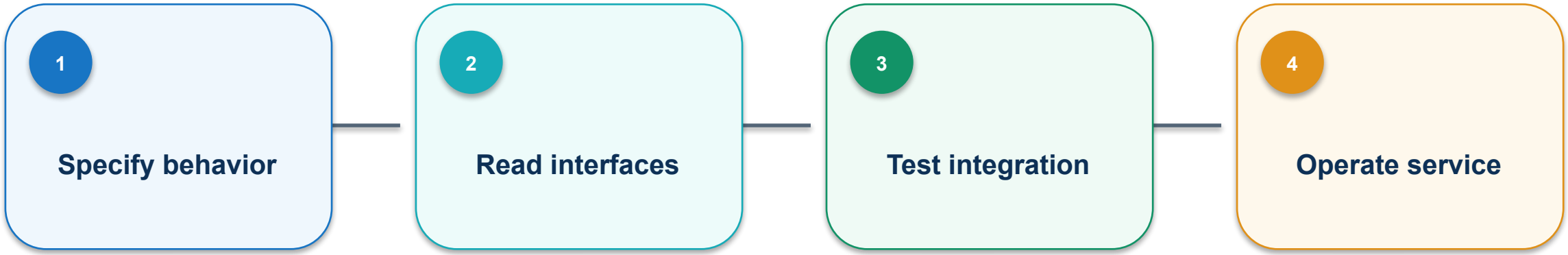
named human

Completion principle: evidence travels with the product hypothesis, changed state and decision—not in an unsupported claim.

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MECHANISM

05 · Programming skill mix: execution path



STATE PASSED BETWEEN STAGES

prompting

api

testing

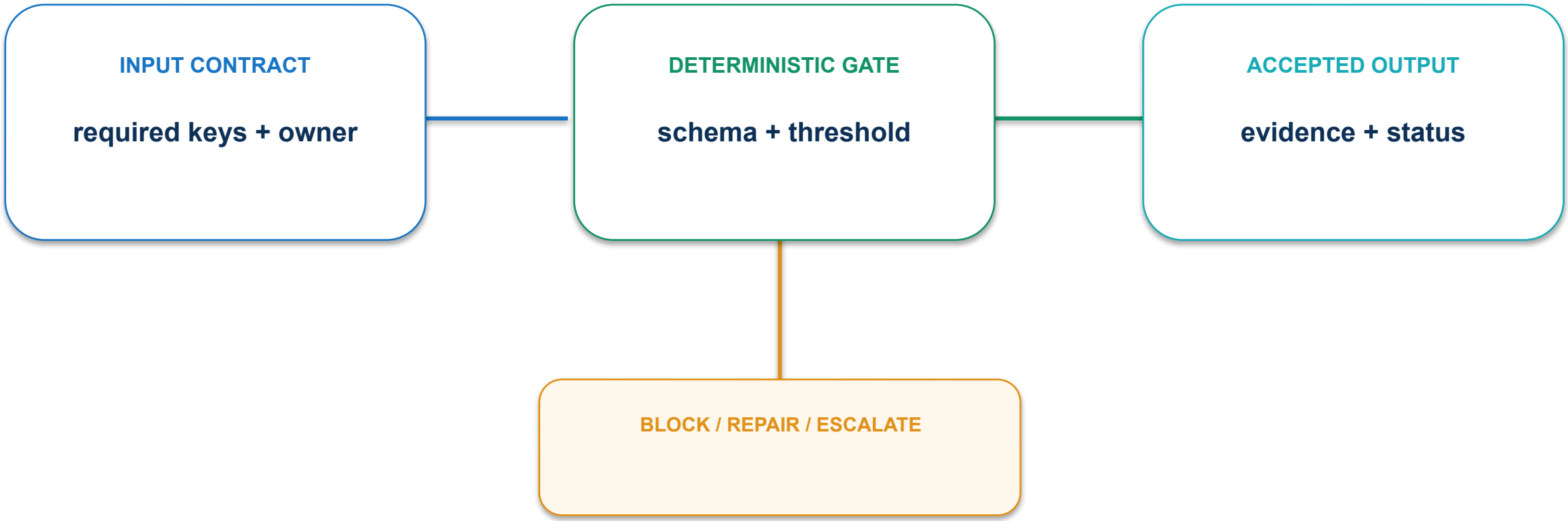
operations

FAILURE MODE

Natural-language prototyping is assumed to replace integration and operational expertise.

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05 · Programming skill mix: contract and control



Plan the product, prompt, data, integration, testing and operations skills explicitly.

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05 · Programming skill mix: evidence and decision

MEASURE

Critical skill roles assigned with accountable owners

ACCEPTANCE EVIDENCE

Skills matrix and gap plan

OBSERVED

machine value

THRESHOLD

approved rule

DECISION

PASS / REPAIR

OWNER

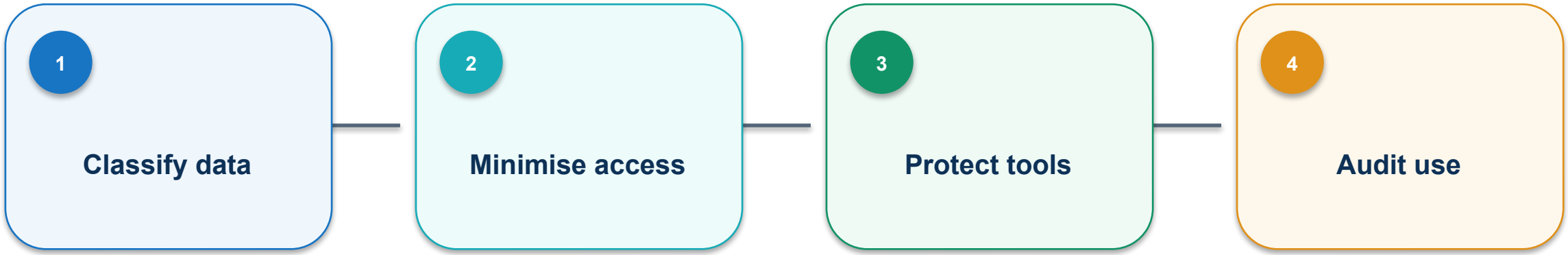
named human

Completion principle: evidence travels with the product hypothesis, changed state and decision—not in an unsupported claim.

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MECHANISM

06 · Security and privacy boundary: execution path



STATE PASSED BETWEEN STAGES

data_class

permission

tool_policy

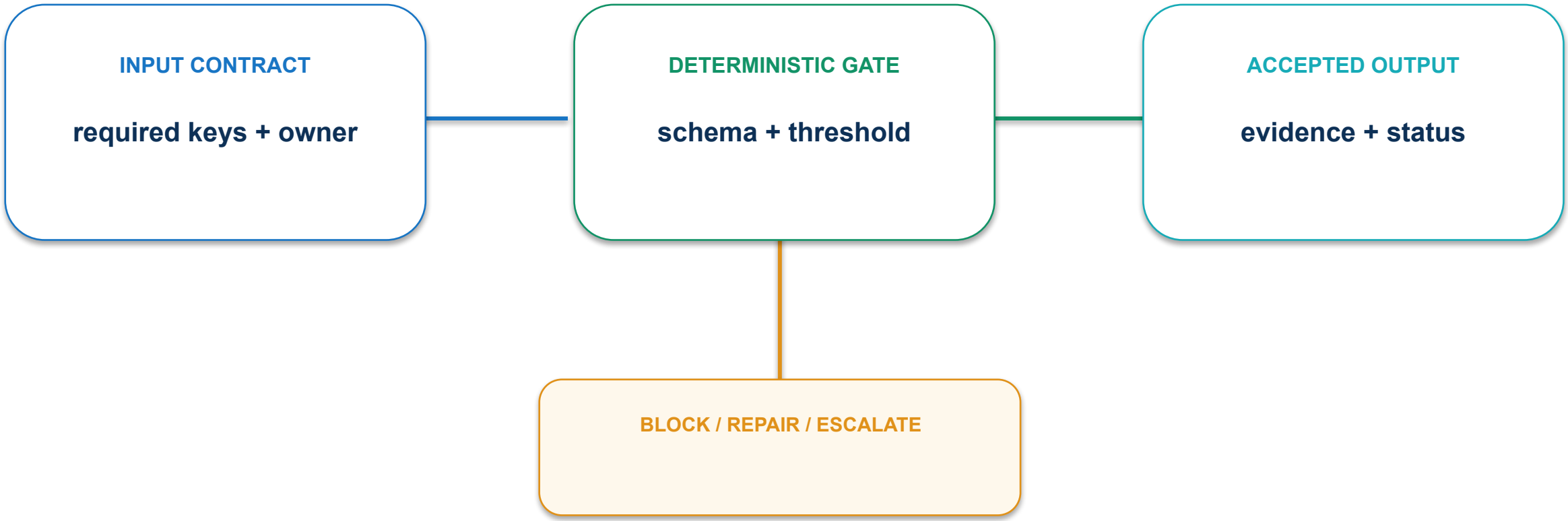
audit

FAILURE MODE

Prompt injection or excess permissions turn untrusted content into consequential actions.

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06 · Security and privacy boundary: contract and control



Layer input controls, tool allowlists, approval and monitoring.

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06 · Security and privacy boundary: evidence and decision

MEASURE

% sensitive paths protected by least privilege

ACCEPTANCE EVIDENCE

Threat model, permission map and tested controls

OBSERVED

machine value

THRESHOLD

approved rule

DECISION

PASS / REPAIR

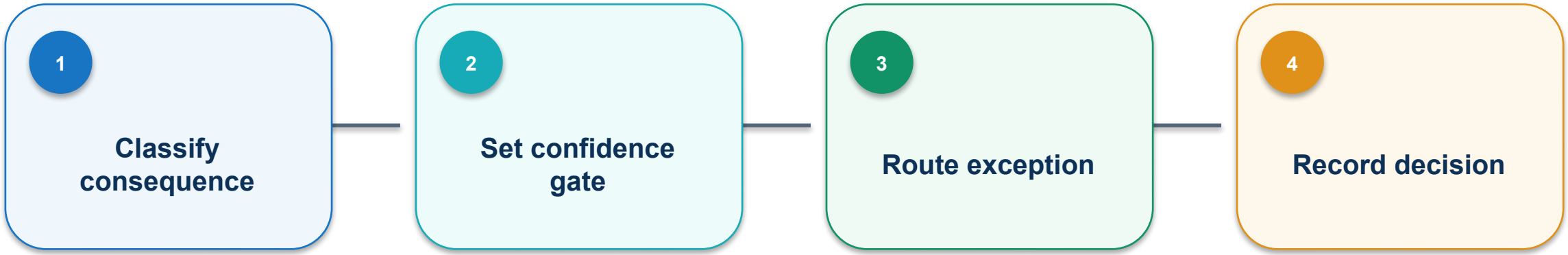
OWNER

named human

Completion principle: evidence travels with the product hypothesis, changed state and decision—not in an unsupported claim.

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07 · Human oversight design: execution path



STATE PASSED BETWEEN STAGES

risk_class

threshold

reviewer

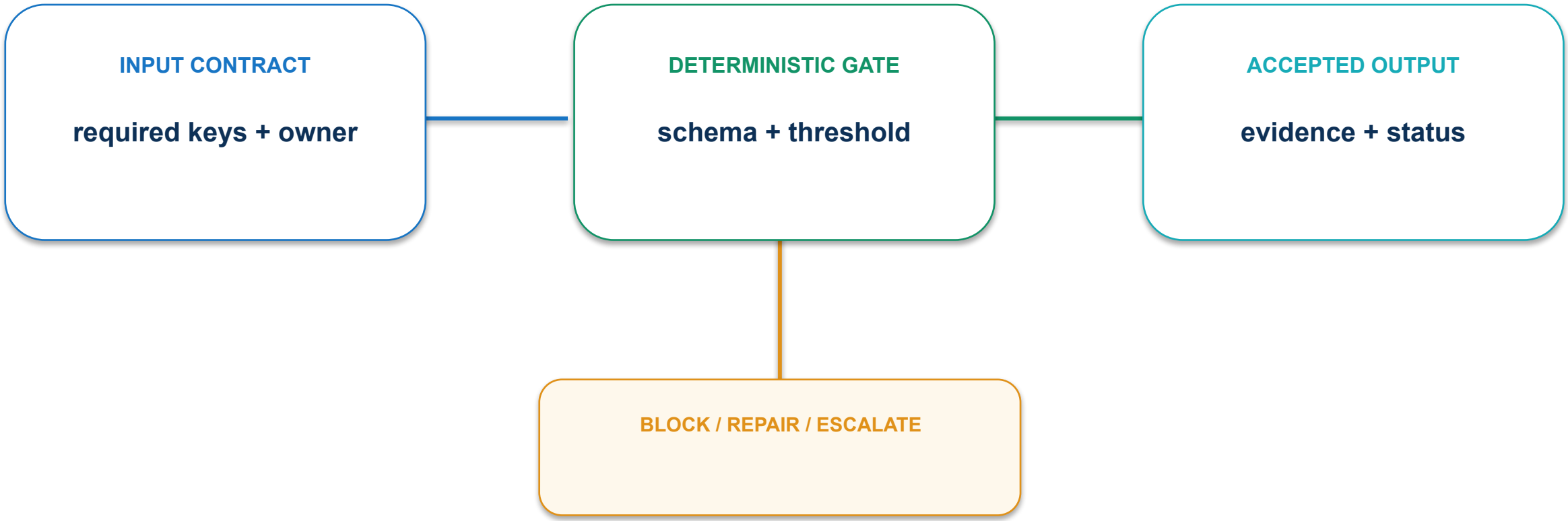
decision

FAILURE MODE

A nominal approval step becomes routine click-through.

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07 · Human oversight design: contract and control



Show the evidence and consequence at the point of decision.

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07 · Human oversight design: evidence and decision

MEASURE

% consequential actions with meaningful human review

ACCEPTANCE EVIDENCE

Approval log tied to current output

OBSERVED

machine value

THRESHOLD

approved rule

DECISION

PASS / REPAIR

OWNER

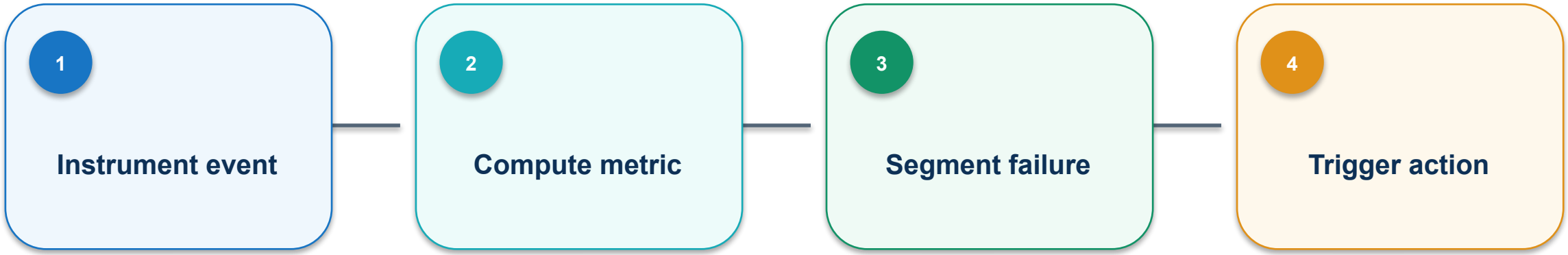
named human

Completion principle: evidence travels with the product hypothesis, changed state and decision—not in an unsupported claim.

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MECHANISM

08 · Observability and product KPIs: execution path



STATE PASSED BETWEEN STAGES

event

metric

segment

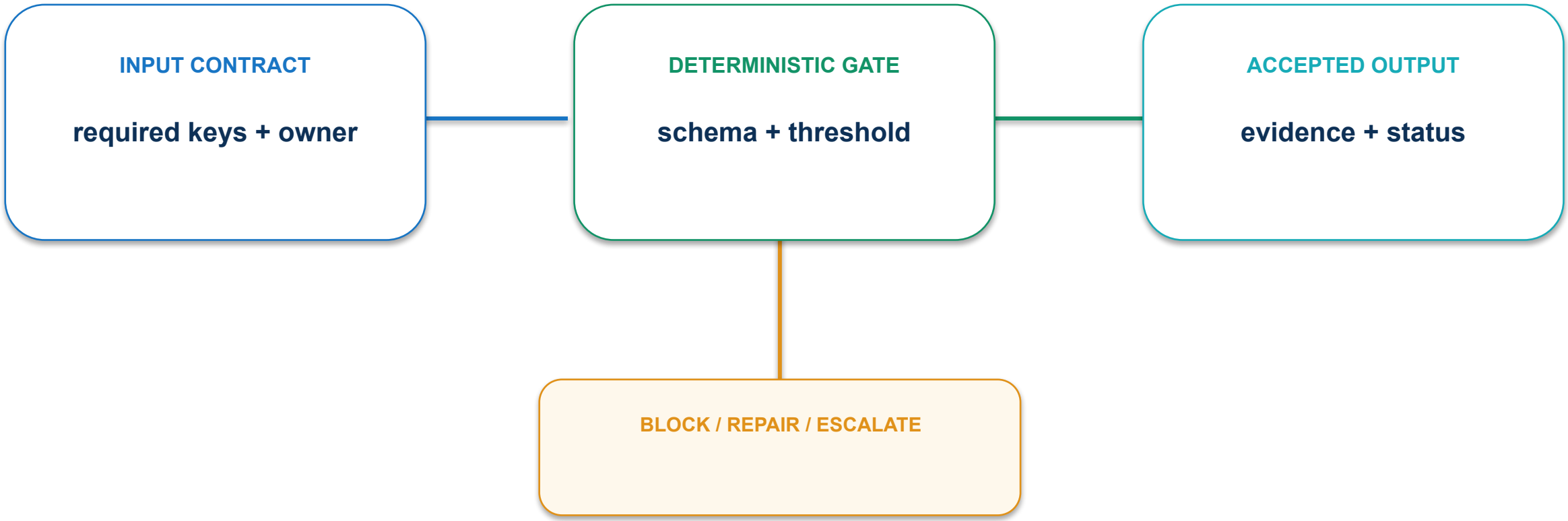
action

FAILURE MODE

Usage growth masks falling answer quality or rising correction work.

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08 · Observability and product KPIs: contract and control



Pair business outcomes with reliability, safety and experience measures.

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08 · Observability and product KPIs: evidence and decision

MEASURE

North-star outcome plus quality, cost and risk guardrails

ACCEPTANCE EVIDENCE

Metric tree and alert ownership

OBSERVED

machine value

THRESHOLD

approved rule

DECISION

PASS / REPAIR

OWNER

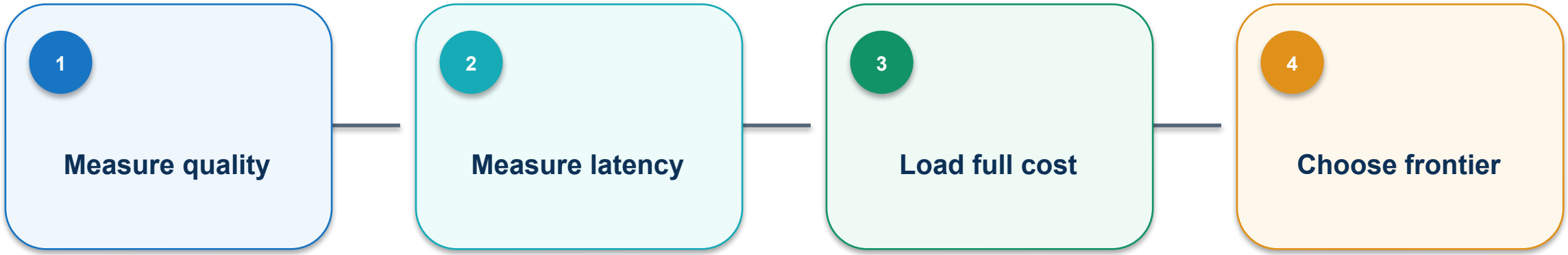
named human

Completion principle: evidence travels with the product hypothesis, changed state and decision—not in an unsupported claim.

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MECHANISM

09 · Quality, latency and cost frontier: execution path



STATE PASSED BETWEEN STAGES

quality

latency

cost

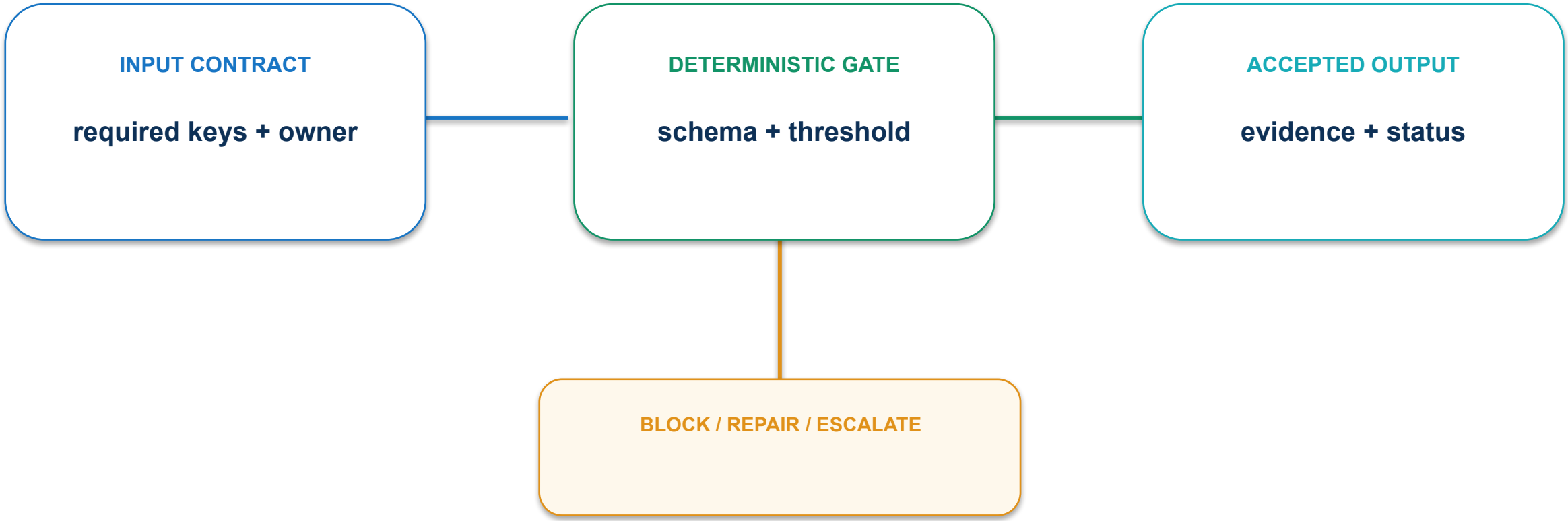
posture

FAILURE MODE

The cheapest or fastest model is chosen without quality-adjusted comparison.

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09 · Quality, latency and cost frontier: contract and control



Compare only configurations meeting minimum quality and risk thresholds.

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09 · Quality, latency and cost frontier: evidence and decision

MEASURE

Accepted outcome cost at target response time

ACCEPTANCE EVIDENCE

Pareto chart and selected operating point

OBSERVED

machine value

THRESHOLD

approved rule

DECISION

PASS / REPAIR

OWNER

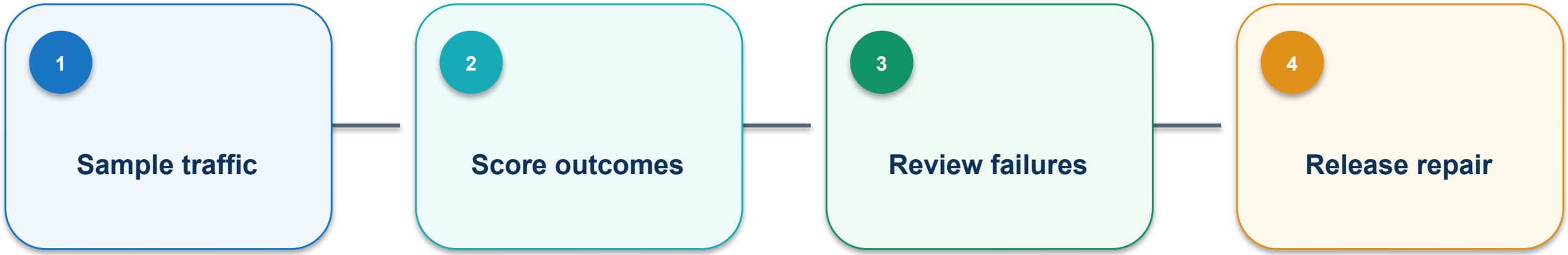
named human

Completion principle: evidence travels with the product hypothesis, changed state and decision—not in an unsupported claim.

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MECHANISM

10 · Production evaluation loop: execution path



STATE PASSED BETWEEN STAGES

sample

score

failure

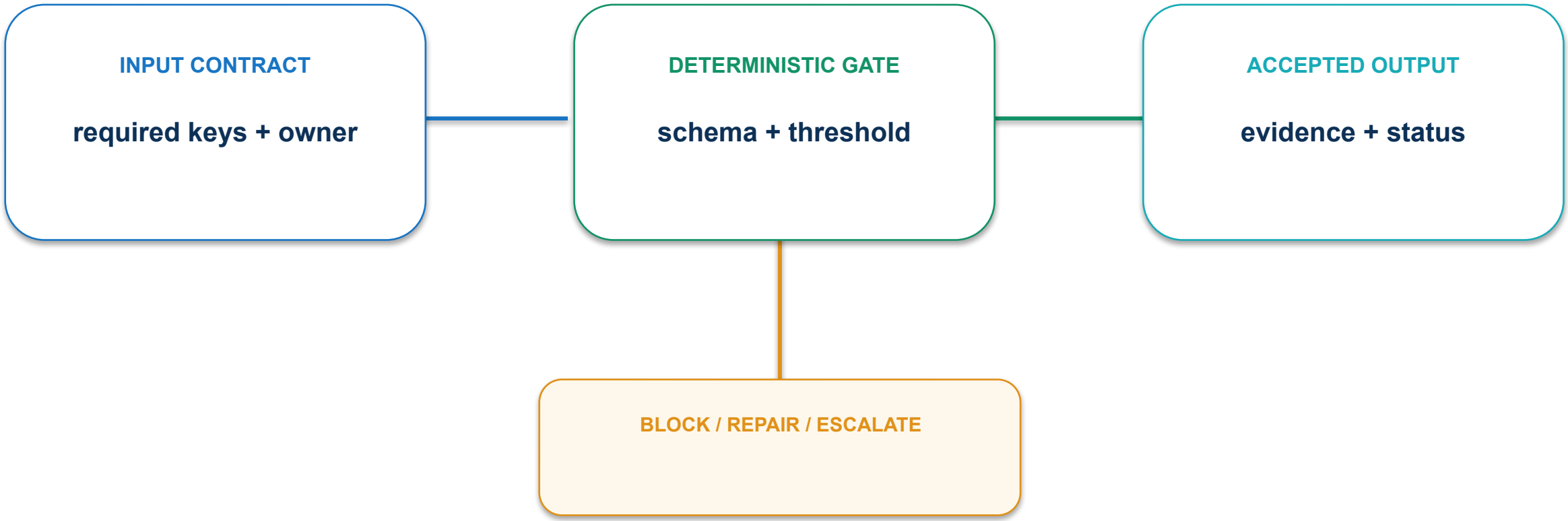
release

FAILURE MODE

Offline evals remain green while live inputs drift.

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10 · Production evaluation loop: contract and control



Combine frozen regression sets with representative live sampling.

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10 · Production evaluation loop: evidence and decision

MEASURE

% production samples scored and actioned

ACCEPTANCE EVIDENCE

Eval dashboard and linked repair record

OBSERVED

machine value

THRESHOLD

approved rule

DECISION

PASS / REPAIR

OWNER

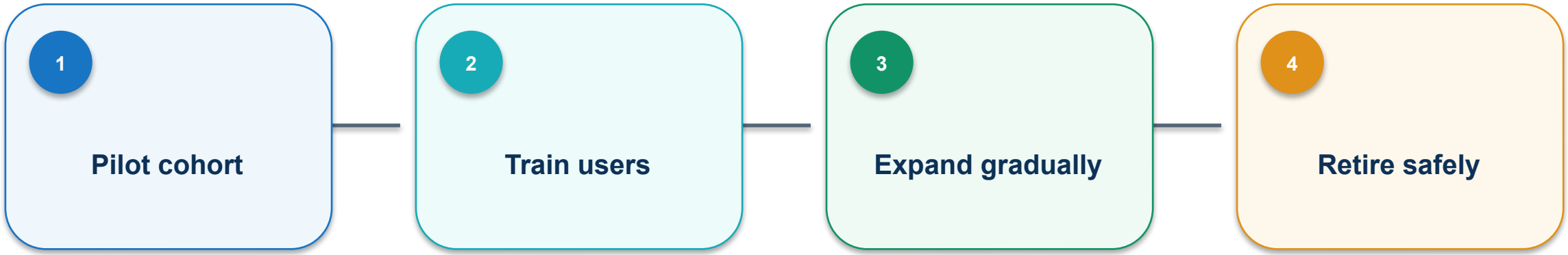
named human

Completion principle: evidence travels with the product hypothesis, changed state and decision—not in an unsupported claim.

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MECHANISM

11 · Rollout and change management: execution path



STATE PASSED BETWEEN STAGES

cohort

enablement

gate

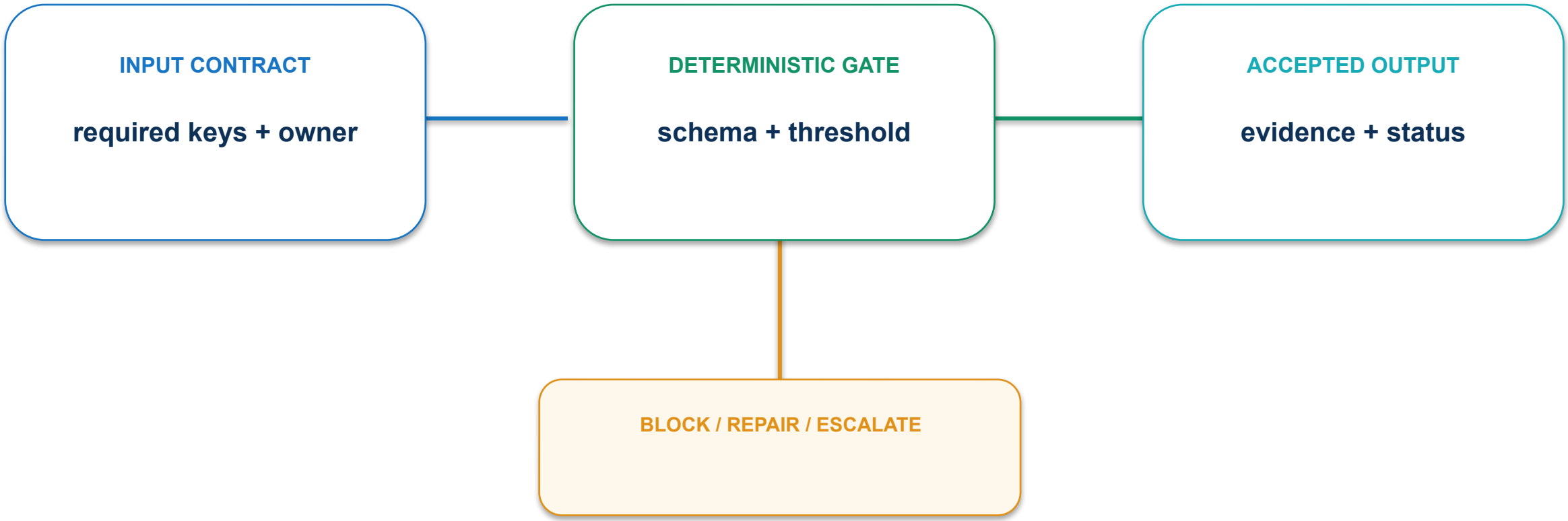
rollback

FAILURE MODE

A big-bang launch creates unmanaged behavior and support load.

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11 · Rollout and change management: contract and control



Use staged rollout, telemetry, support playbooks and rollback criteria.

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11 · Rollout and change management: evidence and decision

MEASURE

Cohorts expanded only after approved thresholds

ACCEPTANCE EVIDENCE

Rollout plan with go/no-go gates

OBSERVED

machine value

THRESHOLD

approved rule

DECISION

PASS / REPAIR

OWNER

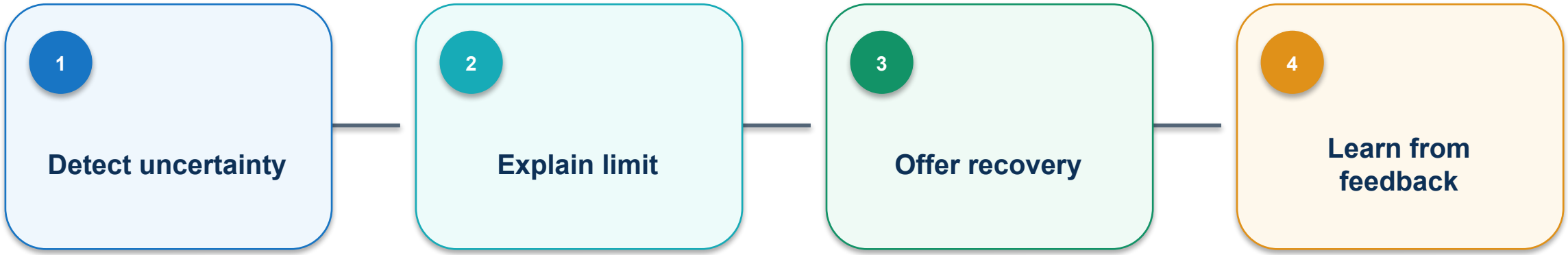
named human

Completion principle: evidence travels with the product hypothesis, changed state and decision—not in an unsupported claim.

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MECHANISM

12 · Feedback and graceful failure: execution path



STATE PASSED BETWEEN STAGES

uncertainty

message

recovery

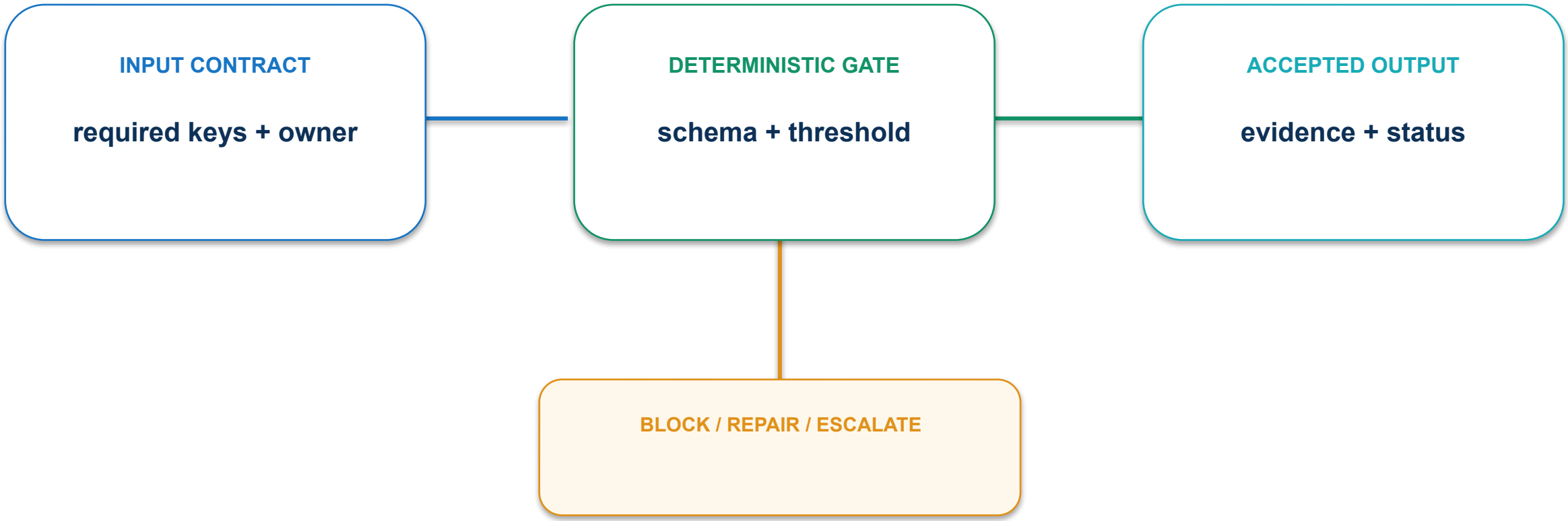
feedback

FAILURE MODE

The product hides uncertainty or traps users in repeated bad answers.

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12 · Feedback and graceful failure: contract and control



Design honest limits, user control and a path to human help.

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12 · Feedback and graceful failure: evidence and decision

MEASURE

Successful recovery rate after agent failure

ACCEPTANCE EVIDENCE

Failure-state prototype and feedback taxonomy

OBSERVED

machine value

THRESHOLD

approved rule

DECISION

PASS / REPAIR

OWNER

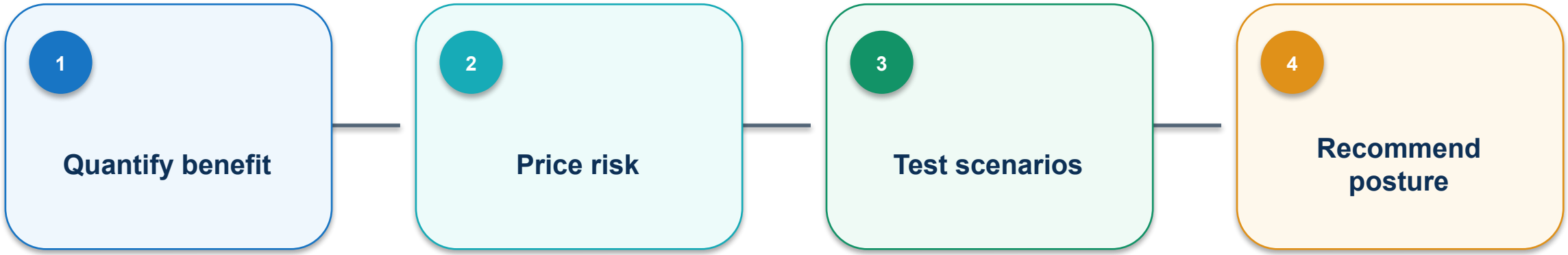
named human

Completion principle: evidence travels with the product hypothesis, changed state and decision—not in an unsupported claim.

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MECHANISM

13 · Benefit-trade-off recommendation: execution path



STATE PASSED BETWEEN STAGES

benefit

risk

scenario

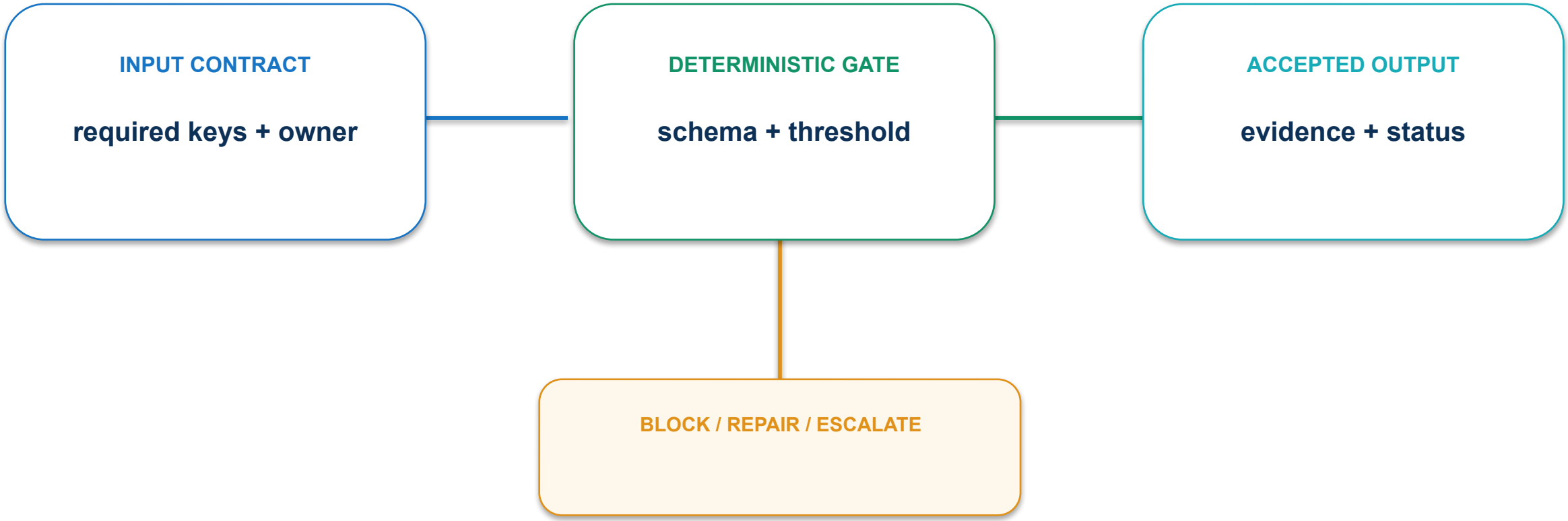
recommendation

FAILURE MODE

The business case counts automation savings but excludes oversight, incidents and change cost.

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13 · Benefit-trade-off recommendation: contract and control



Use risk-adjusted scenarios and issue a pilot, expand, hold or stop decision.

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13 · Benefit-trade-off recommendation: evidence and decision

MEASURE

Net value remains positive under downside scenarios

ACCEPTANCE EVIDENCE

Executive decision memo with conditions and owner

OBSERVED

machine value

THRESHOLD

approved rule

DECISION

PASS / REPAIR

OWNER

named human

Completion principle: evidence travels with the product hypothesis, changed state and decision—not in an unsupported claim.

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Deployment boundary: product, platform and accountable owner

1

PRODUCT

Set value and experience target

2

PLATFORM

Operate agent and telemetry

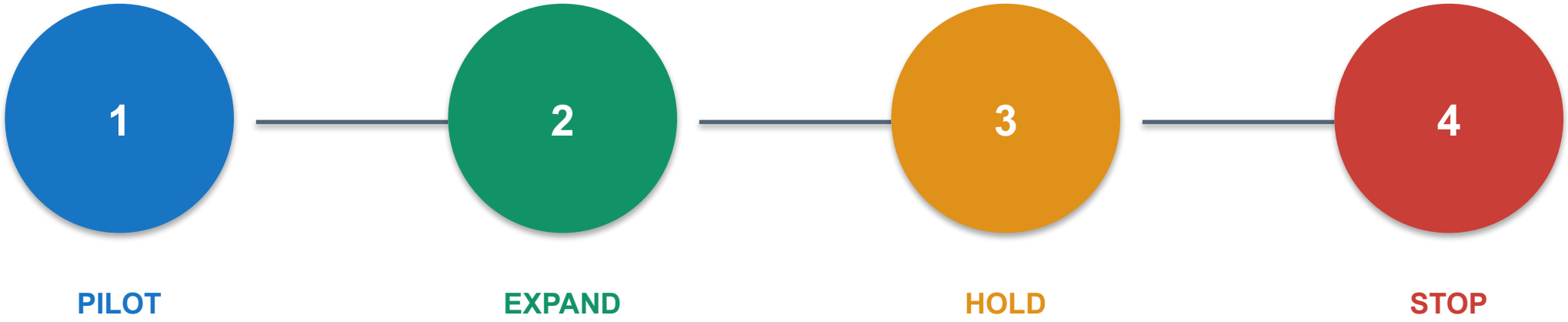
3

OWNER

Approve rollout and exceptions

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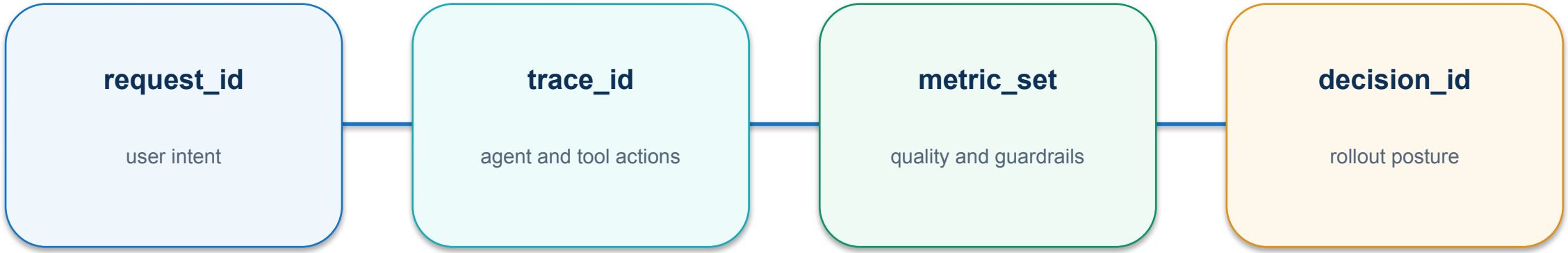
Adoption state: PILOT, EXPAND, HOLD and STOP



The adoption posture records evidence, conditions, owner and review date.

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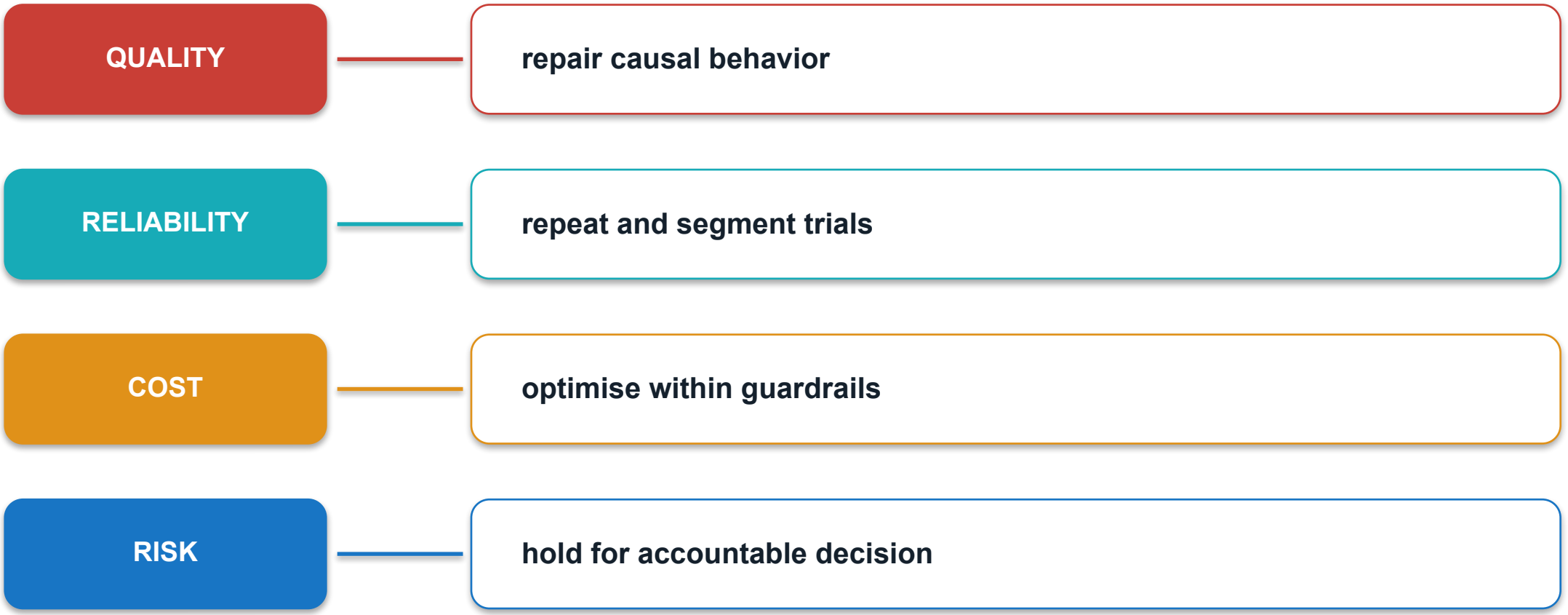
Production evidence chain: request → trace → metric → decision



A production recommendation is defensible when every metric resolves to a trace and owner.

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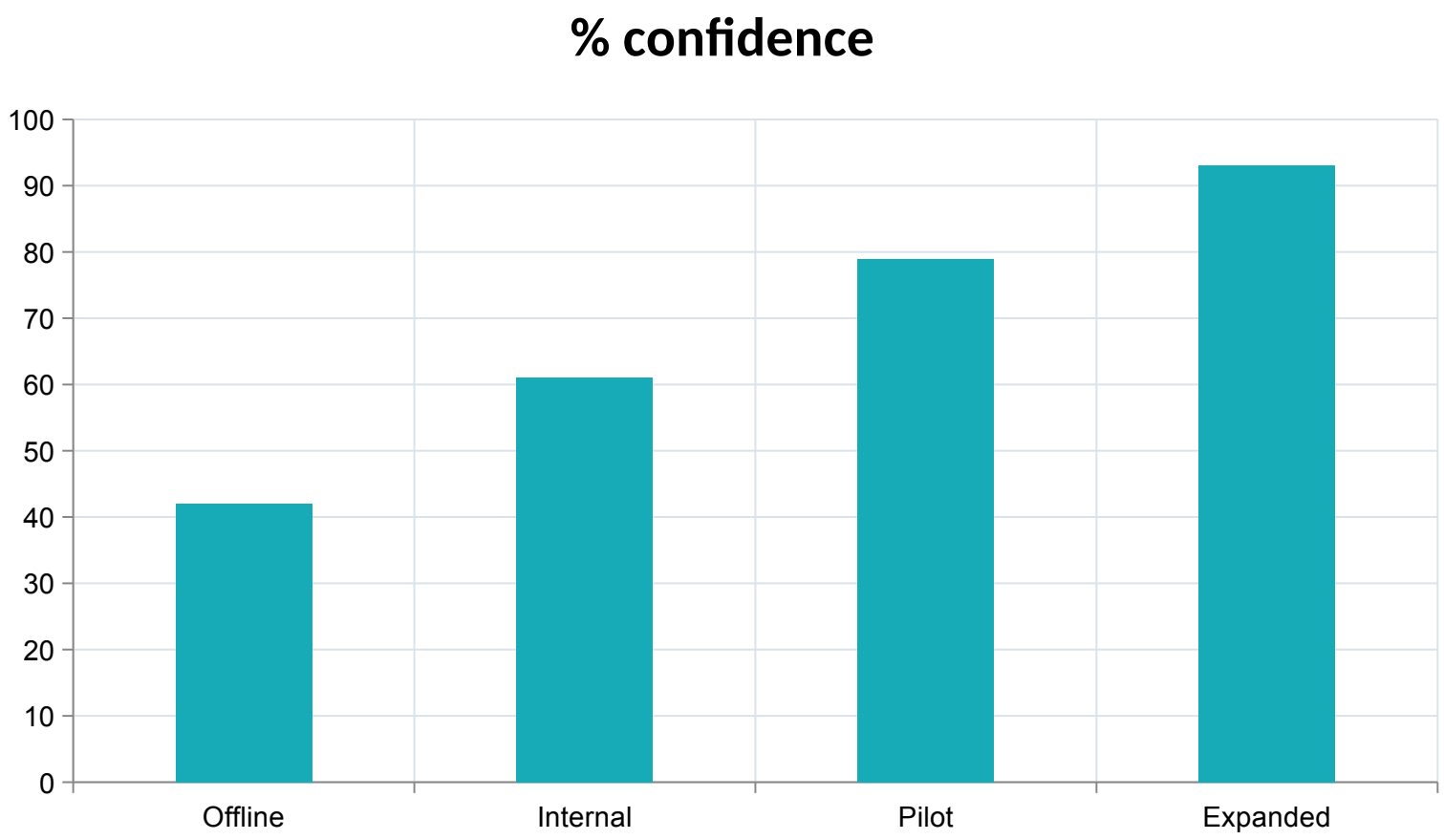
Deployment taxonomy: quality, reliability, cost and risk



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EDITABLE NATIVE CHART

Staged rollout increases decision confidence



INSIGHT

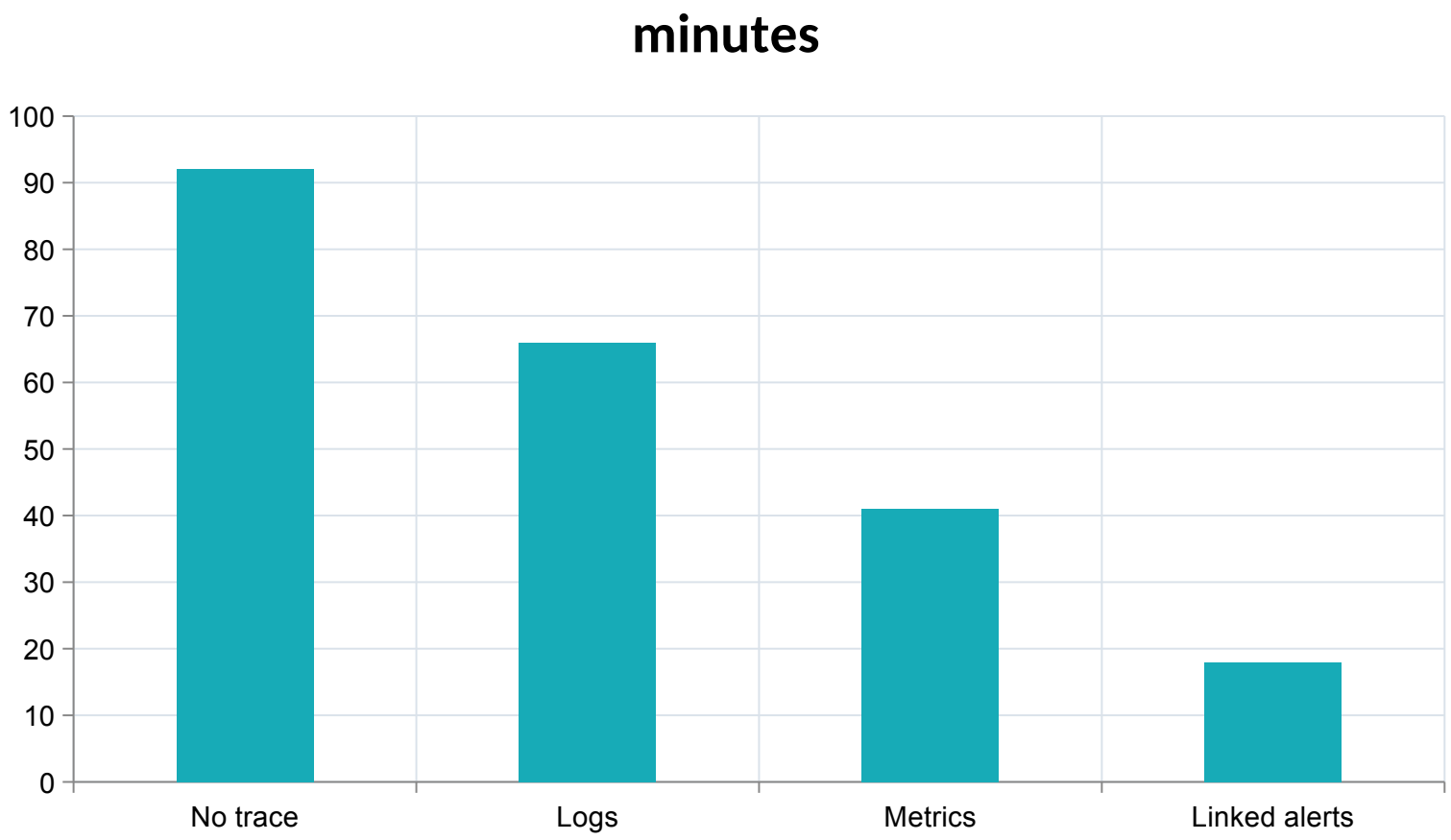
Staged rollout converts production evidence into a safer expansion decision.

Illustrative synthetic values for mechanism explanation.

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EDITABLE NATIVE CHART

Observability shortens recovery time



INSIGHT

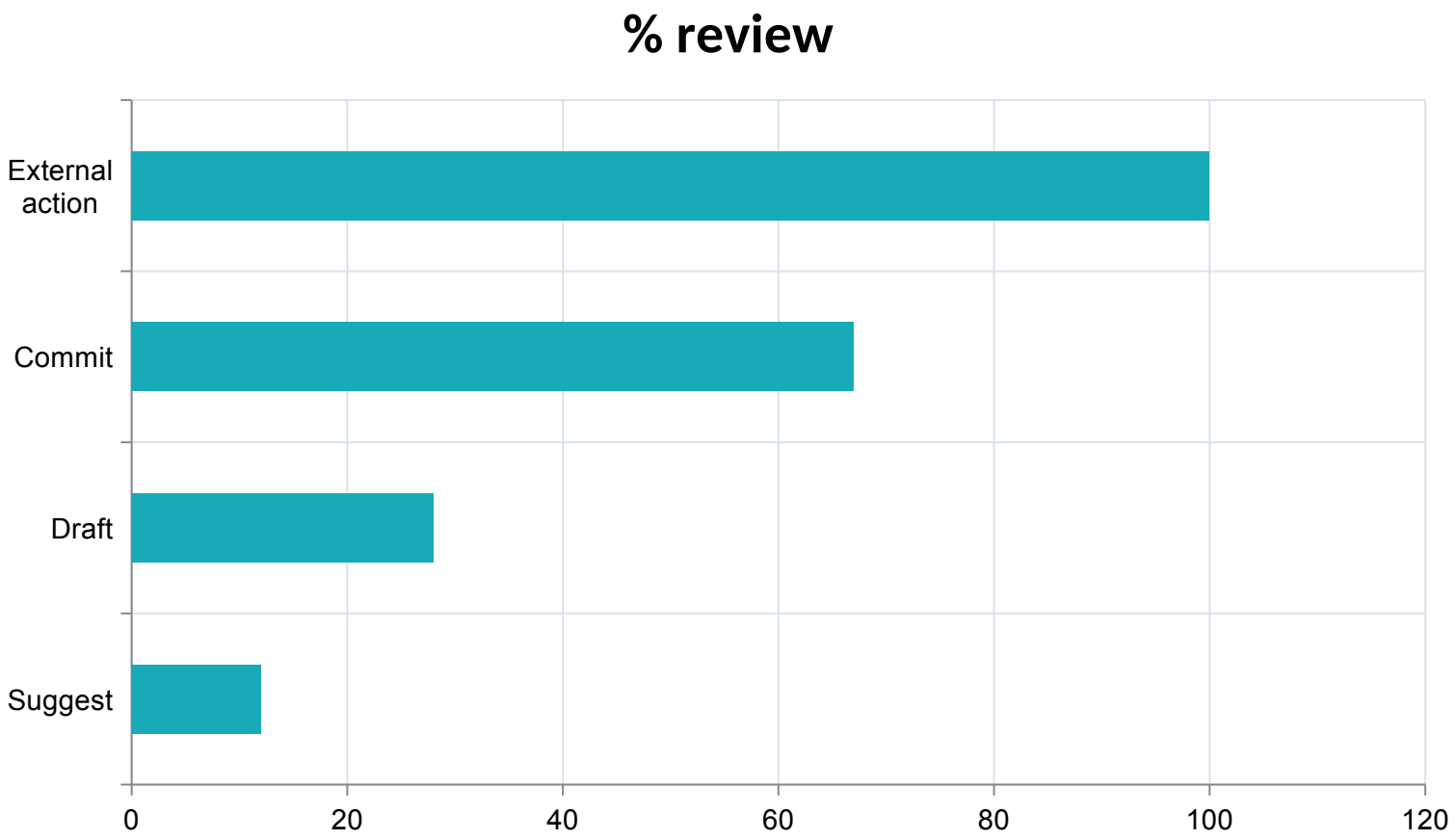
Linked traces, metrics and alerts shorten the path from symptom to accountable action.

Illustrative synthetic values for mechanism explanation.

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EDITABLE NATIVE CHART

Human oversight rises with consequence



INSIGHT

Human oversight should increase with the consequence and reversibility of the action.

Illustrative synthetic values for mechanism explanation.

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ACTIVITY BRIEF

Activity 09 · Select a Deployment Posture

SCENARIO

Compare channel and architecture options for an agentic product using task, integration and risk evidence.

EXPECTED OUTPUT

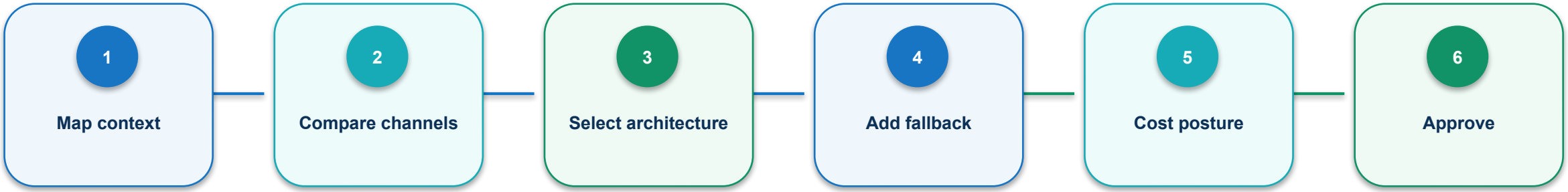
Deployment matrix, reference architecture and fallback path

Detailed procedures, prompts, troubleshooting and acceptance checks are in the Learner Guide and activity PDF.

Self-contained folder: activities/activity-09-deployment-decision/

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Activity 09 · agent workflow contract



Consequential actions require a current diff, passing checks and a named decision.

HUMAN GATE

Approval resumes only when the reviewed diff and current state still match.

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Activity 09 · evidence and failure gates

PASS EVIDENCE

The selected option meets task, integration, safety and recovery thresholds; caveats are explicit.

FAIL CLOSED WHEN

a protected boundary is crossed; evidence is missing; approval is absent; or the current diff no longer matches review.

README.pdf

detailed procedure

mock-data/

synthetic source state

templates/

learner working files

evidence/

acceptance record

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ACTIVITY BRIEF

Activity 10 · Conduct a Risk and Control Review

SCENARIO

Threat-model an agent workflow and assign layered controls, evidence and owners.

EXPECTED OUTPUT

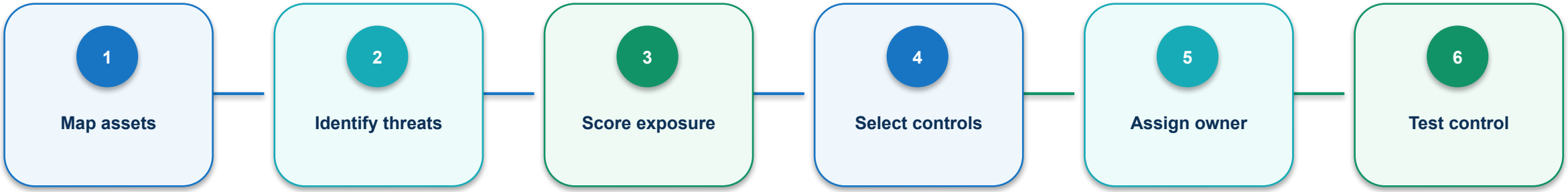
Risk register, permission map and control test plan

Detailed procedures, prompts, troubleshooting and acceptance checks are in the Learner Guide and activity PDF.

Self-contained folder: activities/activity-10-risk-control-review/

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Activity 10 · agent workflow contract



Consequential actions require a current diff, passing checks and a named decision.

HUMAN GATE

Approval resumes only when the reviewed diff and current state still match.

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Activity 10 · evidence and failure gates

PASS EVIDENCE

High risks have preventive and detective controls, named owners and testable evidence.

FAIL CLOSED WHEN

a protected boundary is crossed; evidence is missing; approval is absent; or the current diff no longer matches review.

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detailed procedure

mock-data/

synthetic source state

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learner working files

evidence/

acceptance record

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ACTIVITY BRIEF

Activity 11 · Design KPI and Feedback Dashboard

SCENARIO

Create a metric tree and monitoring view that balances business value, quality, experience, cost and risk.

EXPECTED OUTPUT

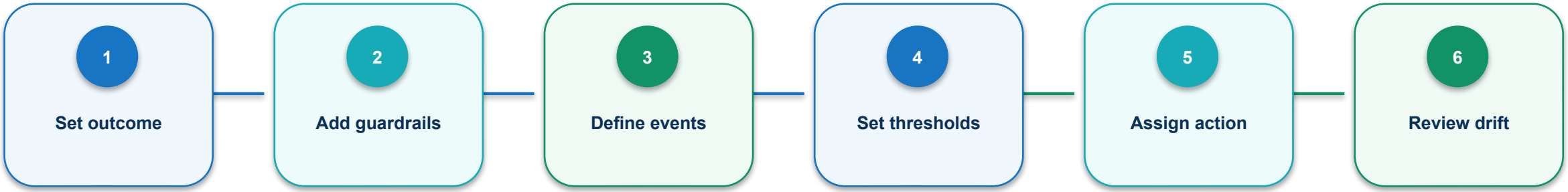
KPI tree, alert table and feedback taxonomy

Detailed procedures, prompts, troubleshooting and acceptance checks are in the Learner Guide and activity PDF.

Self-contained folder: activities/activity-11-kpi-feedback-dashboard/

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Activity 11 · agent workflow contract



Each node consumes named keys, emits a status, and leaves retrievable execution evidence.

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Activity 11 · evidence and failure gates

PASS EVIDENCE

Metrics have formulas, data sources, thresholds, segments and action owners; no vanity-only measure drives release.

FAIL CLOSED WHEN

a protected boundary is crossed; evidence is missing; approval is absent; or the current diff no longer matches review.

README.pdf

detailed procedure

mock-data/

synthetic source state

templates/

learner working files

evidence/

acceptance record

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ACTIVITY BRIEF

Activity 12 · Deliver a Governed Product Recommendation

SCENARIO

Synthesize value, evaluation, readiness and risk evidence into an executive pilot, expand, hold or stop decision.

EXPECTED OUTPUT

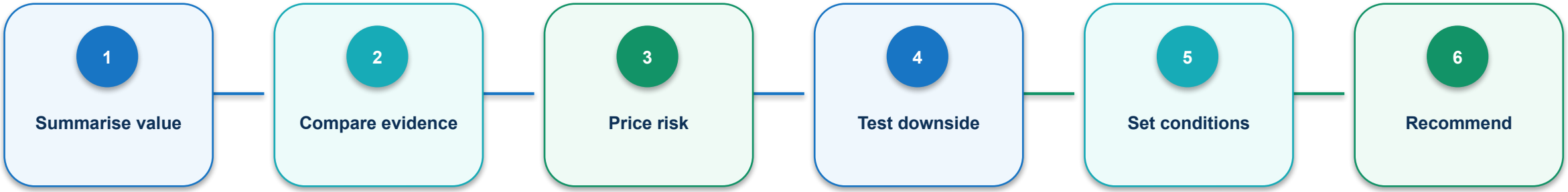
One-page decision memo and 90-day rollout scorecard

Detailed procedures, prompts, troubleshooting and acceptance checks are in the Learner Guide and activity PDF.

Self-contained folder: activities/activity-12-executive-recommendation/

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Activity 12 · agent workflow contract



Consequential actions require a current diff, passing checks and a named decision.

HUMAN GATE

Approval resumes only when the reviewed diff and current state still match.

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Activity 12 · evidence and failure gates

PASS EVIDENCE

The recommendation is traceable to evidence, survives a downside scenario, and has measurable exit criteria.

FAIL CLOSED WHEN

a protected boundary is crossed; evidence is missing; approval is absent; or the current diff no longer matches review.

README.pdf

detailed procedure

mock-data/

synthetic source state

templates/

learner working files

evidence/

acceptance record

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Topic 3 technical checkpoint

1. Which state field prevents a silent downstream assumption?

2. What evidence proves the mechanism completed successfully?

3. Which failure must stop rather than retry?

4. Who owns the consequential decision?

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CHECKPOINT

Topic 3 transfer challenge

1. Change one input constraint and predict the affected nodes.
2. Name the identifier that preserves provenance across the change.
3. Define one machine threshold and one human decision.
4. Describe the minimum evidence required to resume the run.

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Course achievement map

1. **LO1 Evaluate agentic AI builders for product research and development, comparing strengths, limitations, resources and skills.**

2. **LO2 Apply optimisation techniques to improve the efficiency, quality and control of agentic product-development workflows.**

3. **LO3 Evaluate the benefits and trade-offs of deploying agentic AI-powered product solutions and recommend a governed path forward.**

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Access the learning platform

LMS

Use the LMS for learner resources and course materials.

1. **Open https://lms-tms.tertiaryinfotech.com/**
2. Use your assigned learner account
3. Confirm the course title and code before downloading files
4. Ask the trainer if access is blocked

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Agentic product controls that must remain intact

GATE

Automation is complete only when governance travels with the product decision.

1. **Every agent role has a bounded goal and authority**
2. Every handoff preserves source and state provenance
3. Every consequential action has a named human owner
4. Every release claim resolves to evaluation evidence

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References and further study

REF

The course synthesises the original deck with both supplied eBooks and current primary guidance.

1. **AI Product Management — Aman Khan**
2. Building AI-Powered Products — Marily Nika
3. Anthropic — Building effective agents
4. Google PAIR — People + AI Guidebook
5. NIST — AI Risk Management Framework
6. Tertiary Courses — current course specification

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Course feedback

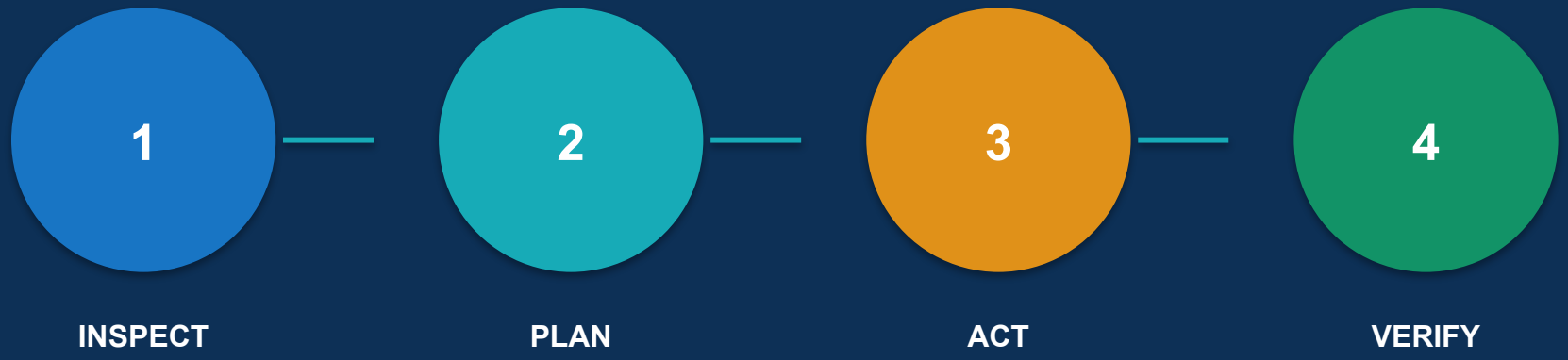
Tell us how the course worked for you before you leave.

1. **Complete the course feedback survey**
2. Tell us what worked well
3. Suggest what we can improve
4. Your certificate of completion is emailed after the course

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Build agentic products that can prove what happened.

Evidence before completion • Human approval before consequence



Dr Alfred Ang | Tertiary Infotech Academy Pte. Ltd.

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